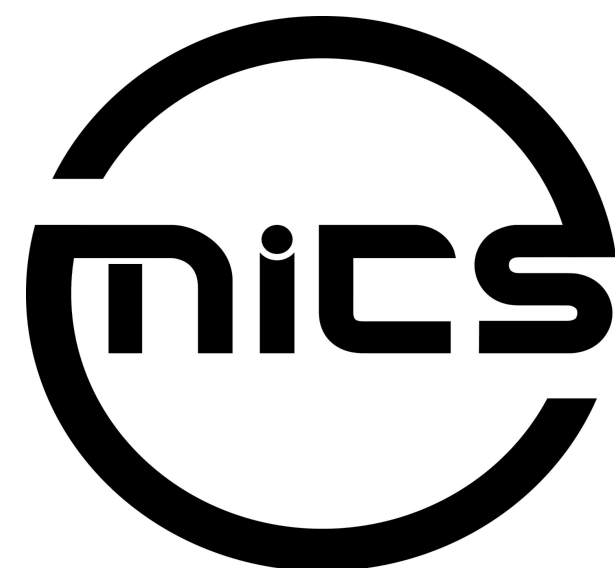


Medical Imaging and Deep Learning

General Overview

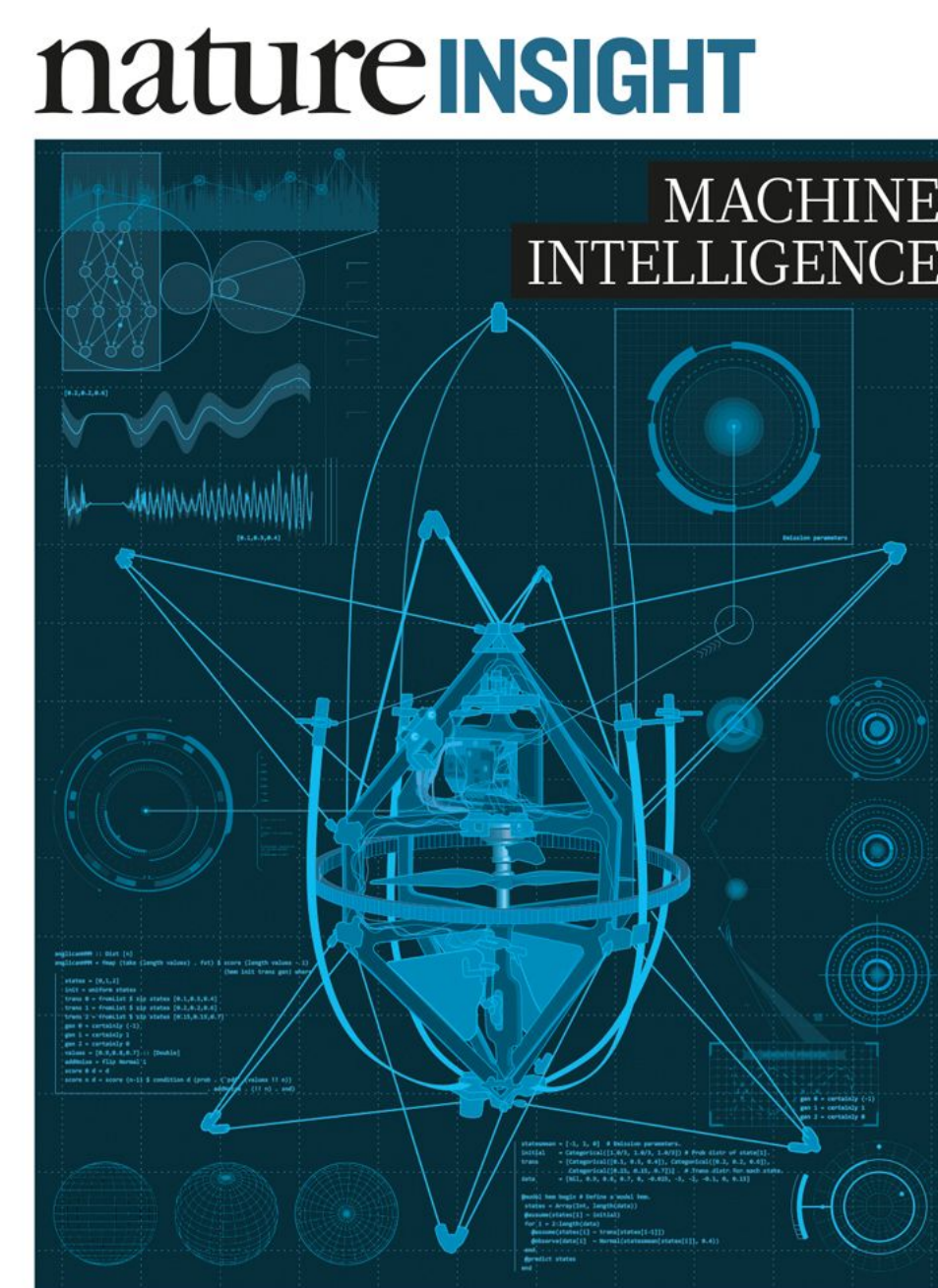
Maria Vakalopoulou
Mathematics and Informatics (MICS)
CentraleSupélec, University Paris-Saclay



Motivation

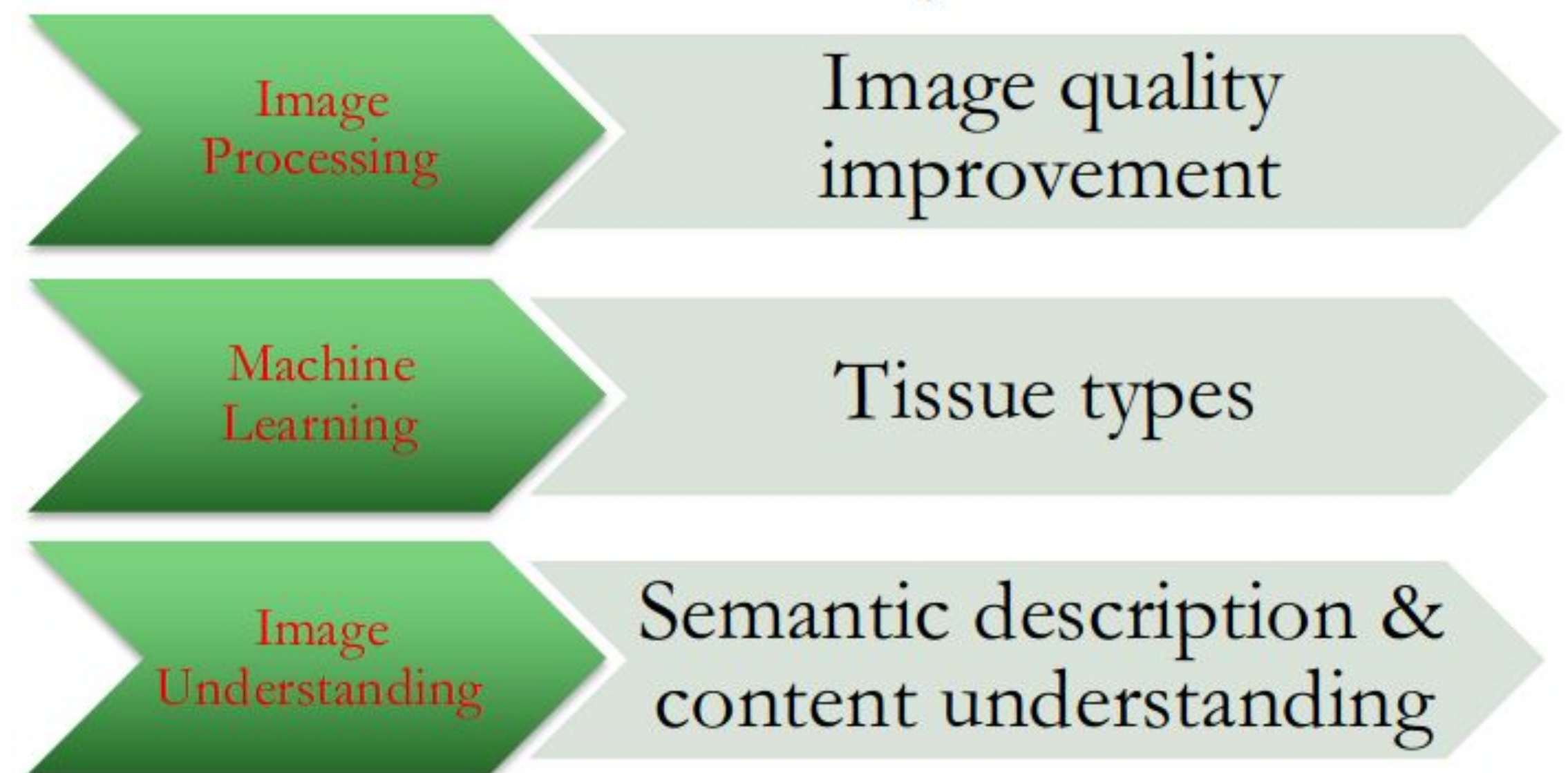
- **Artificial Intelligence for Health Care**

Geoff Hinton: “Medical schools should stop training radiologists now” (AI vs MD, New Yorker, April 2017)



Medical Imaging

- The most direct way to see inside the human (or animal) body is cut it open (i.e. surgery)
- We can see inside the human body in ways that are less invasive or (completely non-invasive)
- We can see metabolic/functional/molecular activities which are not visible to naked eye



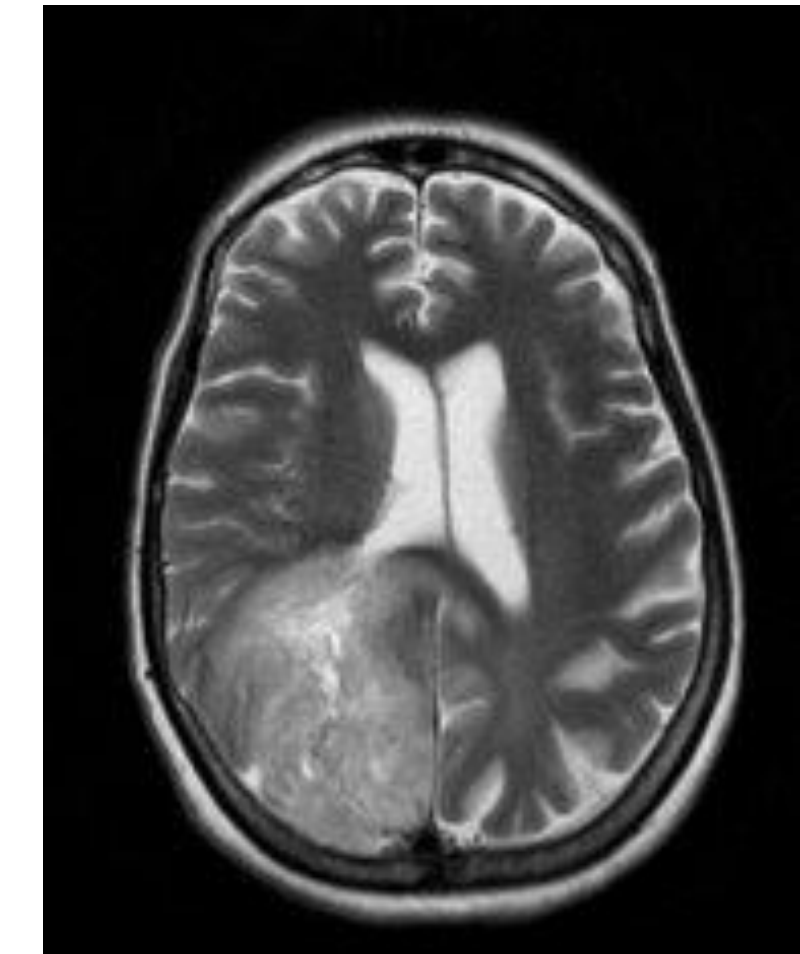
Medical Imaging

- **Medical Care (Clinical Practice)**

- Detection of lesions/ anomalies
- Screening
- Quantification (extraction of biomarkers)
- Follow up pathology

65-year old female

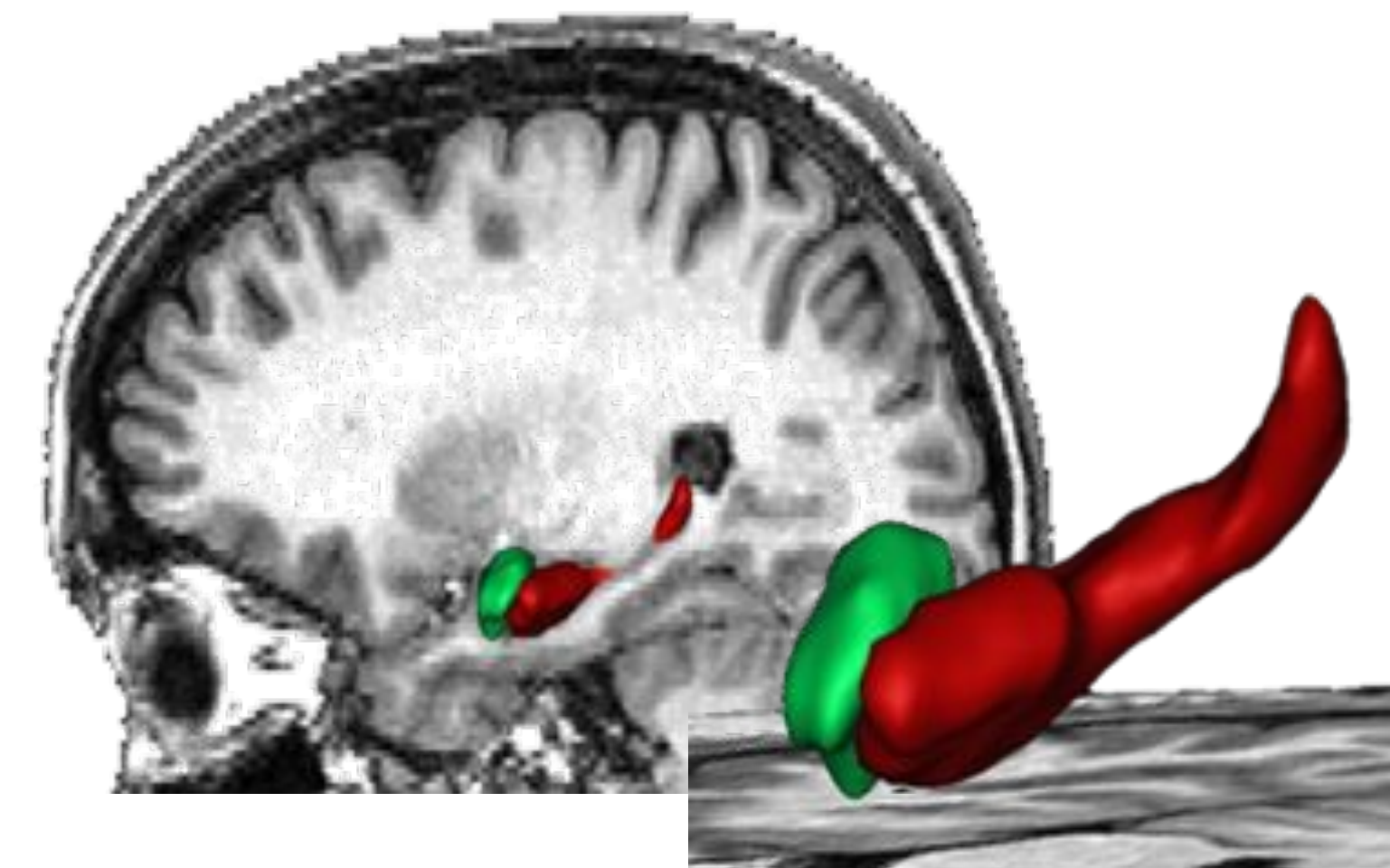
Presented with
left-side weakness and
headache



Brain tumor and Glioblastoma

- **Research (Medical Image Computing)**

- Automate the clinical process
- Prognosis e.g. evaluation of a specific treatment
- Understand the alternation between specific diseases



Medical Imaging

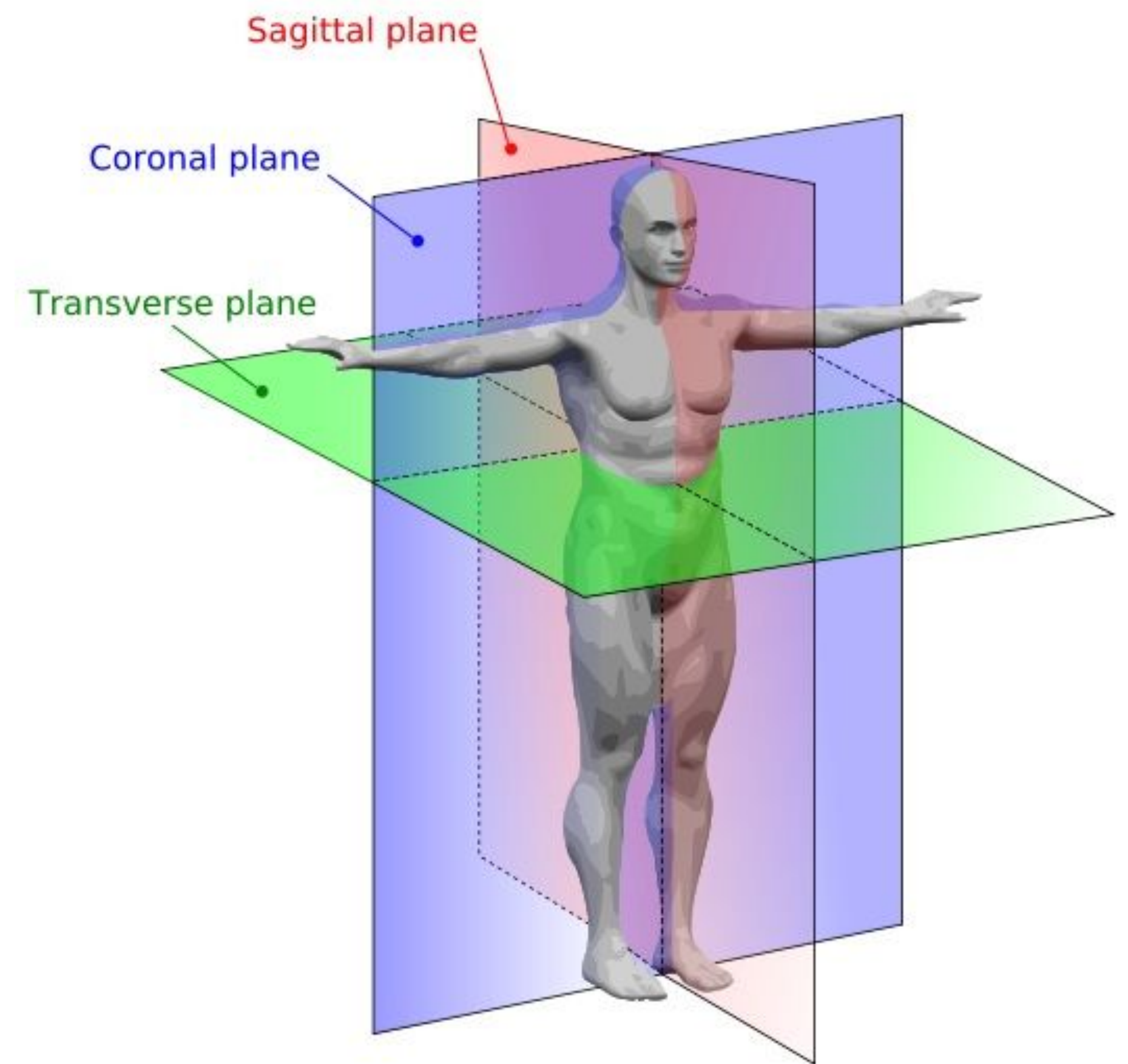
Non-invasive visualization of internal organs, tissues, etc:

- **Representations**

- 2D signal $f(x,y)$
- 3D signal $f(x,y,z)$

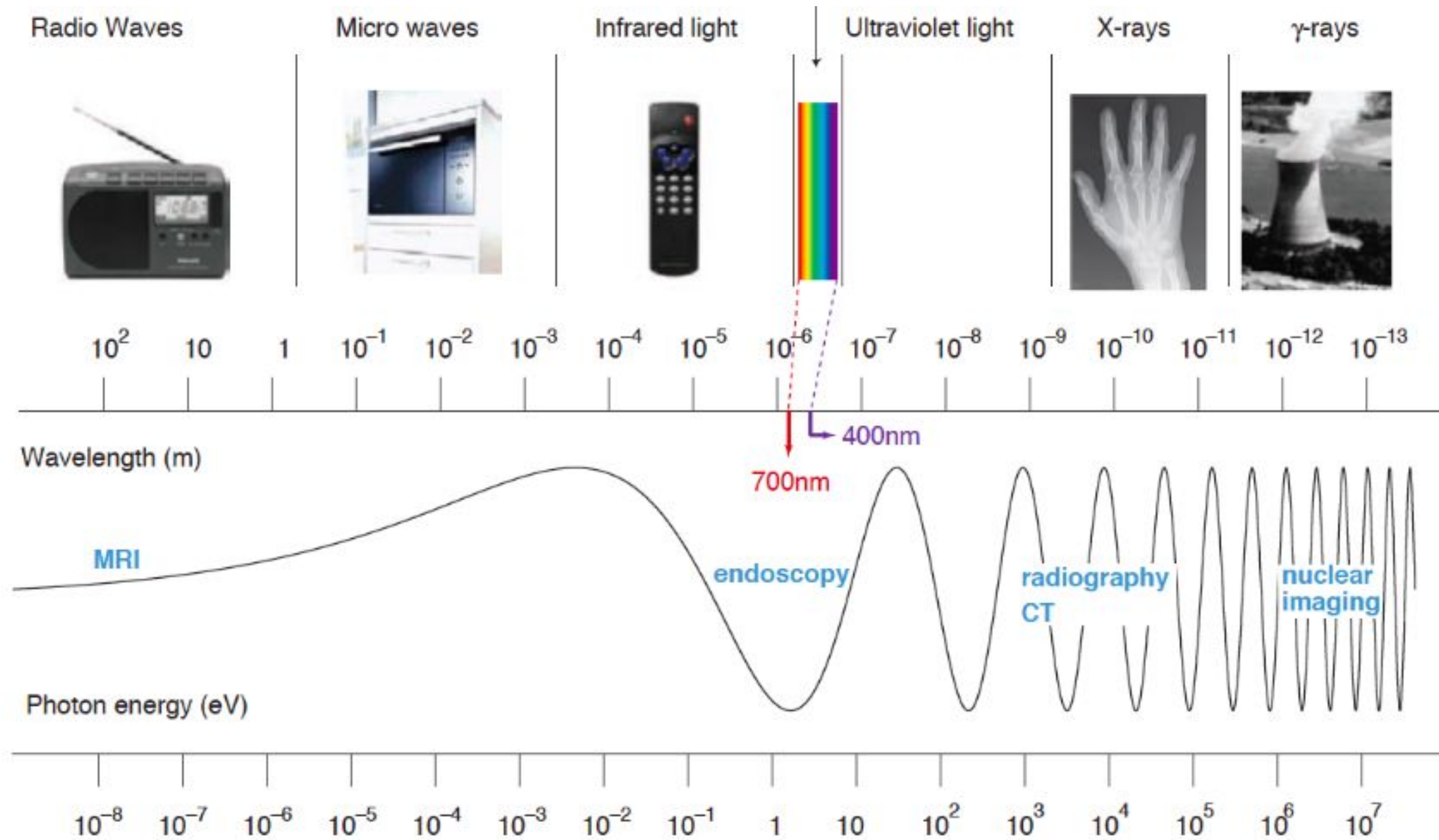
- **Major Modalities**

- Projection X-ray (Radiography)
- X-ray Computed Tomography (CT)
- Nuclear Medicine (PET, ...)
- Ultrasound
- Magnetic Resonance Imaging (MRI)
- Mammography



Medical Imaging

Main Modalities



ELECTROMAGNETIC SPECTRUM (P. Suetens)

Medical Image Analysis

- Because of the rapid technical advances in medical imaging technology and the introduction of new clinical application, the field has become a **highly active research field**.
- Improvements in **image quality**, changing **clinical requirements**, advances in **computer hardware** and **algorithmic progress** (deep learning) in processing all have a direct impact on the state of the art in medical image analysis
- Medical images are often **multidimensional** (2D,3D,4D,nD) have a large dynamic range, are produced on different imaging modalities.
 - A high resolution MR image of the brain, for instance, may consist of more than 300 slices of 512x512 voxels each
 - The data, even a single image in the case of histopathology, can easily exceed some GBs

Main Modalities

X-ray radiography



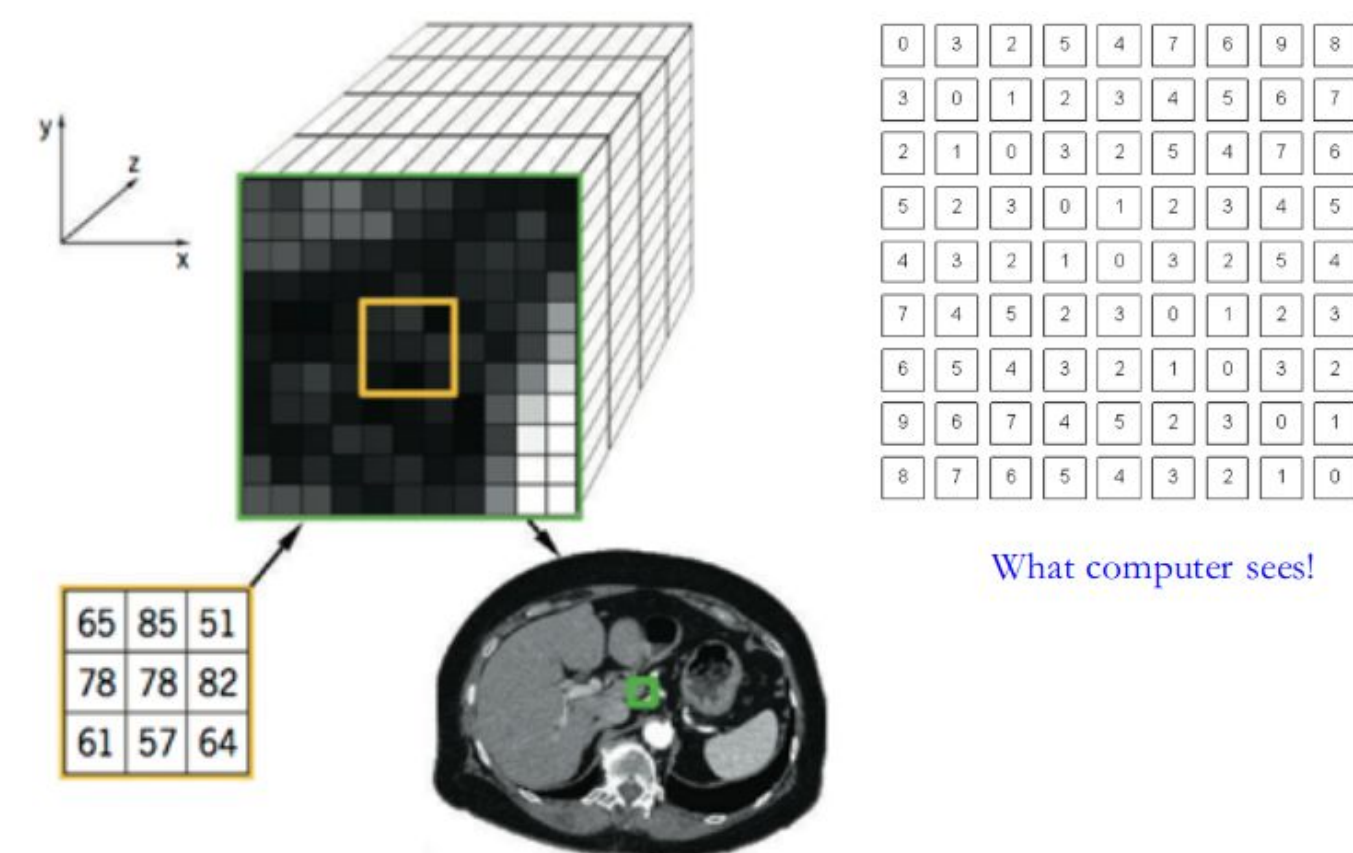
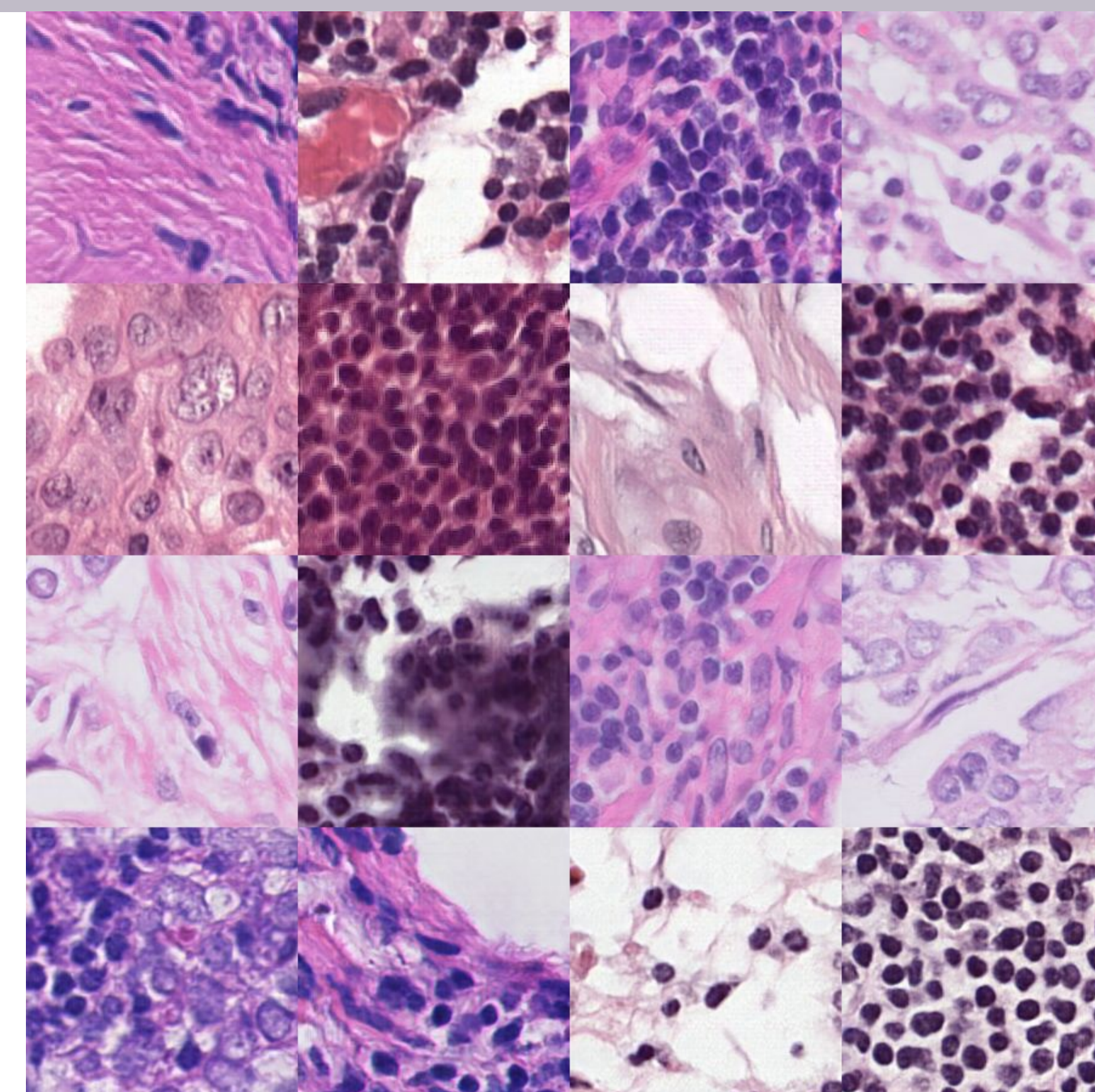
MRI



PET / CT



Ultrasound (echography)

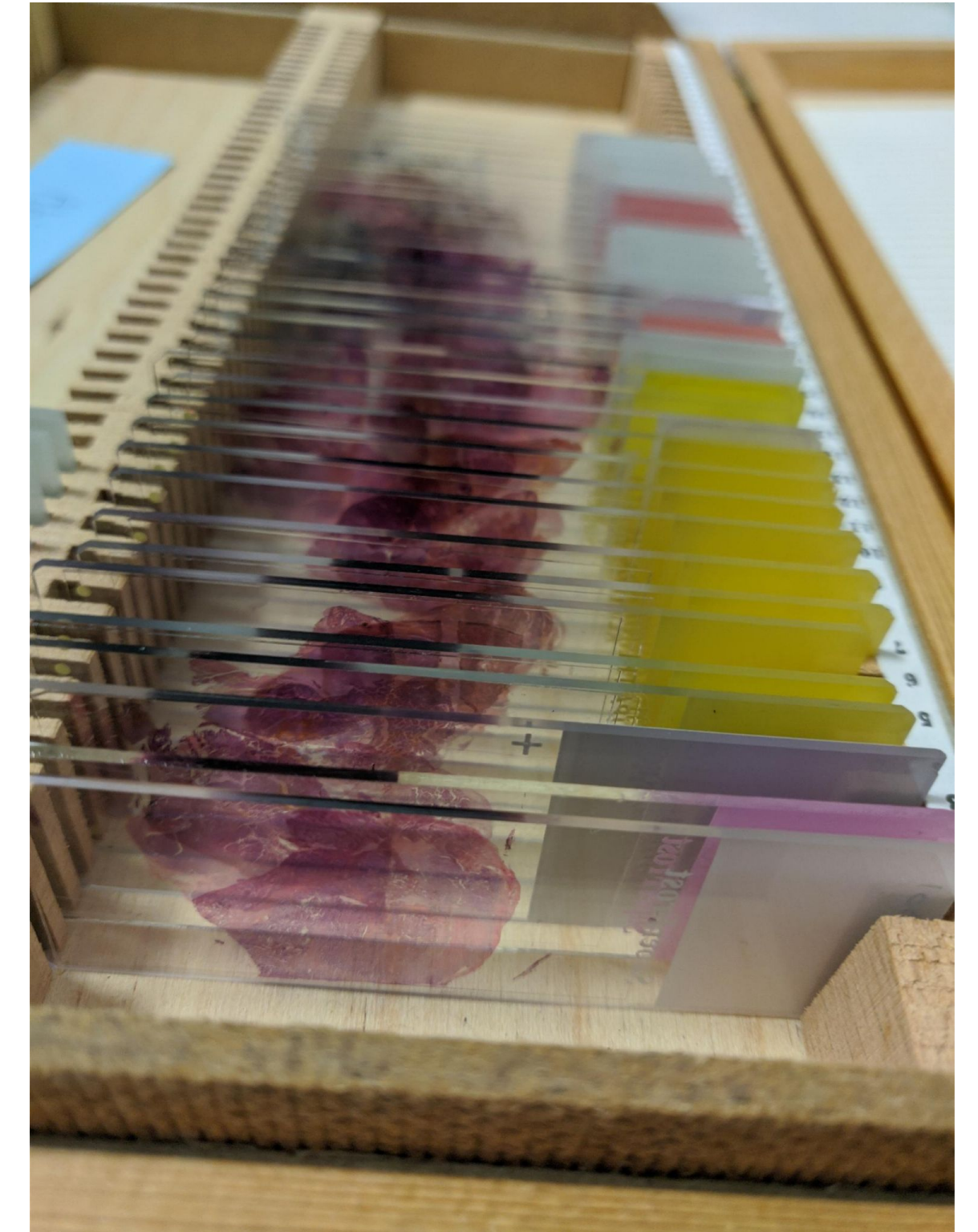


Comparison of Imaging Methods

	Chest	Abdomen	Head/Neck	Cardiovascular	Skeletal/muscular
CT	gold standard	Need contrast for excellency, widely used	Good for trauma	Gold standard	Gold standard
US	no use except heart or P.Effusion	Problems with gas	Poor	Poor	Elastography
Nuclear	Extensive use in heart and therapy in lung	CT or MRI is merged	PET	Perfusion	bone marrow
MRI	growing cardiac applications	Increased role of MRI	Gold standard	Will replace ct in near future	Excellent

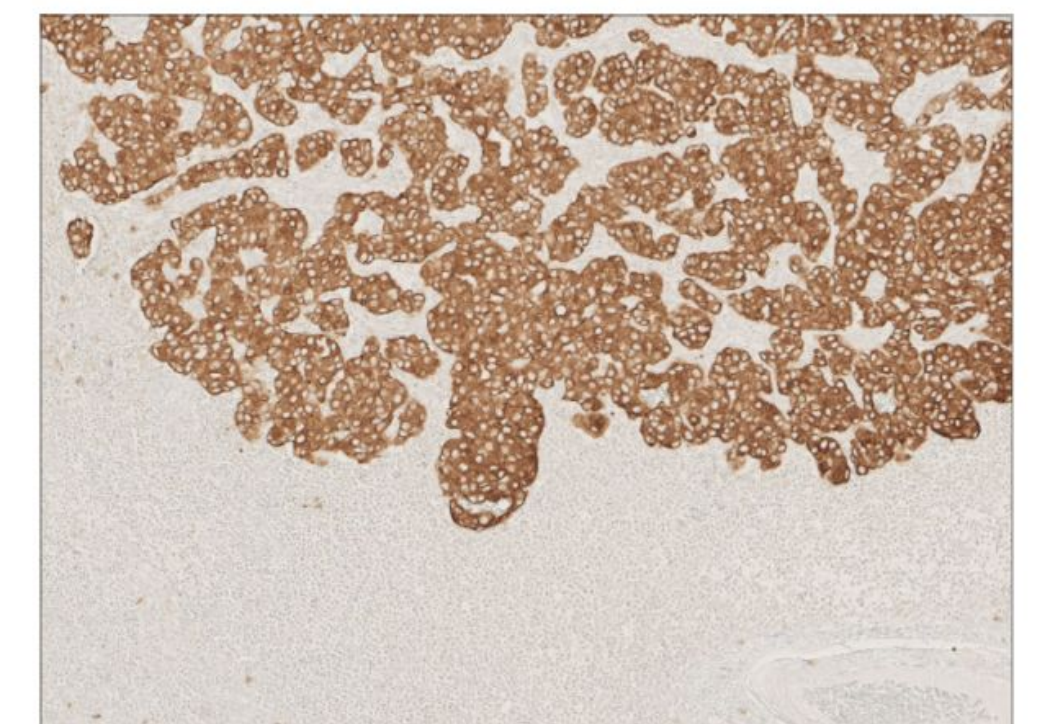
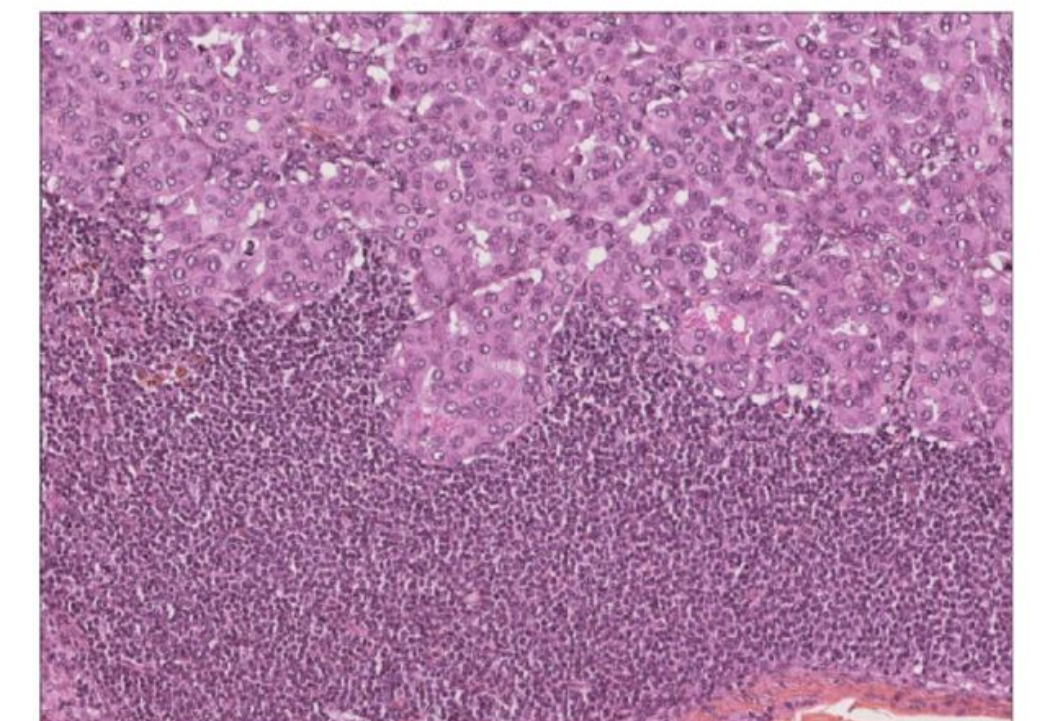
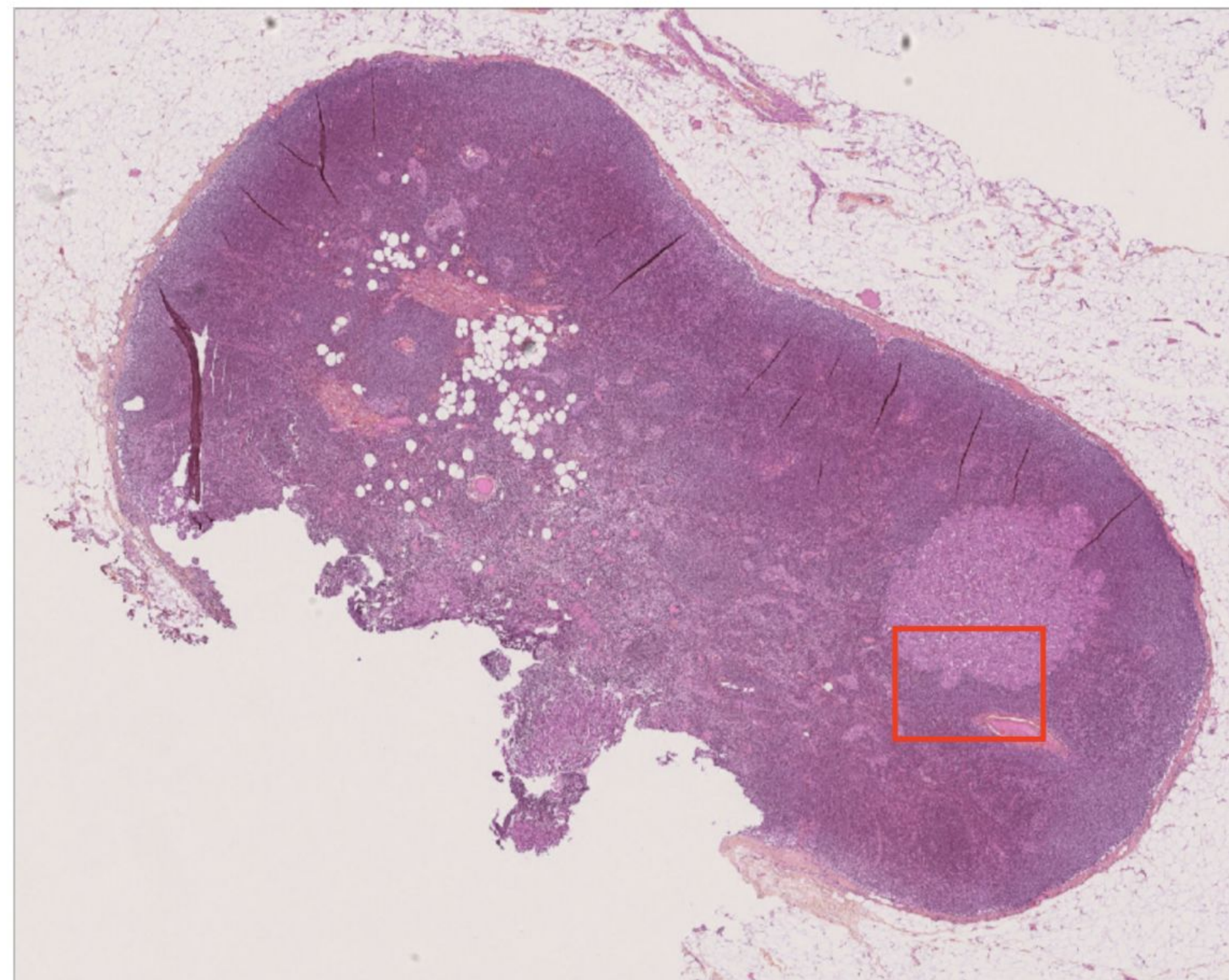
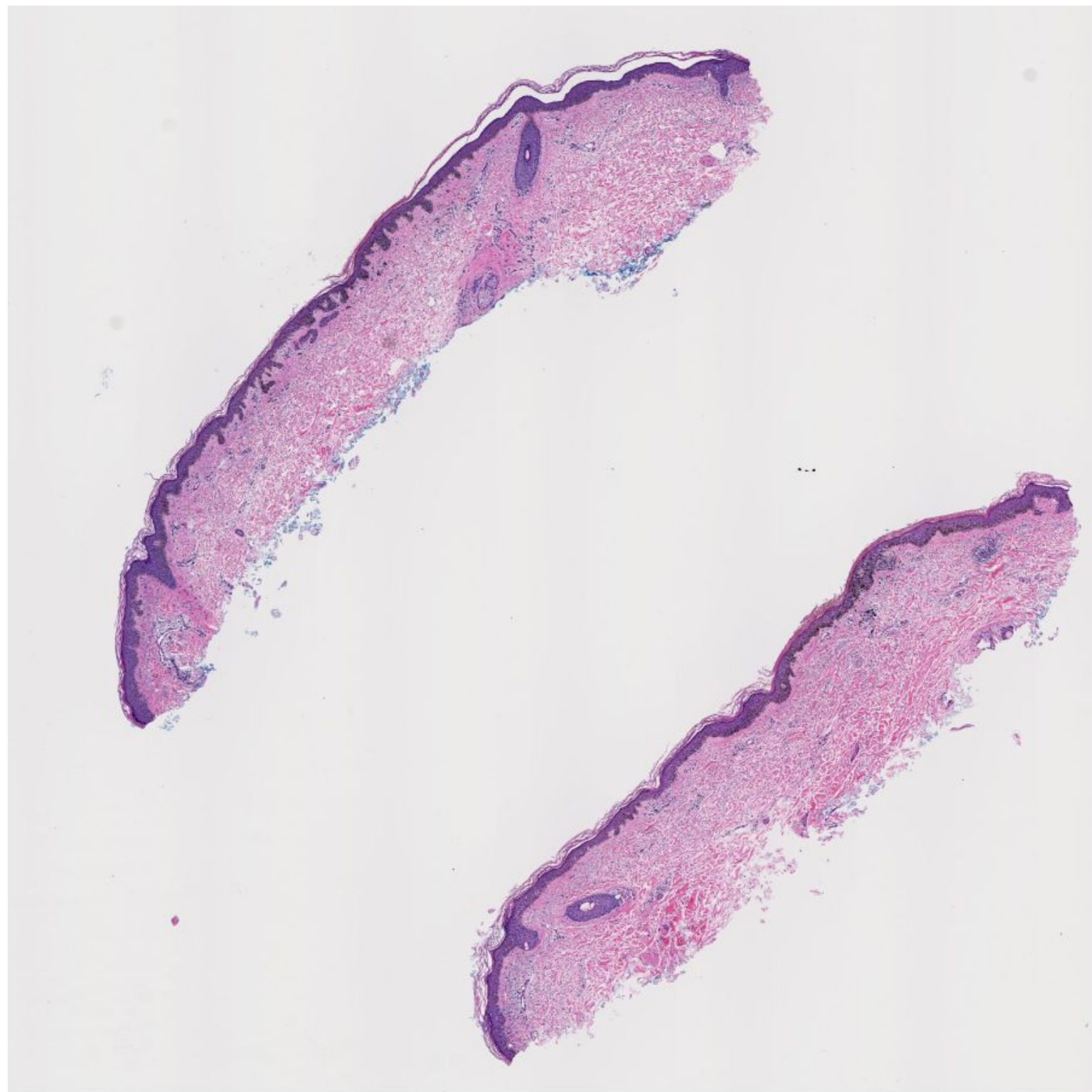
Digital Pathology (focus on Histopathology)

- Digital slides are created from glass slides using specialized scanning machines
- An integrated camera and a motorized stage to move the slide around while parts of the tissue are imaged.
- Two different types of scanning tile-based scanning and line-based scanning
 - Tile scanners capture square field-of-view images covering the entire tissue area on the slide
 - Line-scanners capture images of the tissue in long, uninterrupted stripes rather than tiles
- Software associated with the scanner stitch the tiles or lines together into a single, seamless image.



Digital Pathology (focus on Histopathology)

- Gigabytes of size images
- Millions of cells
- Cells are captured in their (usually tumor) microenvironment
- Different types of staining using different biomarkers



Commonly used software

Practical Info for processing medical imaging

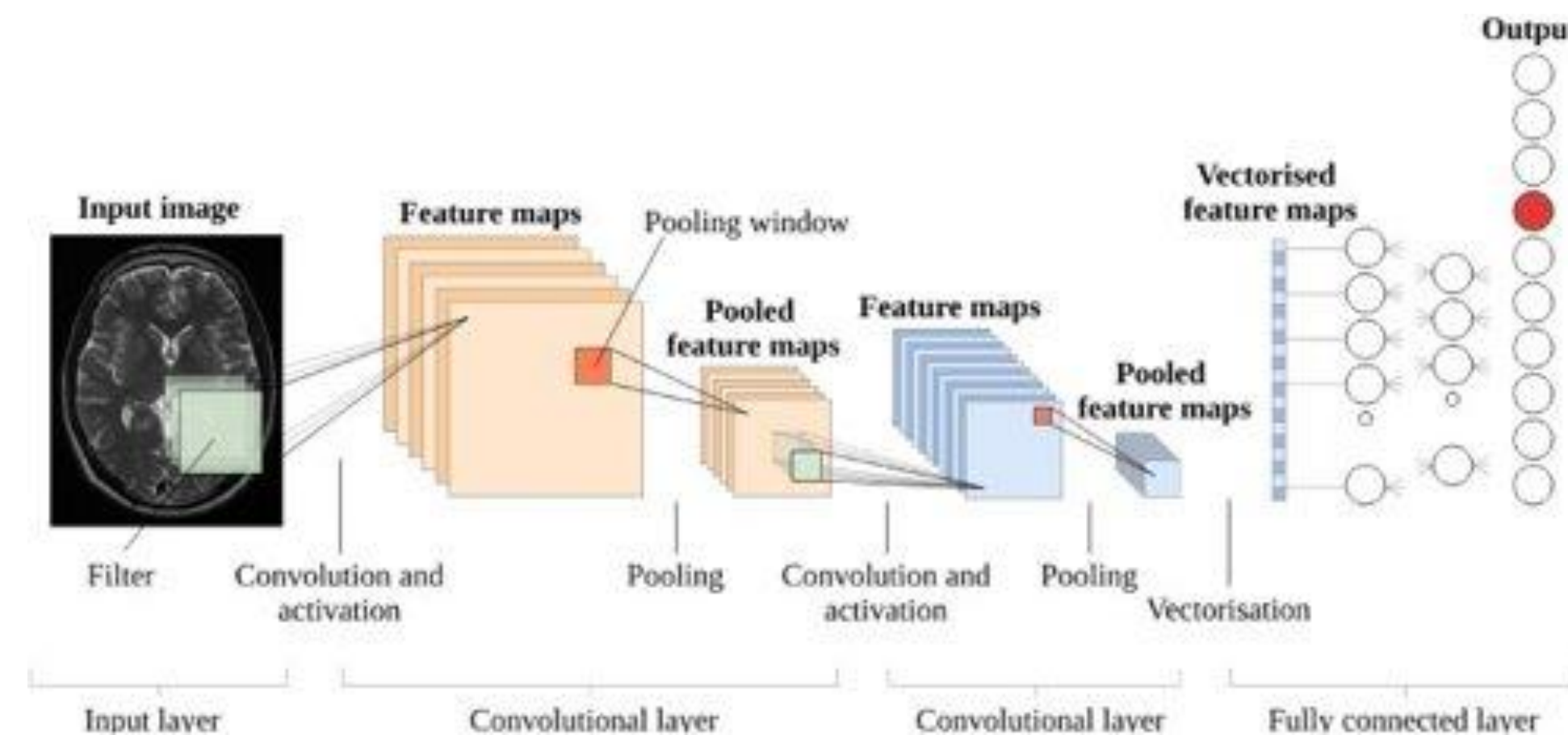
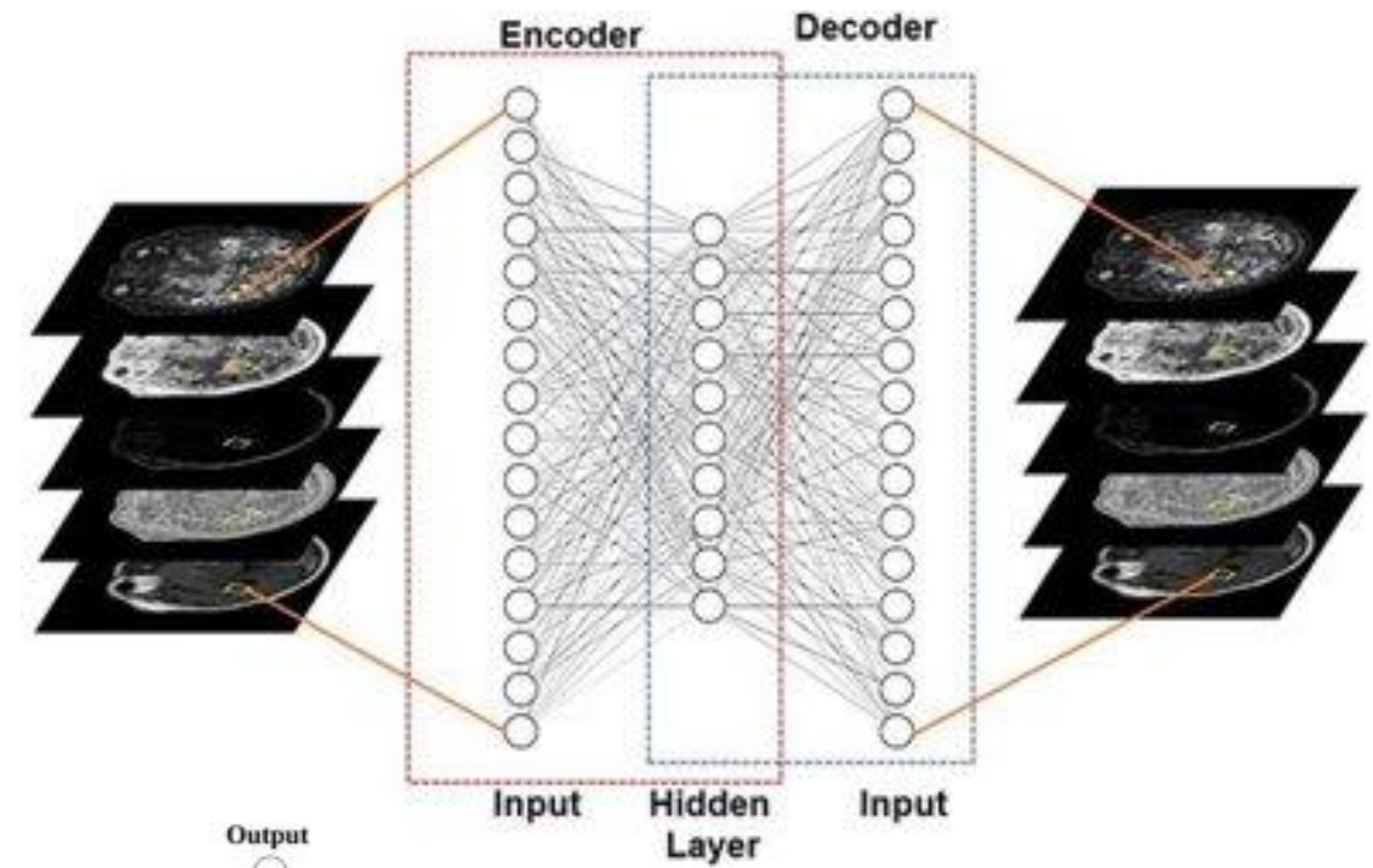
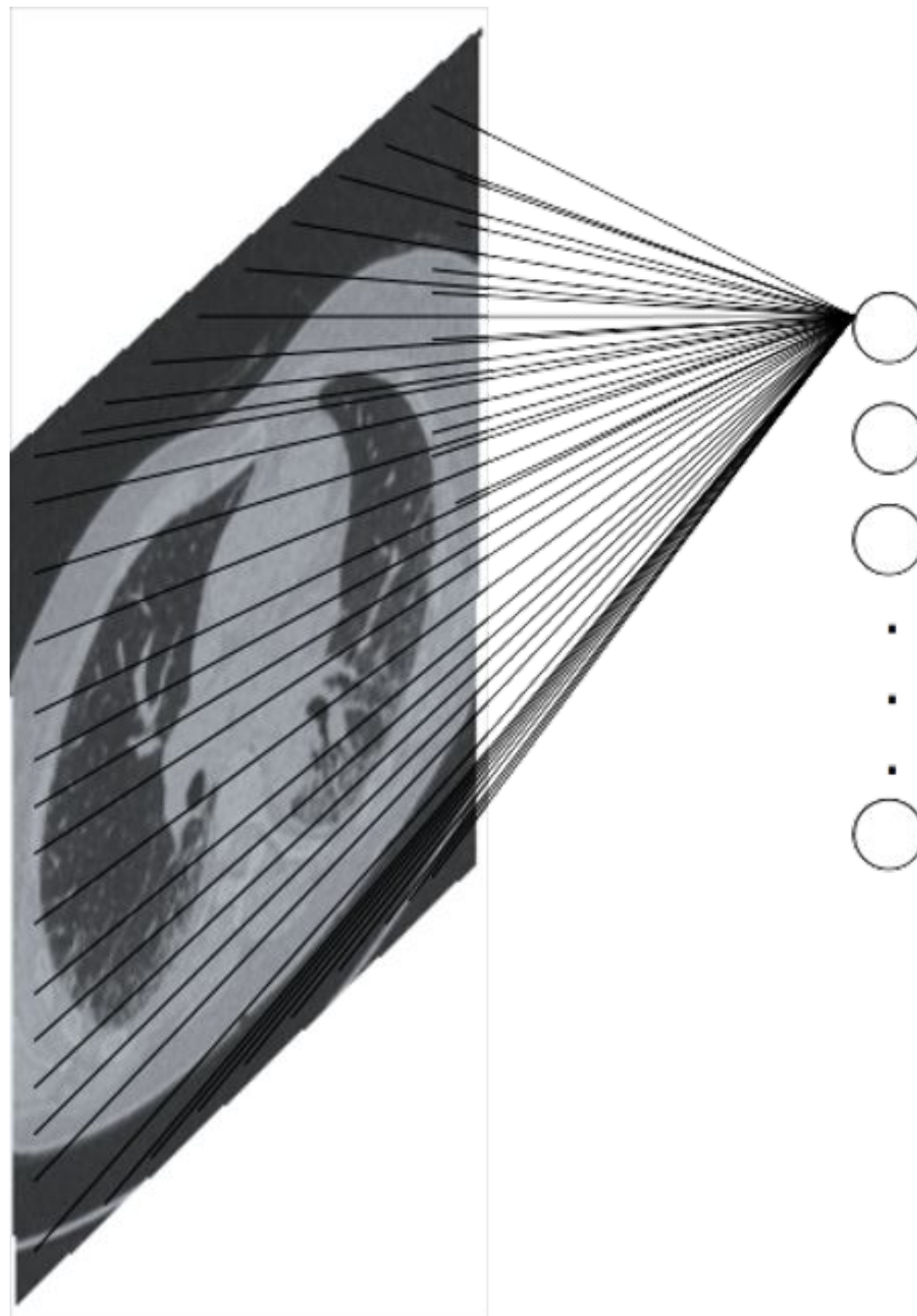
- Software (with GUI) you can use for analysis of medical imaging
 - ITKSnap, Slicer, MTIK, ImageJ/Fiji, Mango,....
- Coding (self): ITK library is usually used
 - C/C++ and Python can be used to call libraries
 - SimpleITK with python is very commonly used
- Image Format
 - DICOM
 - Analyse (.img/hdr)
 - Nifti
 -

ITK Library

National Library of Medicine (of NIH) Insight Segmentation and Registration Toolkit (ITK)

- ITK is an open-source, cross platform system that provides developers with an extensive suite of software tools for image analysis
- Very very good documentation
- Tools for:
 - Image processing
 - Segmentation
 - Registration
 - ...

Medical Imaging and Deep Learning



Challenges

Artificial Intelligence for Health Care

- Limited annotated datasets
- Small samples of anomalies
- Variation in annotation even between doctors
- Variation in appearance due to different sensors, different parameters
- Bias in the dataset, the AI algorithms are built on
- No explanatory power. AI works well, but it is very difficult to explain why

Challenges

Artificial Intelligence for Health Care

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[More and more publicly available datasets e.g. Luna, Brats, TCGA, Bowl 2017, ...]

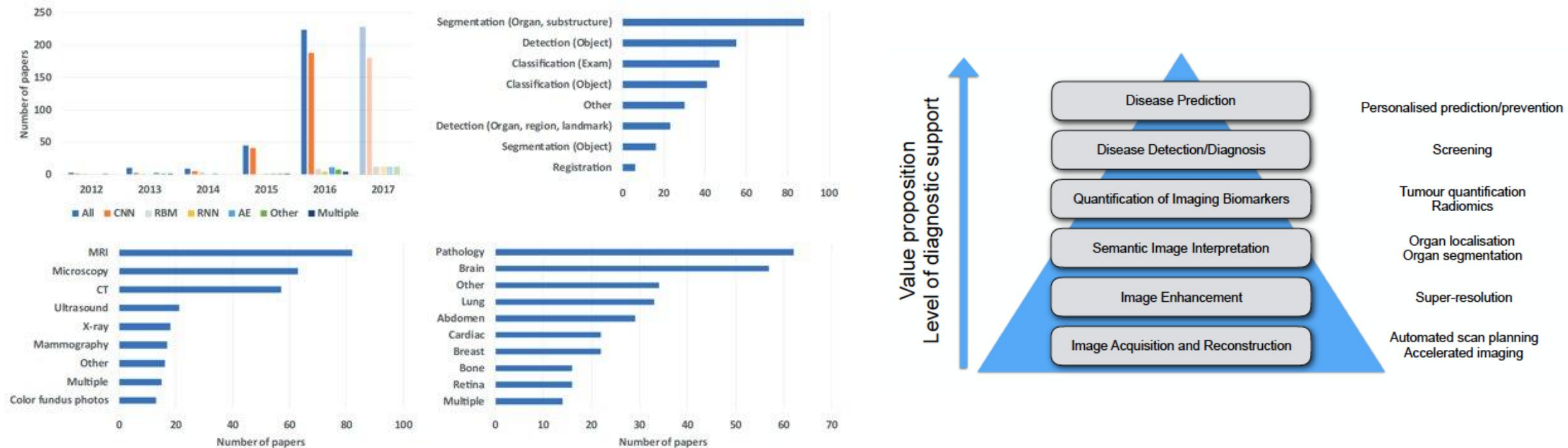
[Different models for different modalities, transfer learning, few shot learning, ...]

[Annotations from different medical experts, exploitation of more than one datasets.]

[Combine models with strong priors, analyse the decisions of the models]

Current Research

- A lot of research focuses currently on deep learning techniques and their use in medical imaging.



Arxiv: 1702.05747

Current Research

A lot of similarities and similar algorithmic choices with computer vision and deep learning with focus on:

- Limited amount of data
- Domain adaptation
- Specific metrics for evaluation and training
- Exploitation of the 3D nature of data
- Objects, organs that are very small in size and difficult to distinguish from the background
- Multimodality and mechanisms for proper fusion of data
- Self-supervision and transfer learning

Semantic Segmentation

- **Evaluation metrics usually applied**

- Most widely used measure for evaluating segmentation in medical imaging
- It is calculated between the reference, and the prediction
- Dice Coefficient

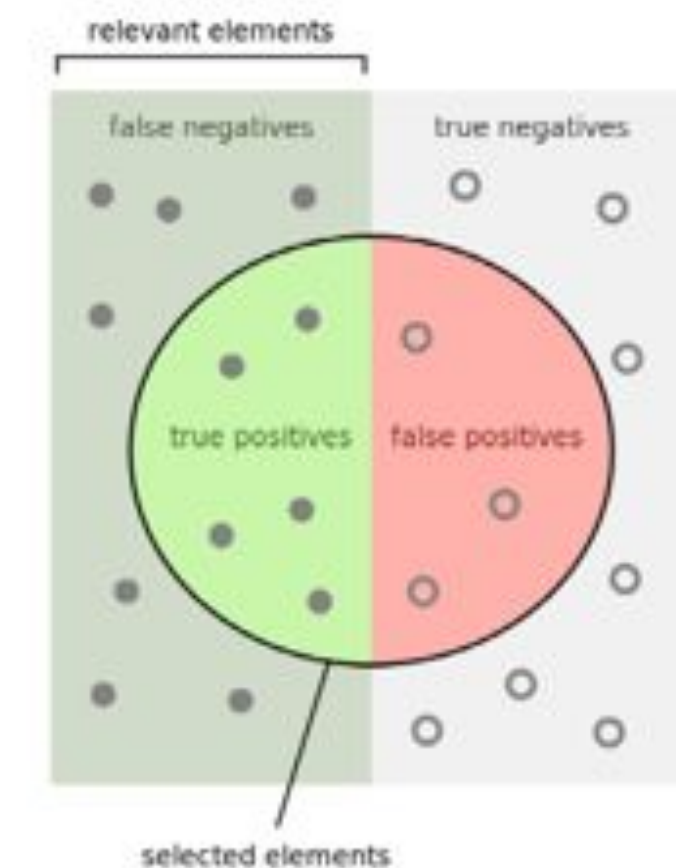
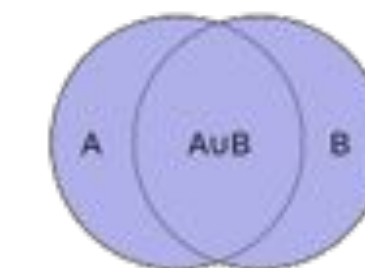
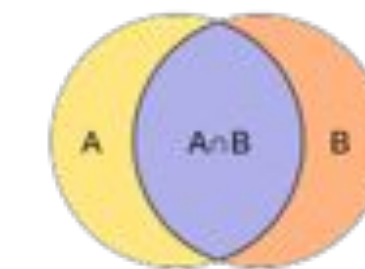
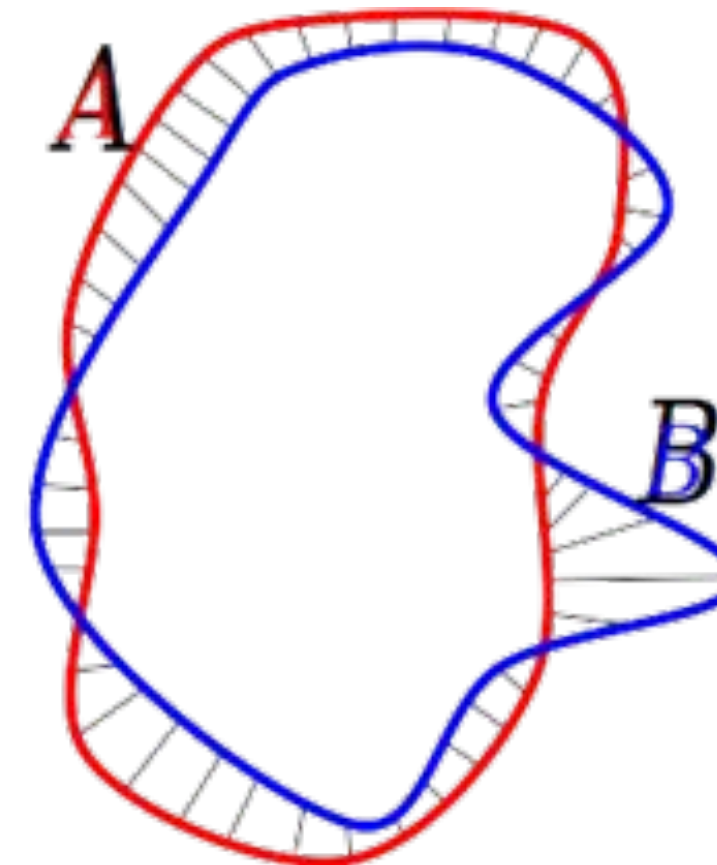
$$DSC = \frac{2|A \cap B|}{|A| + |B|} \quad \begin{array}{l} \bullet |A| = TP + FN \\ \bullet |B| = TP + FP \\ \bullet |A \cap B| = TP \end{array}$$

$$DSC = \frac{2TP}{2TP + FP + FN} = F_1$$

- Hausdorff Distance

$$HD = \max(h(A, B), h(B, A))$$

$$h(A, B) = \max_{a \in A} \min_{b \in B} \|a - b\|$$



Semantic Segmentation

- **Loss functions**

- Dice Loss [Milletari et al., 2016]
$$D = \frac{2 \sum_i^N p_i g_i}{\sum_i^N p_i^2 + \sum_i^N g_i^2} \quad \frac{\partial D}{\partial p_j} = 2 \left[\frac{g_j \left(\sum_i^N p_i^2 + \sum_i^N g_i^2 \right) - 2 p_j \left(\sum_i^N p_i g_i \right)}{\left(\sum_i^N p_i^2 + \sum_i^N g_i^2 \right)^2} \right]$$

- Generalized Dice Loss [Sudre et al., 2017]

$$\text{GDL} = 1 - 2 \frac{\sum_{l=1}^2 w_l \sum_n r_{ln} p_{ln}}{\sum_{l=1}^2 w_l \sum_n r_{ln} + p_{ln}}, \quad \frac{\partial \text{GDL}}{\partial p_i} = -2 \frac{(w_1^2 - w_2^2) \left[\sum_{n=1}^N p_n r_n - r_i \sum_{n=1}^N (p_n + r_n) \right] + N w_2 (w_1 + w_2) (1 - 2 r_i)}{\left[(w_1 - w_2) \sum_{n=1}^N (p_n + r_n) + 2 N w_2 \right]^2}$$

- Focal Loss [Li et al. 2018]

Propose to reshape the loss function to down-weight easy examples and thus focus the training on hard negatives

Binary Cross Entropy

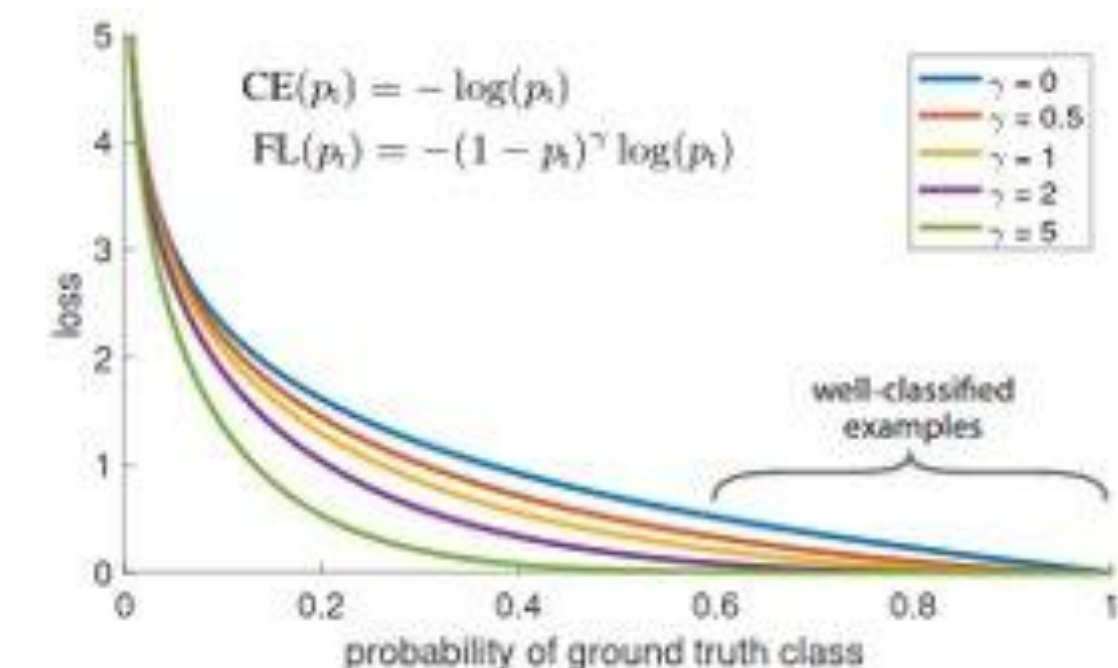
$$\text{CE}(p, y) = \begin{cases} -\log(p) & \text{if } y = 1 \\ -\log(1 - p) & \text{otherwise.} \end{cases}$$

Focal Loss $\text{FL}(p_i) = -(1 - p_i)^\gamma \log(p_i).$

where $\gamma \geq 0$.

when an example is misclassified and p is small

the modulating factor is near 1 and the loss is unaffected. As $p \rightarrow 1$, the factor goes to 0 and the loss of the well-classified examples is down-weighted



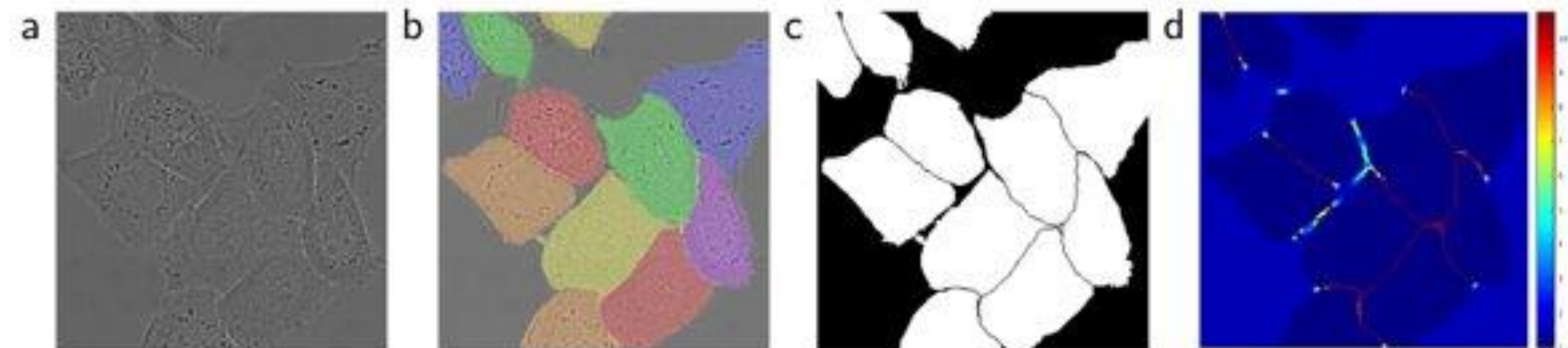
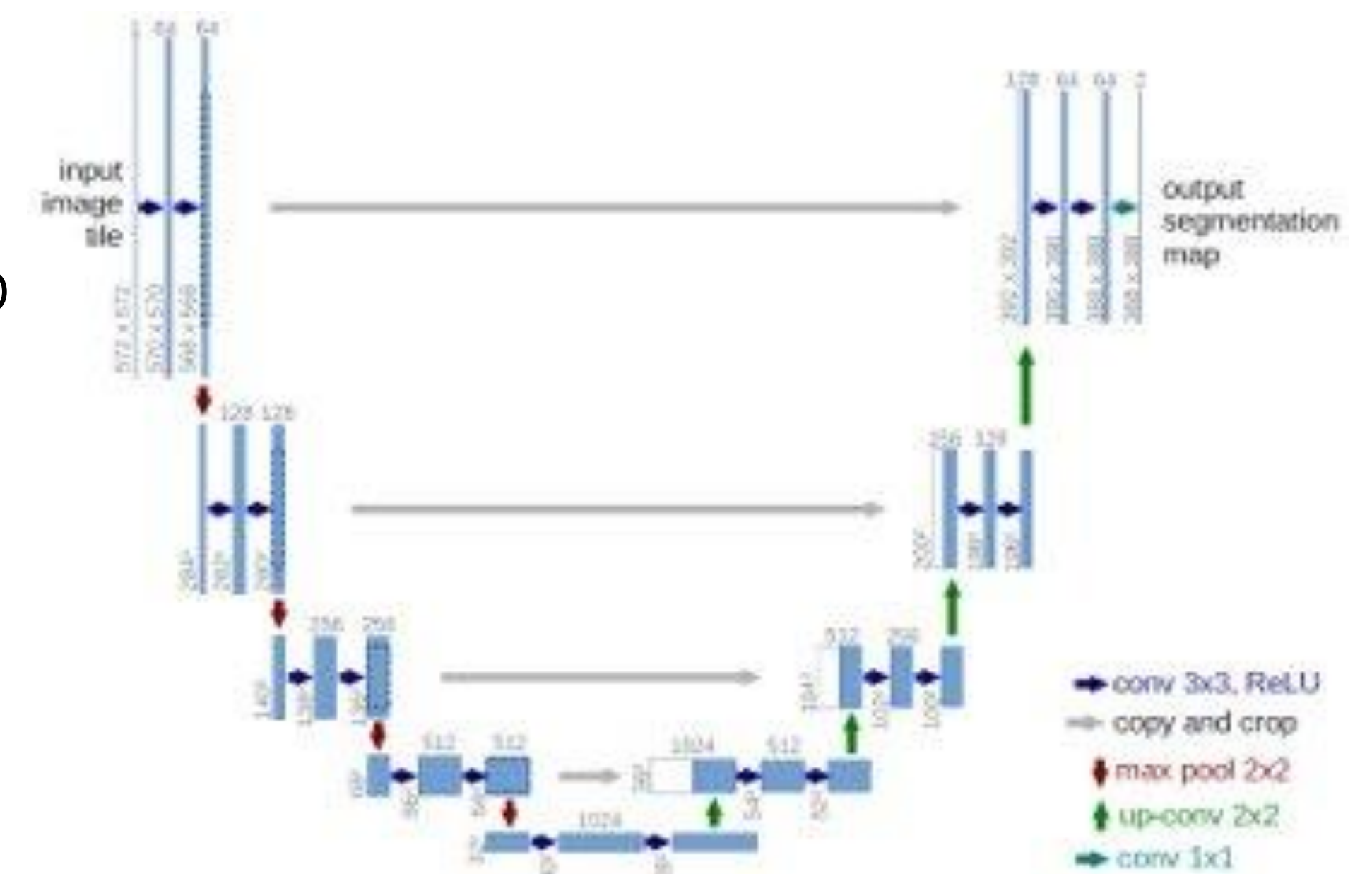
Semantic Segmentation

- **Ronneberger et al.** “U-Net: Convolutional Networks for Biomedical Image Segmentation”
In: **MICCAI 2015**

- Fully convolutional architecture
- Depth of 5 layers with convolutions, max pooling, up convolution or deconvolution and skip connections
- Loss Function:

$$E = \sum_{\mathbf{x} \in \Omega} w(\mathbf{x}) \log(p_{\ell(\mathbf{x})}(\mathbf{x}))$$

$$w(\mathbf{x}) = w_c(\mathbf{x}) + w_0 \cdot \exp\left(-\frac{(d_1(\mathbf{x}) + d_2(\mathbf{x}))^2}{2\sigma^2}\right)$$

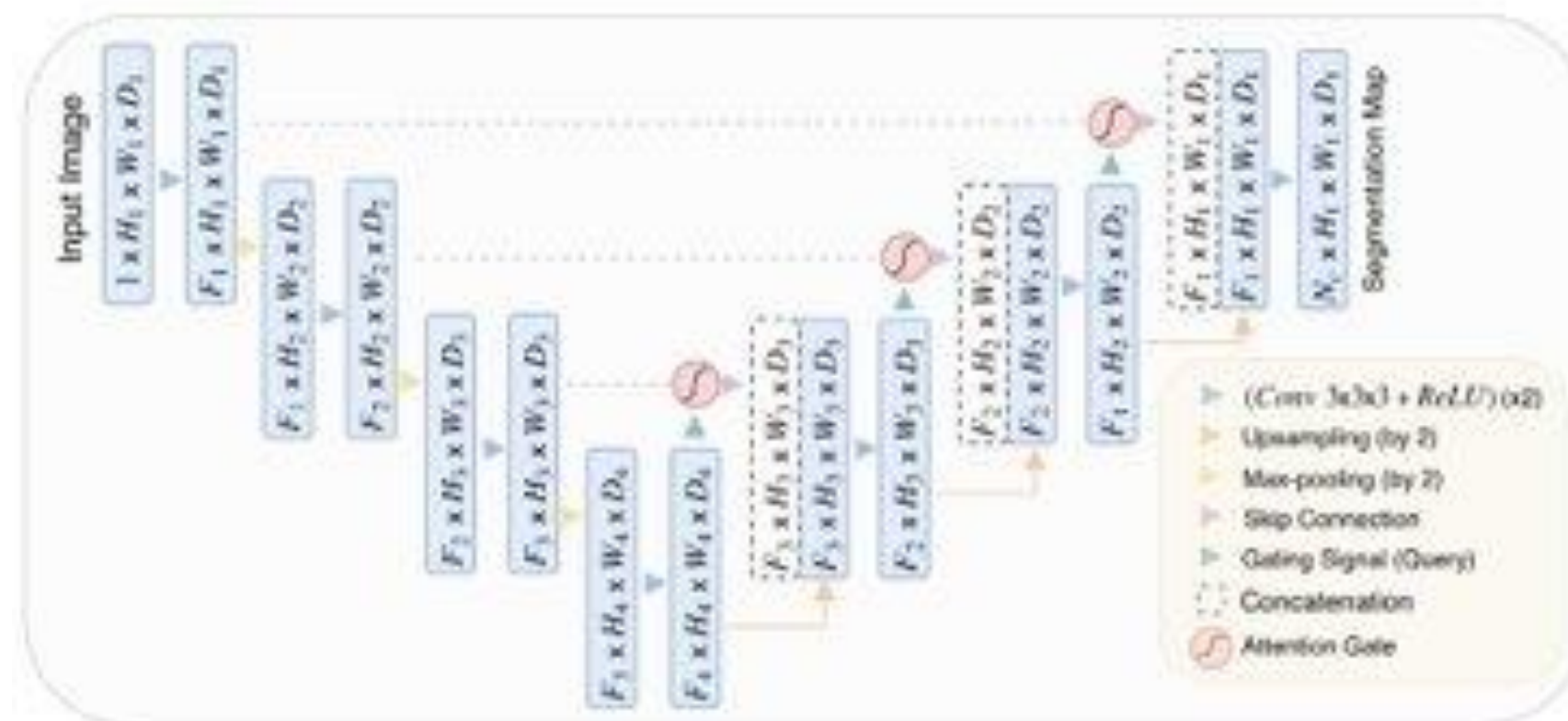


Semantic Segmentation

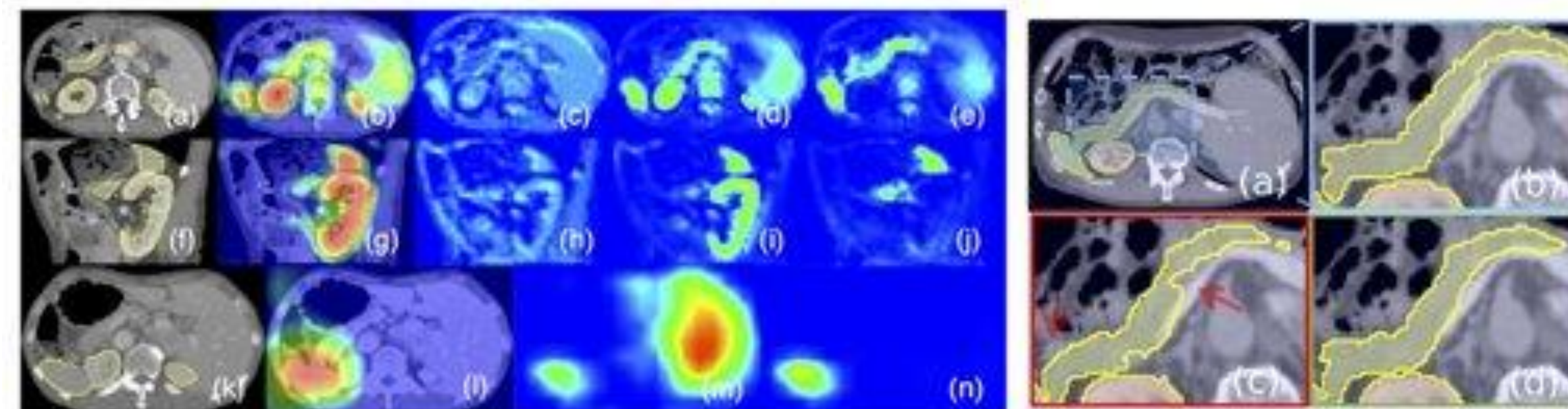
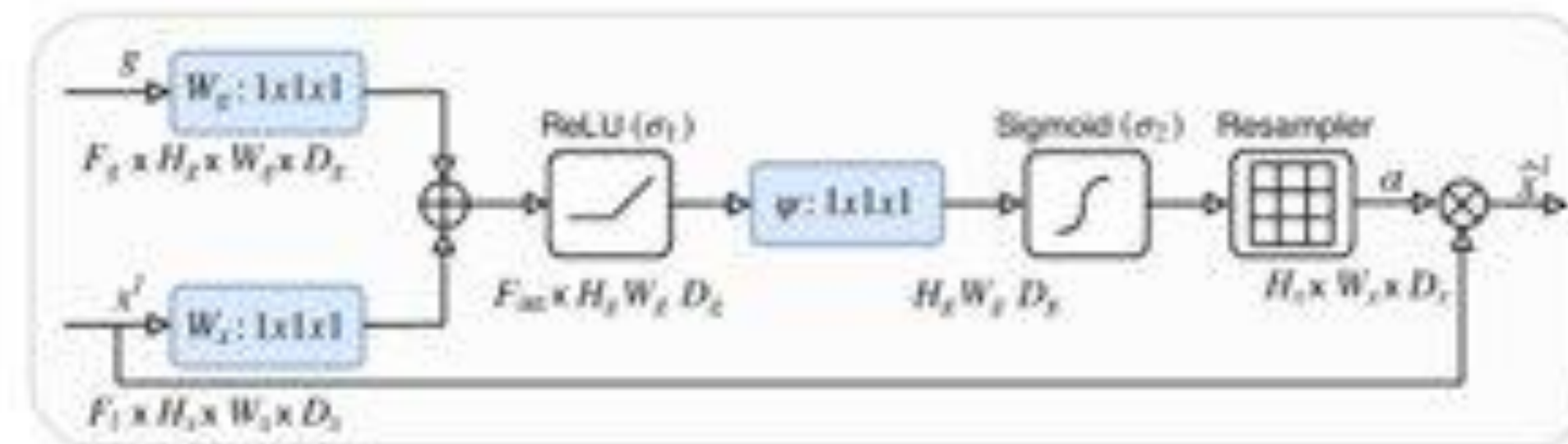
- **Okta et al.** “Attention U-Net: Learning Where to Look for the Pancreas” **In: arXiv 2018**

$$q_{att}^l = \psi^T \left(\sigma_1 \left(W_x^T x_i^l + W_g^T g_i + b_g \right) \right) + b_\psi$$

$$\alpha_i^l = \sigma_2 \left(q_{att}^l(x_i^l, g_i; \Theta_{att}) \right),$$



	Method	Dice Score	Precision	Recall	S2S Dist (mm)
BFT	U-Net [24]	0.690±0.132	0.680±0.109	0.733±0.190	6.389±3.900
	Attention U-Net	0.712±0.110	0.693±0.115	0.751±0.149	5.251±2.551
AFT	U-Net [24]	0.820±0.043	0.824±0.070	0.828±0.064	2.464±0.529
	Attention U-Net	0.831±0.038	0.825±0.073	0.840±0.053	2.305±0.568
SCR	U-Net [24]	0.815±0.068	0.815±0.105	0.826±0.062	2.576±1.180
	Attention U-Net	0.821±0.057	0.815±0.093	0.835±0.057	2.333±0.856



Semantic Segmentation

- **Cicek et al.** “3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation”
In: **MICCAI 2016**

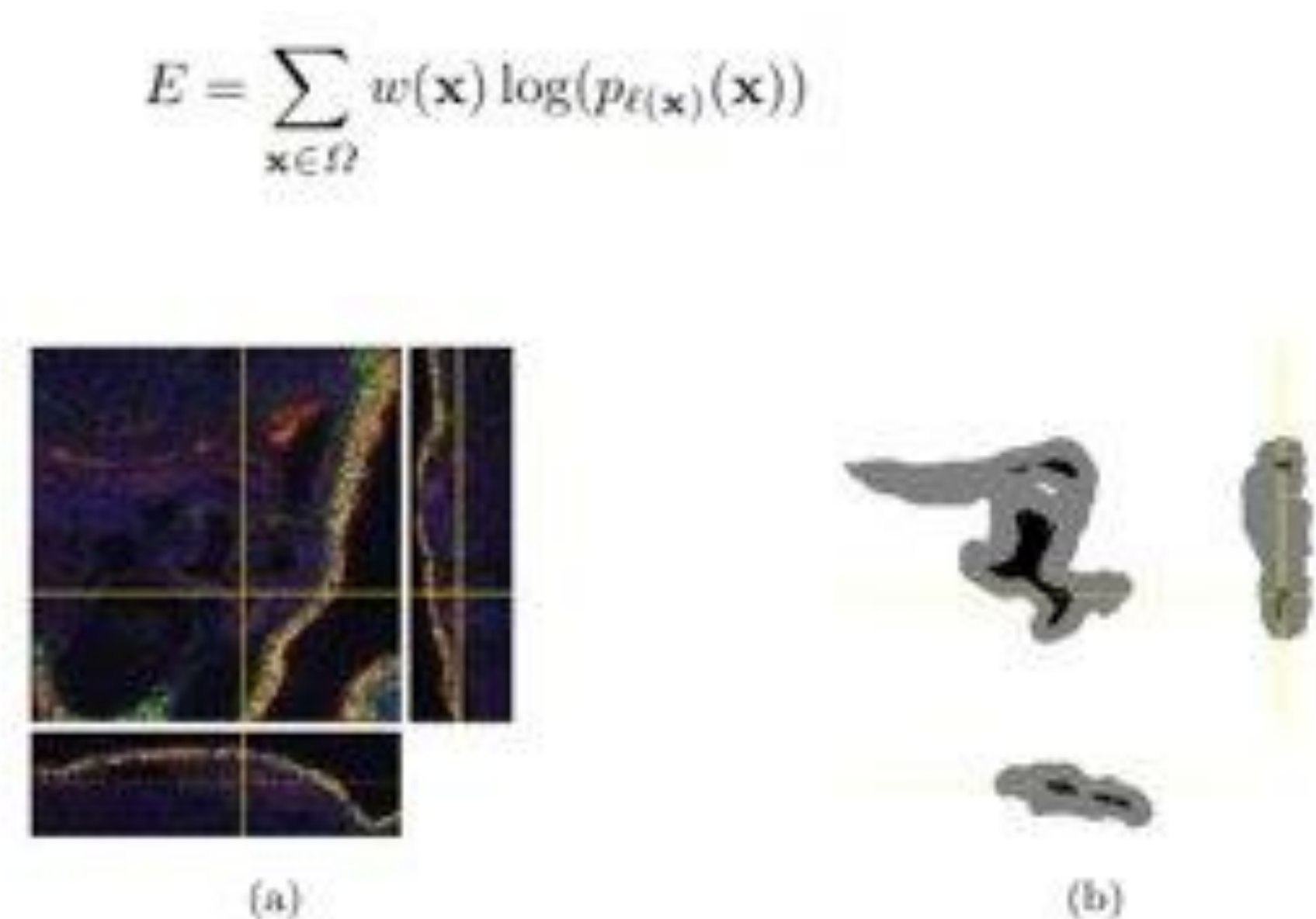


Fig. 3: (a) The confocal recording of our 3rd *Xenopus* kidney. (b) Resulting dense segmentation from the proposed 3D u-net with batch normalization.

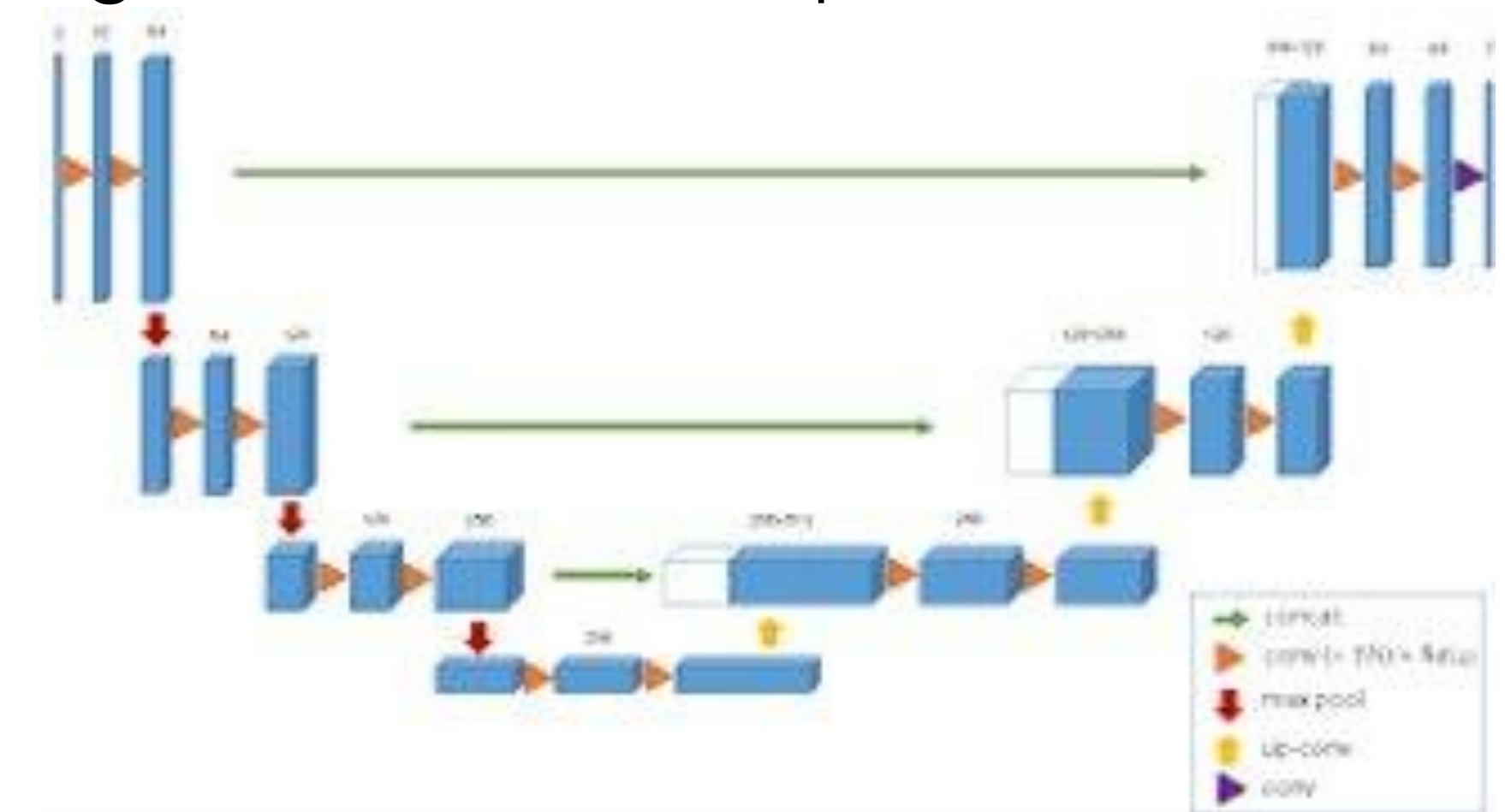


Table 1: Cross validation results for semi-automated segmentation (IoU)

test slices	3D w/o BN	3D with BN	2D with BN
subset 1	0.822	0.855	0.785
subset 2	0.857	0.871	0.820
subset 3	0.846	0.863	0.782
average	0.842	0.863	0.796

Table 2: Effect of # of slices for semi-automated segmentation (IoU)

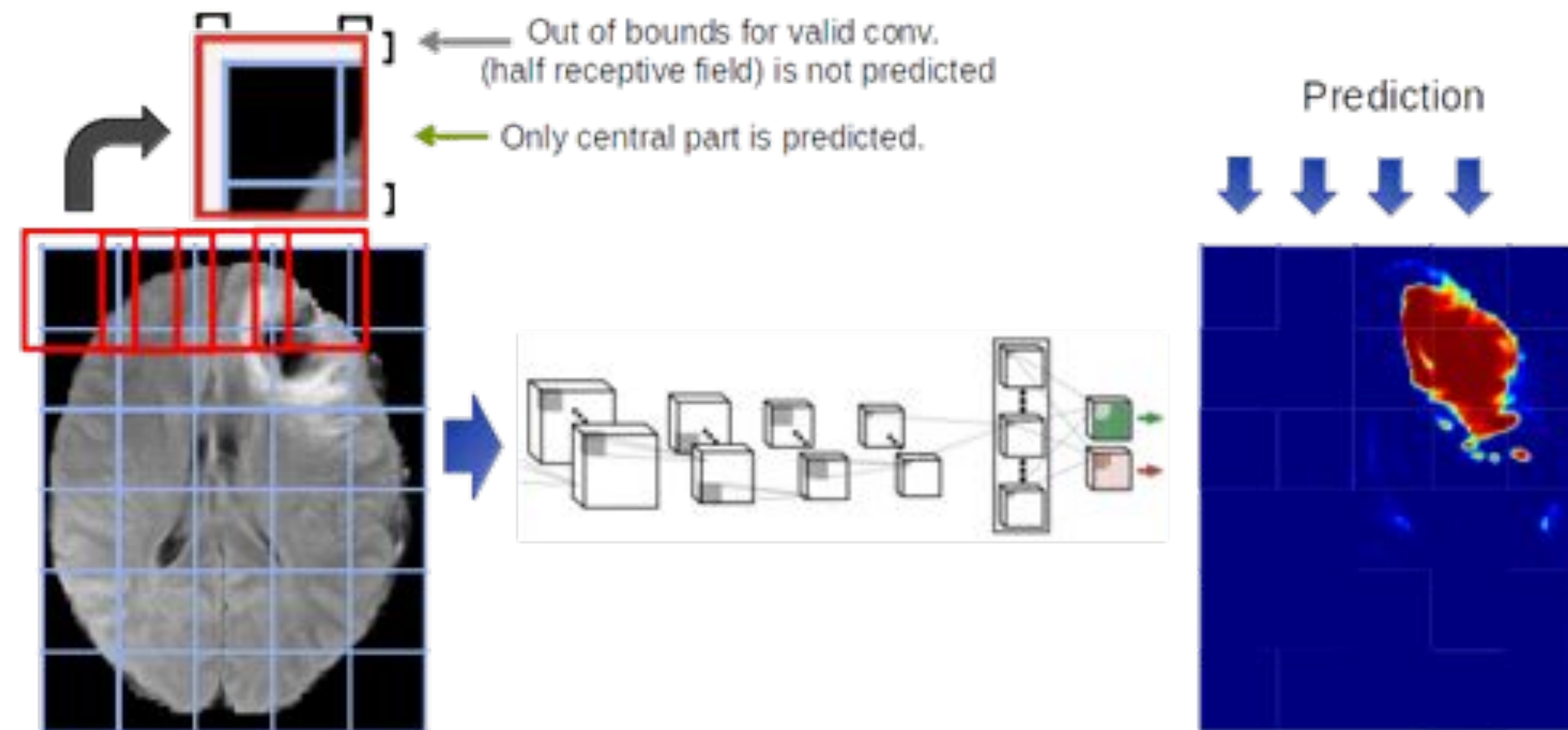
GT slices	GT voxels	IoU S1	IoU S2	IoU S3
1,1,1	2.5%	0.331	0.483	0.475
2,2,1	3.3%	0.676	0.579	0.738
3,3,2	5.7%	0.761	0.808	0.835
5,5,3	8.9%	0.856	0.849	0.872

Table 3: Cross validation results for fully-automated segmentation (IoU)

test volume	3D w/o BN	3D with BN	2D with BN
1	0.655	0.761	0.619
2	0.734	0.798	0.698
3	0.779	0.554	0.325
average	0.723	0.704	0.547

Semantic Segmentation

- **Tiling for 3D networks**



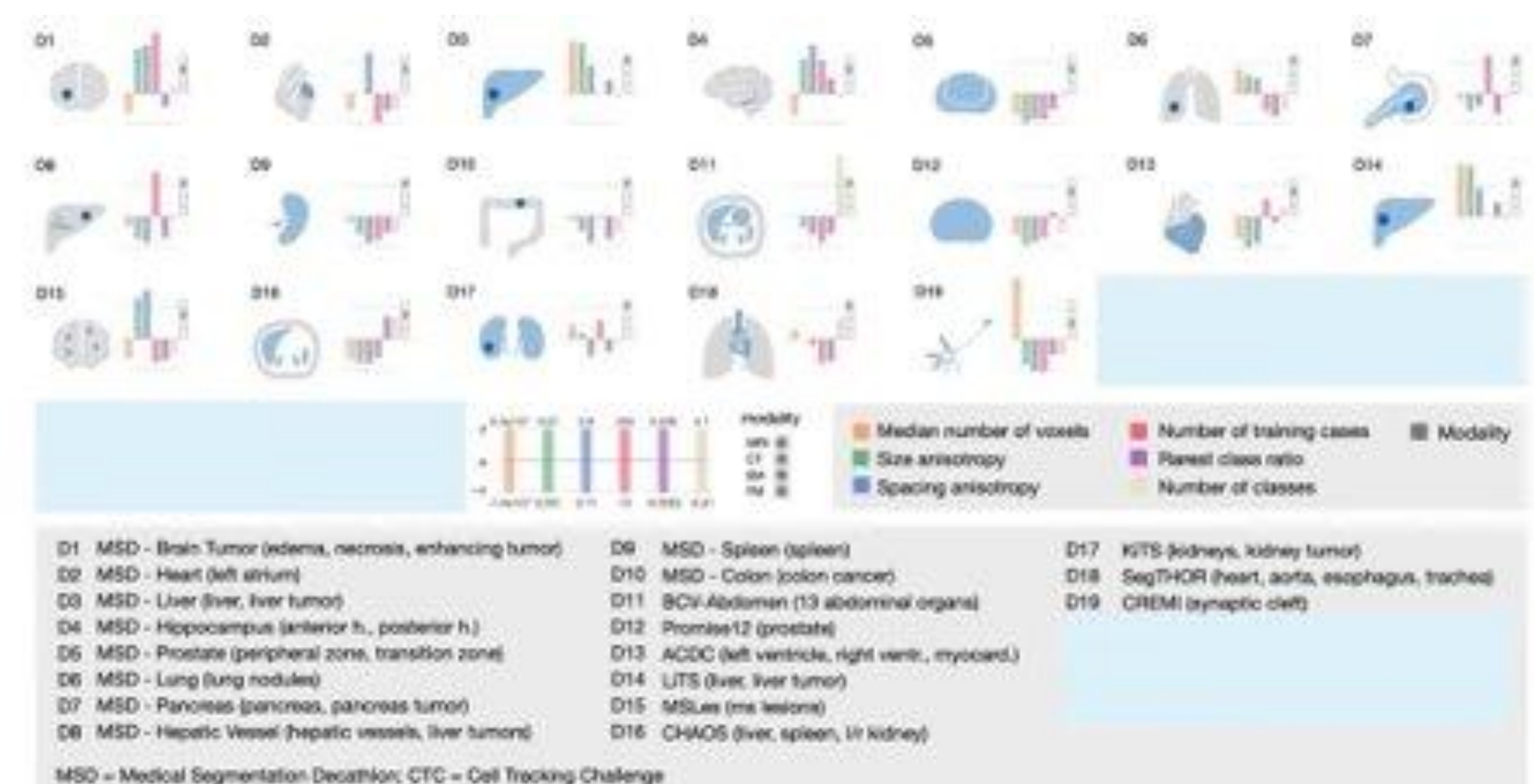
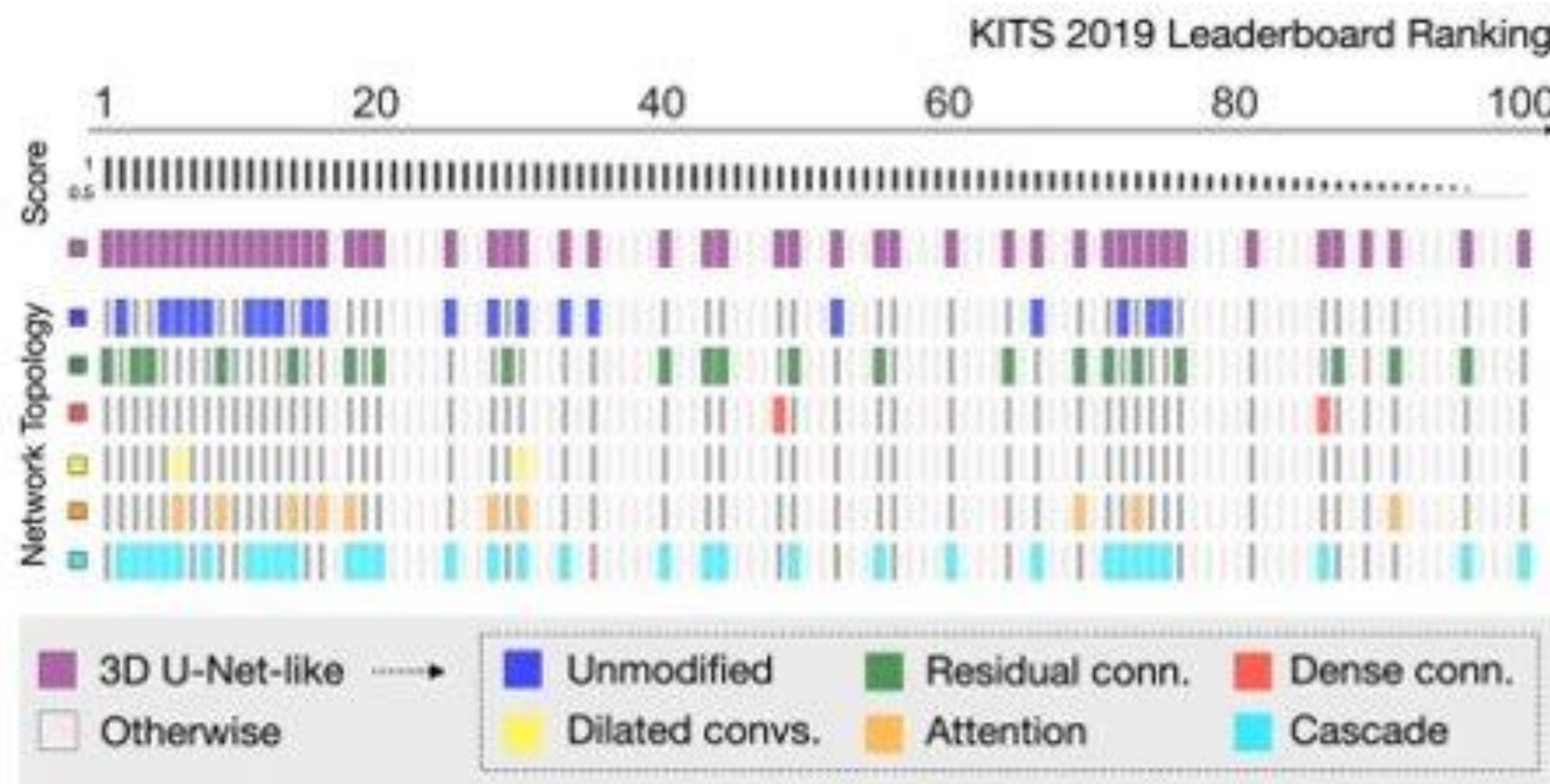
Bigger input tiles (or segments):

- Require more memory (FMs expand)
- But fewer redundant computations (less overlap)

Tile size during testing can be different from training

Semantic Segmentation

- **Isensee et al.** “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation” **In: Nature Methods 2021**



Semantic Segmentation

- **Isensee et al.** “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation” **In: Nature Methods 2021**

Overall observations

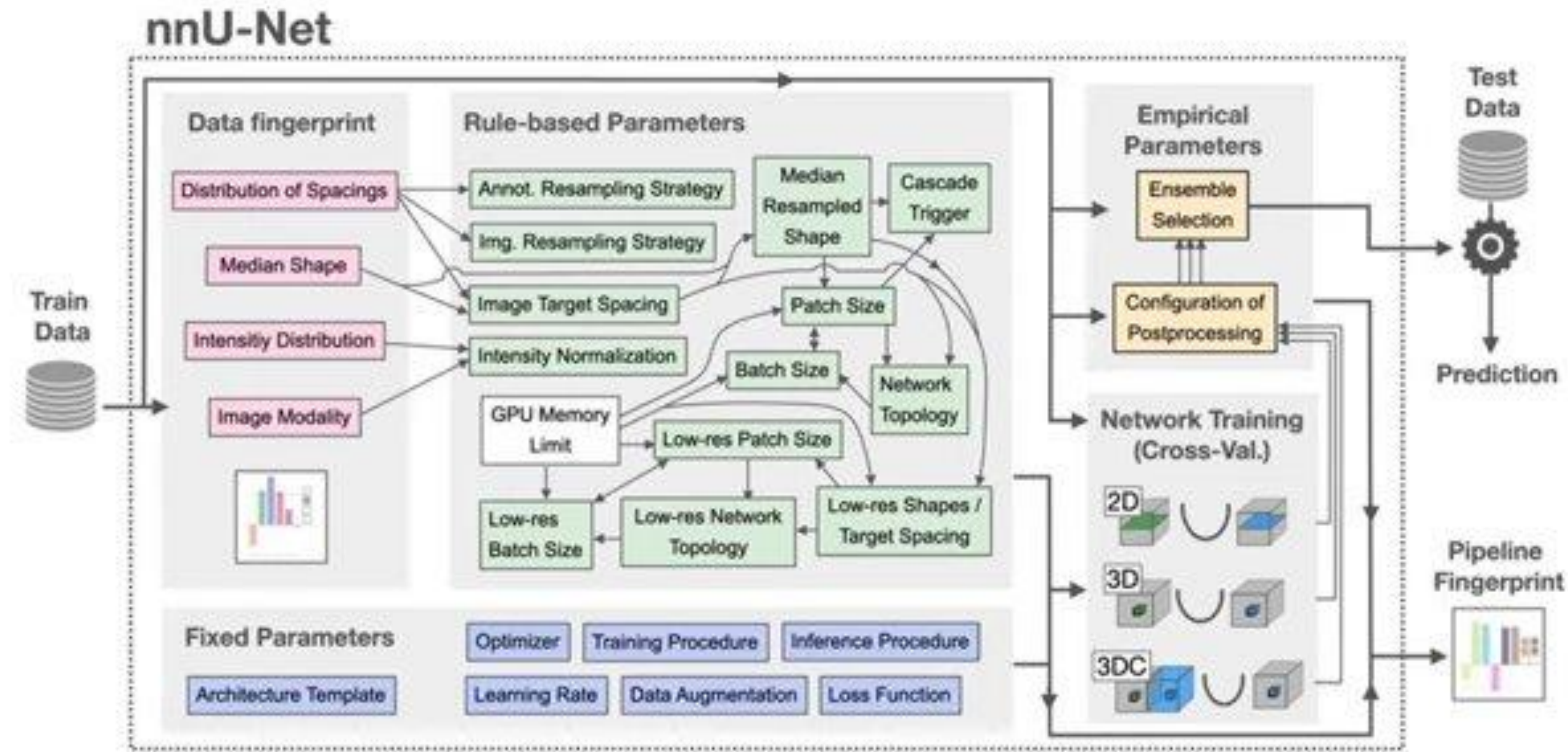
- Time consuming trial and error process
- Success depends on experience of the researcher
- Needs to be repeated on every dataset

nnU-Net: Proposed recipe

- **Fixed Parameters:** Collect design decisions that do not require adaptation between datasets and identify a robust common configuration
- **Rule-based Parameters:** For as many of the remaining decisions as possible formulate explicit dependencies between specific dataset properties ("dataset fingerprint") and design choices ("pipeline fingerprint") in the form of heuristic rules to allow for almost instant adaptation on application.
- **Empirical Parameters:** Learn only the remaining decisions empirically from the data

Semantic Segmentation

- **Isensee et al.** “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation” **In: Nature Methods 2021**



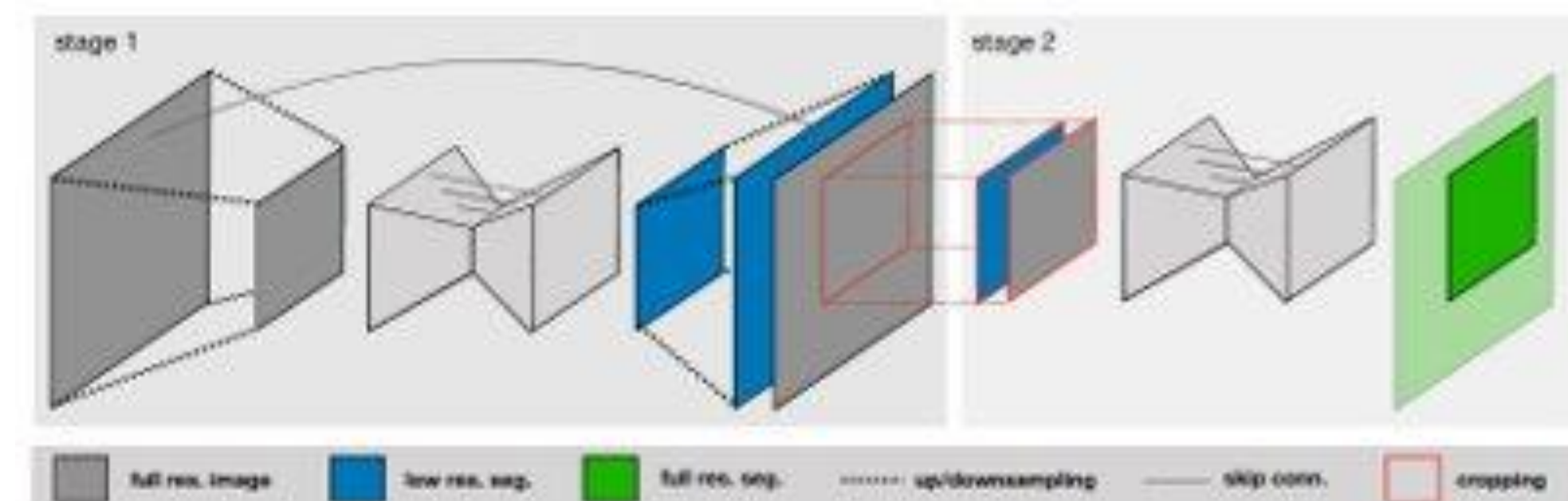
Semantic Segmentation

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$$\mathcal{L}_{total} = \mathcal{L}_{dice} + \mathcal{L}_{CE}$$

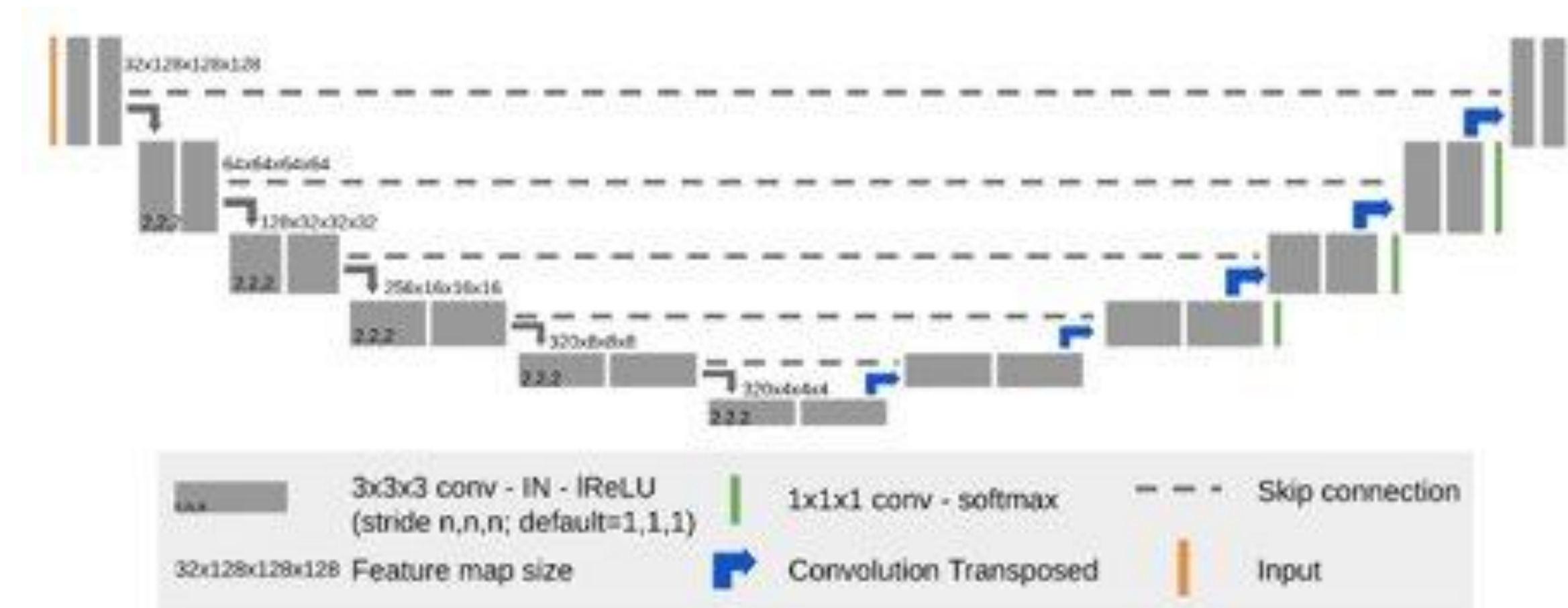
		2D U-Net	3D U-Net	3D U-Net lowres
Brain Tumour	median patient shape	169x138	138x169x138	-
	input patch size	192x160	128x128x128	-
	batch size	89	2	-
	num pool per axis	5, 5	5, 5, 5	-
Heart	median patient shape	320x232	115x320x232	58x160x116
	input patch size	320x256	80x192x128	64x160x128
	batch size	33	2	2
	num pool per axis	6, 6	4, 5, 5	4, 5, 5
Liver	median patient shape	512x512	482x512x512	121x128x128
	input patch size	512x512	128x128x128	128x128x128
	batch size	10	2	2
	num pool per axis	6, 6	5, 5, 5	5, 5, 5
Hippocampus	median patient shape	50x35	36x50x35	-
	input patch size	56x40	40x56x40	-
	batch size	366	9	-
	num pool per axis	3, 3	3, 3, 3	-
Prostate	median patient shape	320x319	20x320x319	-
	input patch size	320x320	20x192x192	-
	batch size	26	4	-
	num pool per axis	6, 6	2, 5, 5	-
Lung	median patient shape	512x512	252x512x512	126x256x256
	input patch size	512x512	112x128x128	112x128x128
	batch size	10	2	2
	num pool per axis	6, 6	4, 5, 5	4, 5, 5
Pancreas	median patient shape	512x512	96x512x512	96x256x256
	input patch size	512x512	96x160x128	96x160x128
	batch size	10	2	2
	num pool per axis	6, 6	4, 5, 5	4, 5, 5

label	Brain Tumour			Heart		Liver		Hippoc.		Prostate		Lung	Pancreas	
	1	2	3	1	1	2	1	2	1	2	1	1	1	2
2D U-Net	78.60	58.65	77.42	91.36	94.37	53.94	88.52	86.70	61.98	84.31	52.68	74.70	35.41	
3D U-Net	80.71	62.22	79.07	92.45	94.11	61.74	89.87	88.20	60.77	83.73	55.87	77.69	42.69	
3D U-Net stage1 only (U-Net Cascade)	-	-	-	90.63	94.09	47.01	-	-	-	-	65.33	79.45	49.65	
3D U-Net (U-Net Cascade) ensemble	-	-	-	92.40	95.38	58.49	-	-	-	-	60.85	79.30	52.12	
2D U-Net+ 3D U-Net ensemble	80.79	61.72	79.16	92.70	94.30	60.24	89.76	88.09	63.78	85.31	55.96	78.26	40.46	
2D U-Net+ 3D U-Net (U-Net Cascade) ensemble	-	-	-	92.64	95.31	60.09	-	-	-	-	61.18	78.79	45.46	
3D U-Net+ 3D U-Net (U-Net Cascade)	-	-	-	92.63	95.43	61.82	-	-	-	-	65.16	79.70	49.14	
test set	67.71	47.73	68.16	92.77	95.24	73.71	90.37	88.95	75.81	89.59	69.20	79.53	52.27	



Semantic Segmentation

- **Isensee et al.** “nnU-Net for Brain Tumor Segmentation” **In: MICCAI Brainlesion 2020**



- Additional Modifications:
 - Region-based training [R]: training on the validated classes
 - Post processing especially on the enhancing tumor (remove predictions smaller than a threshold)
 - Big batch size
 - More data augmentation [DA]
 - Increase probability for rotation, scaling
 - Brightness augmentation
 - Elastic deformations

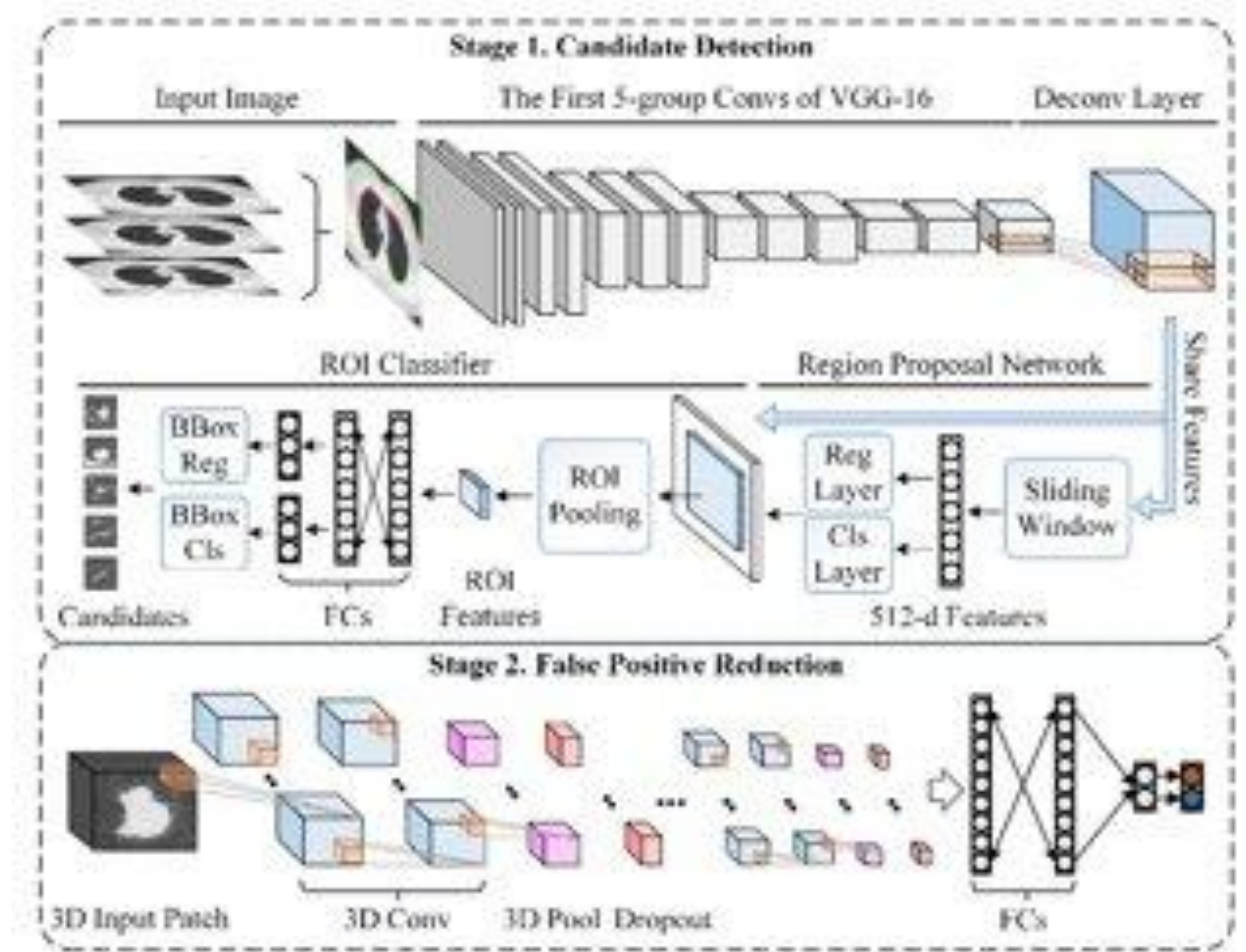
Semantic Segmentation

- **Isensee et al.** “nnU-Net for Brain Tumor Segmentation” **In: MICCAI Brainlesion 2020**
- Additional Modifications:
 - Batch normalization [BN] (instead of instance normalization)
 - Batch dice [BD] Calculate the dice for the entire mini-batch and not per sample. It helps with the cases with very small number of segmentations

Model	Training set				Validation set			
	rank based		BraTS ranking		rank based		BraTS ranking	
	on mean Dice				on mean Dice			
	value	rank	value	rank	value	rank	value	rank
BL	86.55	8	0.3763	8	84.18	6	0.4079	8
BL*	86.09	9	0.3767	9	83.76	9	0.4236	9
BL*+R	86.73	5	0.3393	5	84.13	7	0.4005	7
BL*+R+DA	87.07	1	0.3243	3	84.73	5	0.3647	5
BL*+R+DA+BN	86.82	3	0.3377	4	85.20	3	0.3577	4
BL*+R+DA+BD	86.79	4	0.3231	2	84.12	8	0.3726	6
BL*+R+DA+BN+BD	86.87	2	0.3226	1	85.11	4	0.3487	3*
BL*+R+DA*+BN	86.68	6	0.3521	6	85.58	1	0.3125	1*
BL*+R+DA*+BN+BD	86.64	7	0.3595	7	85.29	2	0.3437	2*

Object Detection

- **Ding et al.** “Accurate pulmonary nodule detection in computed tomography images using deep convolutional neural networks” **In: MICCAI 2017**
- Lung Nodule Detection (Luna):
 - A big number of false positives for nodule detection due to very small size.
 - 2 stage process!
 - Faster R-CNN to detect the boxes of possible nodules
 - 3D network to decide if the detection is good or not!



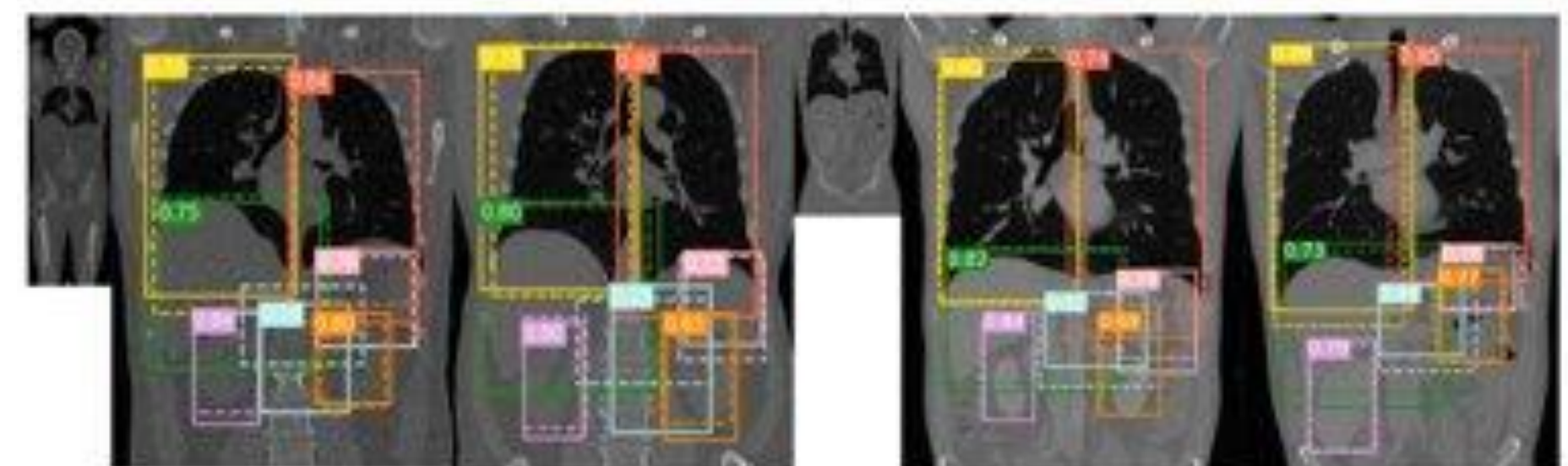
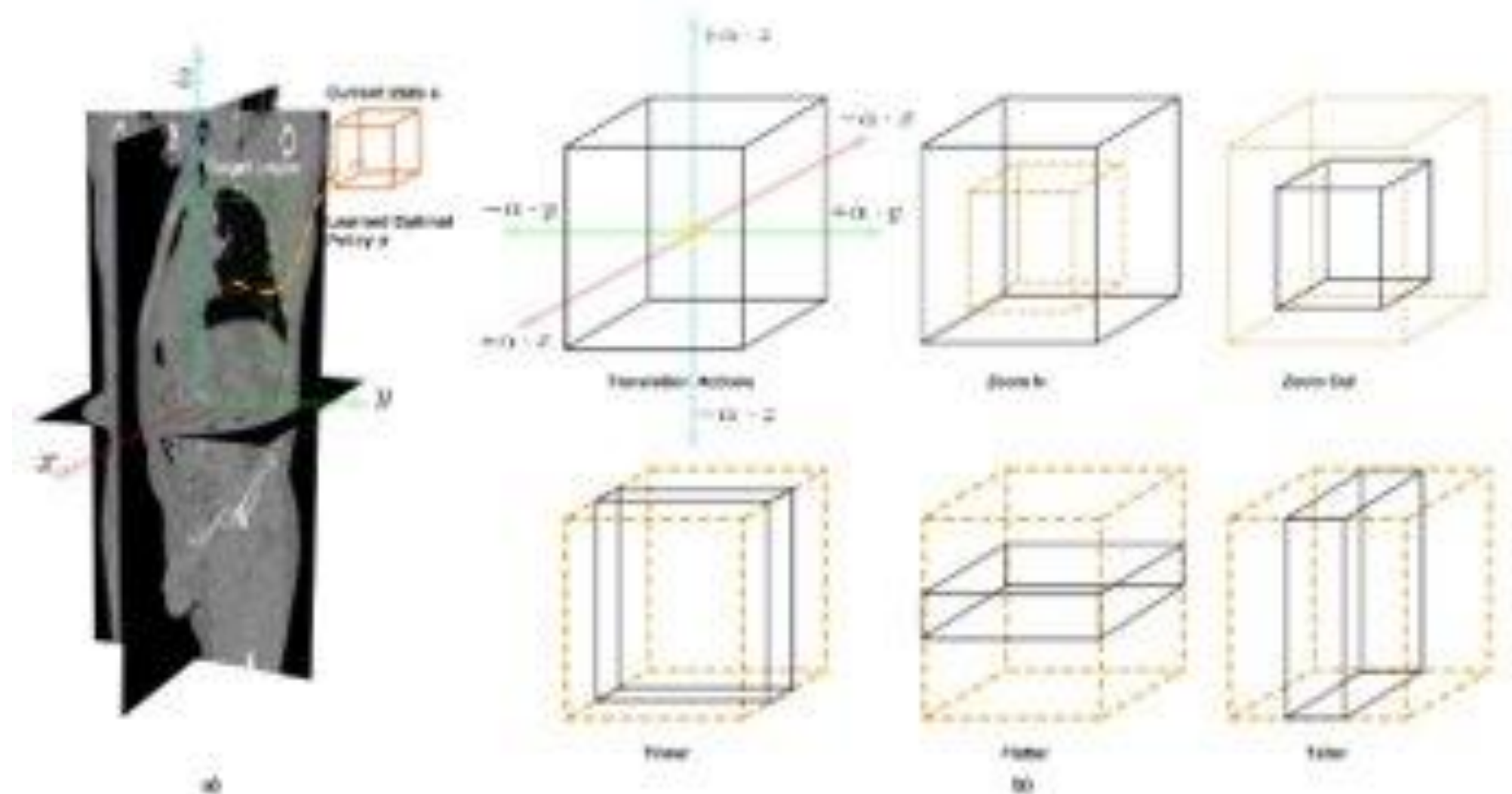
Object Detection

- **Navarro et al.** “Deep reinforcement learning for organ localization in CT” **In: MIDL 2020**
 - The RL environment and action space is the 3D CT scan.
 - The action space consists of 6 actions for translation, 2 for scaling the whole box and 3 to scale the box in each of the three directions.
 - Based on Q-learning algorithm

$$Q^*(s, a) = \max_{\pi} \mathbb{E}[R_t \mid s_t = s, a_t = a, \pi]$$

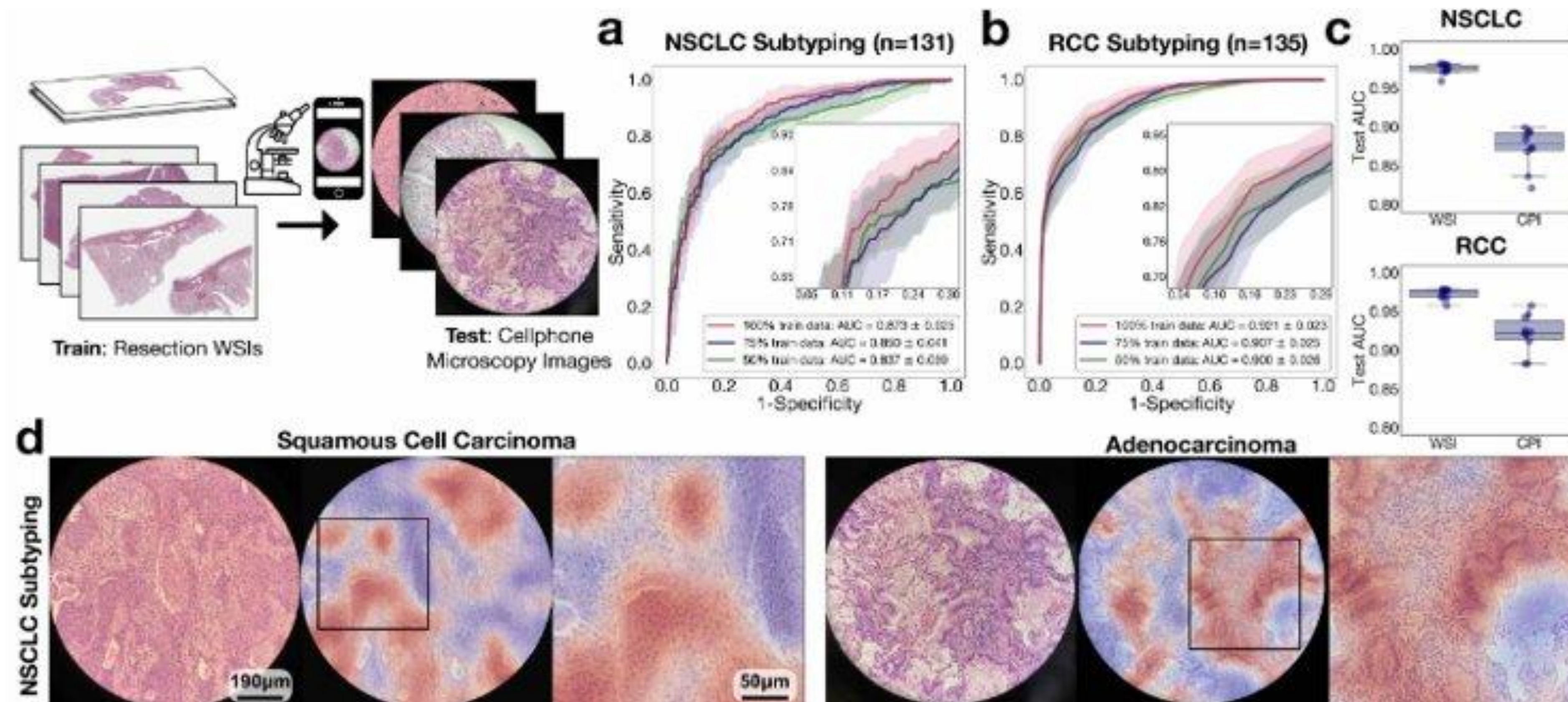
- R_t : expected reward, π policy

	Avg IoU	Wall dist [mm]	Centroid dist [mm]
Right Lung	0.77	3.46 ± 5.28	6.06 ± 10.25
Left Lung	0.73	4.91 ± 7.38	10.32 ± 17.09
Right Kidney	0.60	2.96 ± 2.91	5.69 ± 5.67
Left Kidney	0.57	4.06 ± 4.98	7.52 ± 9.02
Liver	0.80	2.41 ± 0.70	3.36 ± 1.34
Spleen	0.60	5.25 ± 7.23	9.20 ± 12.03
Pancreas	0.32	12.26 ± 13.60	20.79 ± 20.38
Global	0.63	5.04 ± 6.01	8.99 ± 10.82
Median	0.60	2.25	3.65



Classification

- **Lu et al.** “Data Efficient and Weakly Supervised Computational Pathology on Whole Slide Images” **In: Nature Biomedical Engineering 2021**



Classification

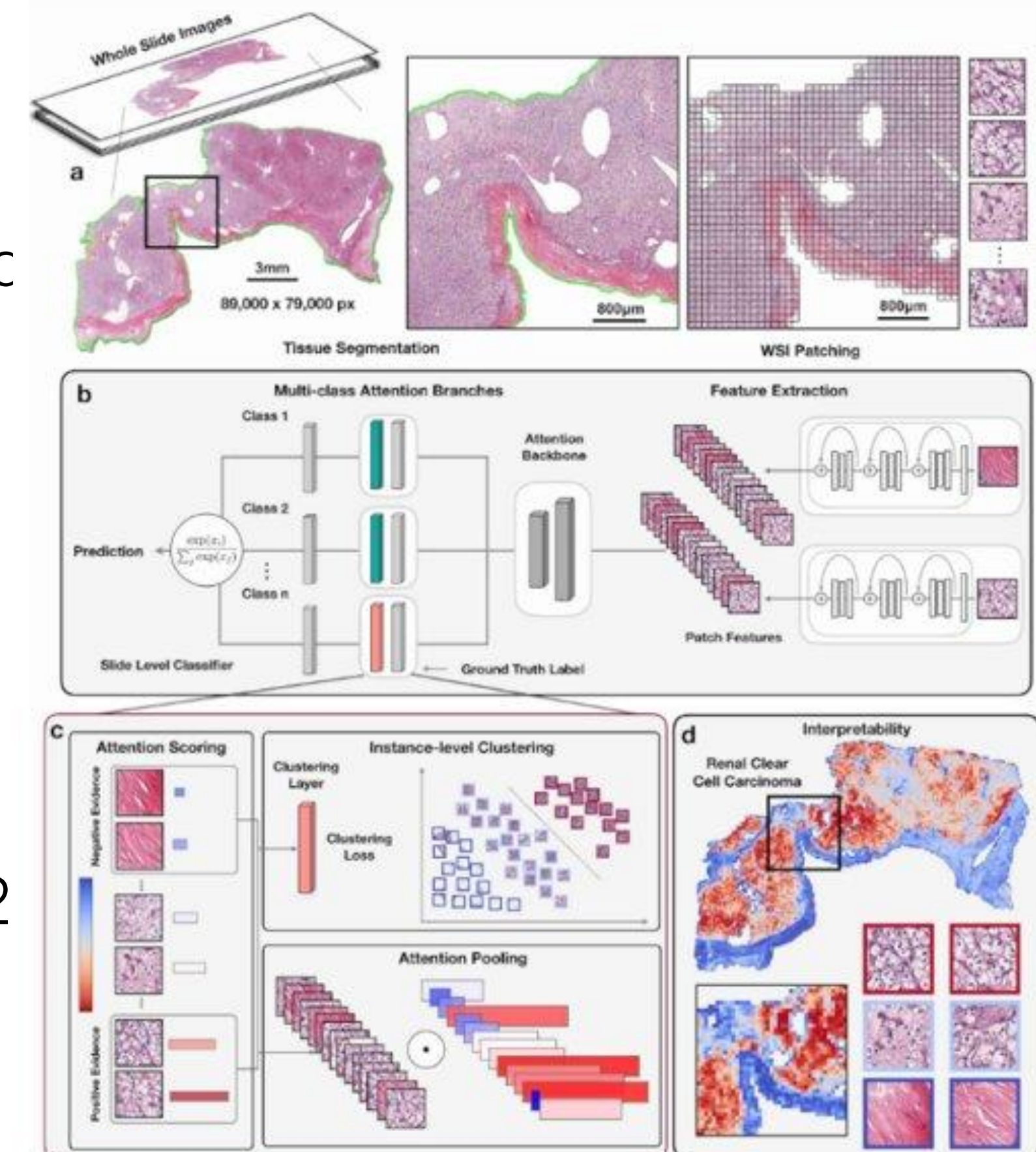
- **Lu et al.** “Data Efficient and Weakly Supervised Computational Pathology on Whole Slide Images” **In: Nature Biomedical Engineering 2021**

Clustering-constrained attention multiple instance learning (CLAM)

- Built on top of multi instance learning setup
- Use of different aggregation functions (e.g. mean operator, generalized mean, log-sum-exp, the quantile function, Noisy-OR, Noisy-And)
- Use of a pretrained ResNet50 network per patch.
- Reduce of the dimensionality from 1024-dimensional patch to 512-dimentional patch
- Attention module using two layers V_a and U_a

$$a_{k,m} = \frac{\exp \{ \mathbf{W}_{a,m} (\tanh (\mathbf{V}_a \mathbf{h}_k^\top) \odot \text{sigm} (\mathbf{U}_a \mathbf{h}_k^\top)) \}}{\sum_{j=1}^N \exp \{ \mathbf{W}_{a,m} (\tanh (\mathbf{V}_a \mathbf{h}_j^\top) \odot \text{sigm} (\mathbf{U}_a \mathbf{h}_j^\top)) \}}$$

- Two different tasks to solve:
 - Instance-level clustering using a fully connected layer with 512 hidden units
 - Smooth top1 SVM loss (ICLR 2018)

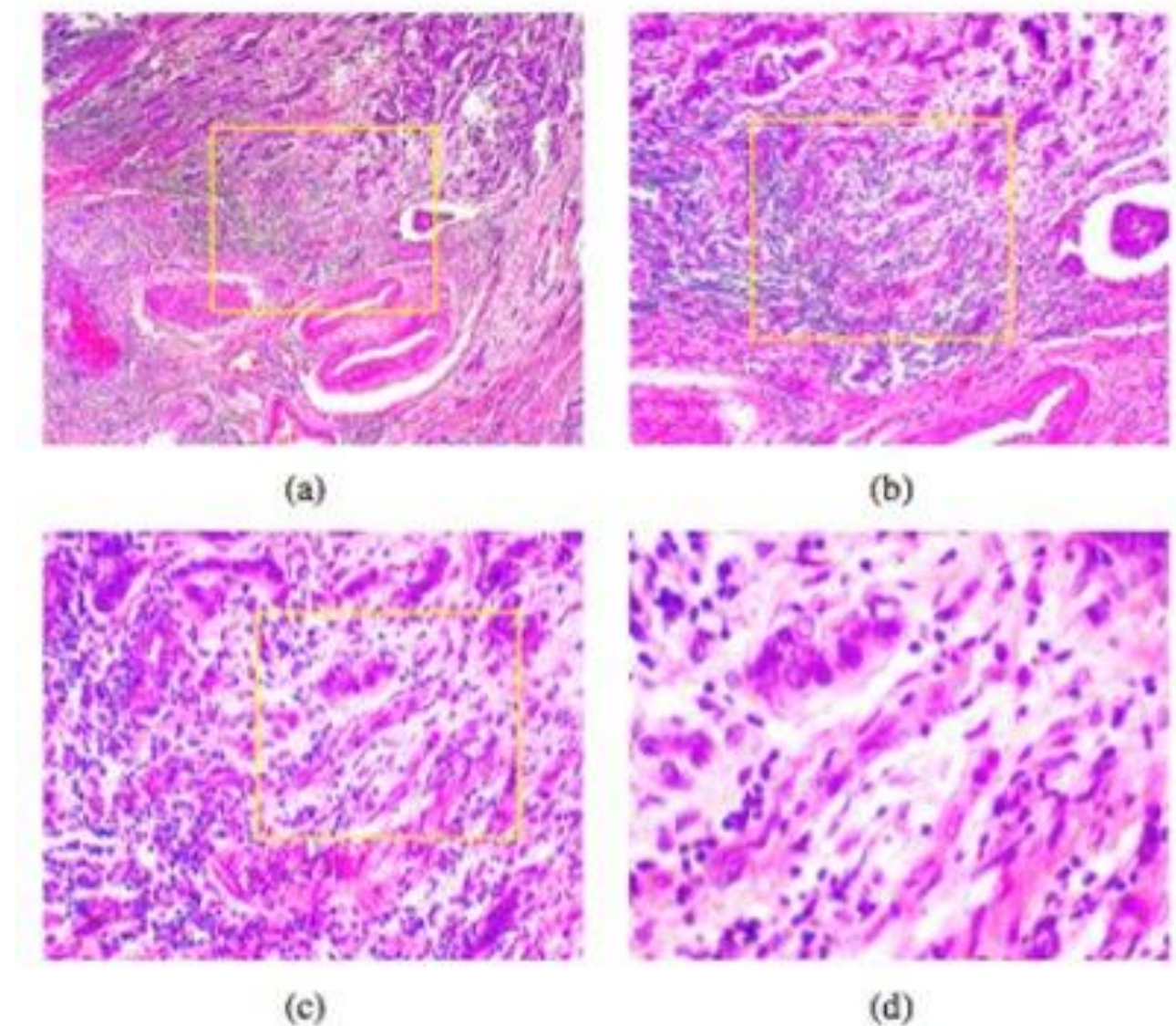


Classification

- **Sahasrabudhe et al.** “Self-Supervised Nuclei Segmentation in Histopathological Images Using Attention” **In: MICCAI 2020**

Main assumption: Given a patch extracted from a WSI viewed at a certain magnification, the level of magnification can be ascertained by looking at the size and texture of the nuclei in the patch.

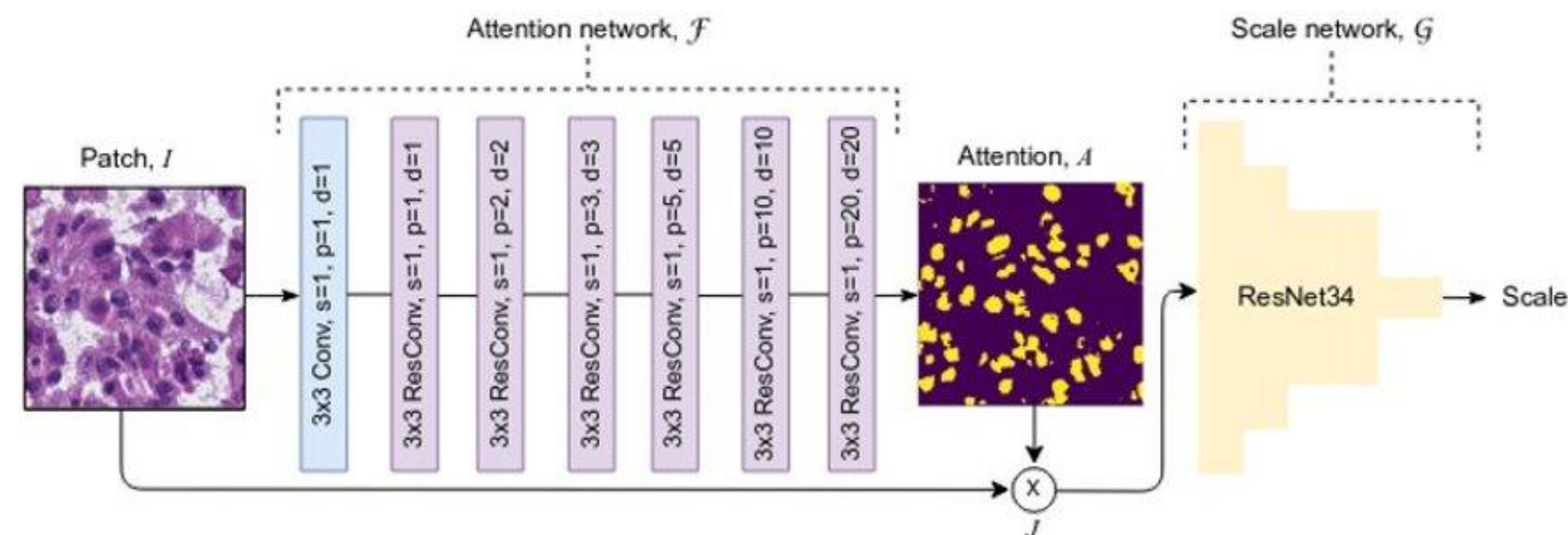
- Train a scale-sensitive network which specifically learns discriminative features for correct scale classification.



A slide of breast malignant tumor (stained with HE) seen in different magnification factors: (a) 40X, (b) 100X, (c) 200X, and (d) 400X.

Classification

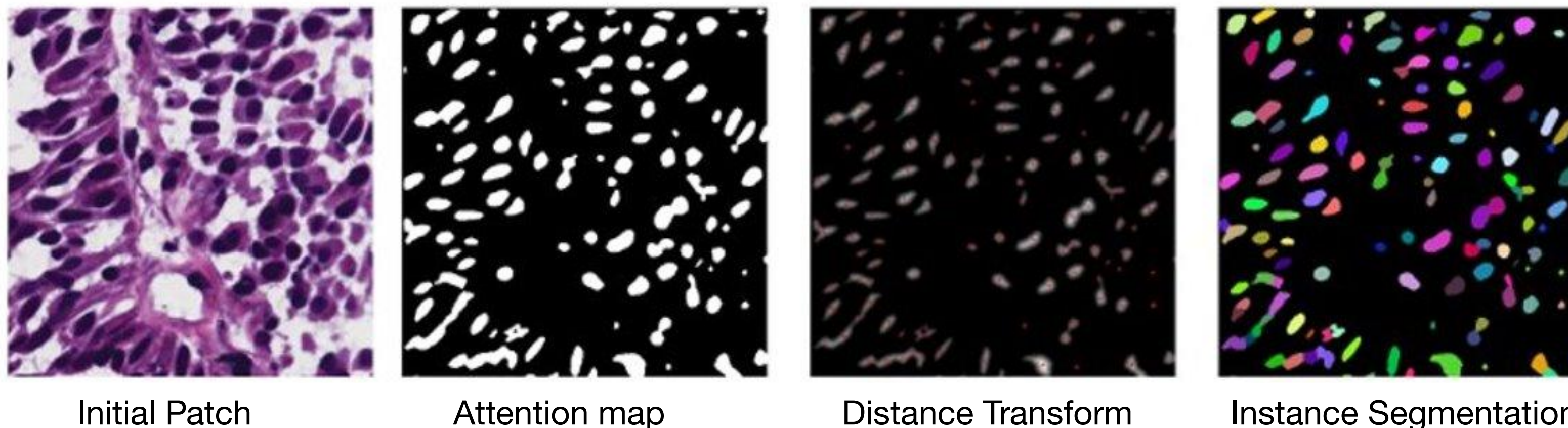
- **Sahasrabudhe et al.** “Self-Supervised Nuclei Segmentation in Histopathological Images Using Attention” **In: MICCAI 2020**



- Our network is composed by two different components:
 - The attention network F [Input: I, Output A]
 - Based on dilated filters
 - $J = A \odot I$, multiplication of the “attended” image and the initial/ patch
 - The scale network G [Input: J, Output: Scale]
 - ResNet34

Classification

- **Sahasrabudhe et al.** “Self-Supervised Nuclei Segmentation in Histopathological Images Using Attention” **In: MICCAI 2020**
- Instance segmentation of the nuclei applying a simple post-processing:
 - Morphological Operations
 - Distance Transform to detect the centers of the bloods
 - Watershed algorithm for the instance segmentation



Test dataset	Method	AJI [13]	AHD	ADC
MoNuSeg test	CNN2 [13] [†]	0.3482	8.6924	0.6928
	CNN3 [13] [†]	0.5083	7.6615	0.7623
	Best Supervised [13] [†]	0.691	-	-
	CellProfiler [13]	0.1232	9.2771	0.5974
	Fiji [13]	0.2733	8.9507	0.6493
	M _{sparse}	0.0312	13.1415	0.2283
	M _{smooth}	0.1929	8.8166	0.4789
	M _{WSI}	0.3025	8.2853	0.6209
	M _{equiv}	0.4938	8.0091	0.7136
	M _{proposed}	0.5354	7.7502	0.7477
TNBC [15]	U-Net [7] [†]	0.514	-	0.681
	SegNet+WS [7] [†]	0.559	-	0.758
	HoverNet [7] [†]	0.590	-	0.749
	CellProfiler	0.2080	-	0.4157
	M _{proposed}	0.2656	-	0.5139
CoNSeP [7]	SegNet [7] [†]	0.194	-	0.796
	U-Net [7] [†]	0.482	-	0.724
	CellProfiler [7]	0.202	-	0.434
	QuPath [7]	0.249	-	0.588
	M _{proposed}	0.1980	-	0.587

Image Registration

- The goal of registration is to define a transformation T which projects one image (source) to the other (target).
 - Two main categories
 - Rigid methods
 - Non-rigid (Deformable methods)
 - A lot of applications
 - Atlas based segmentation
 - Fusion of different modalities
 - Monitoring of diseases
 - Intra and inter patient

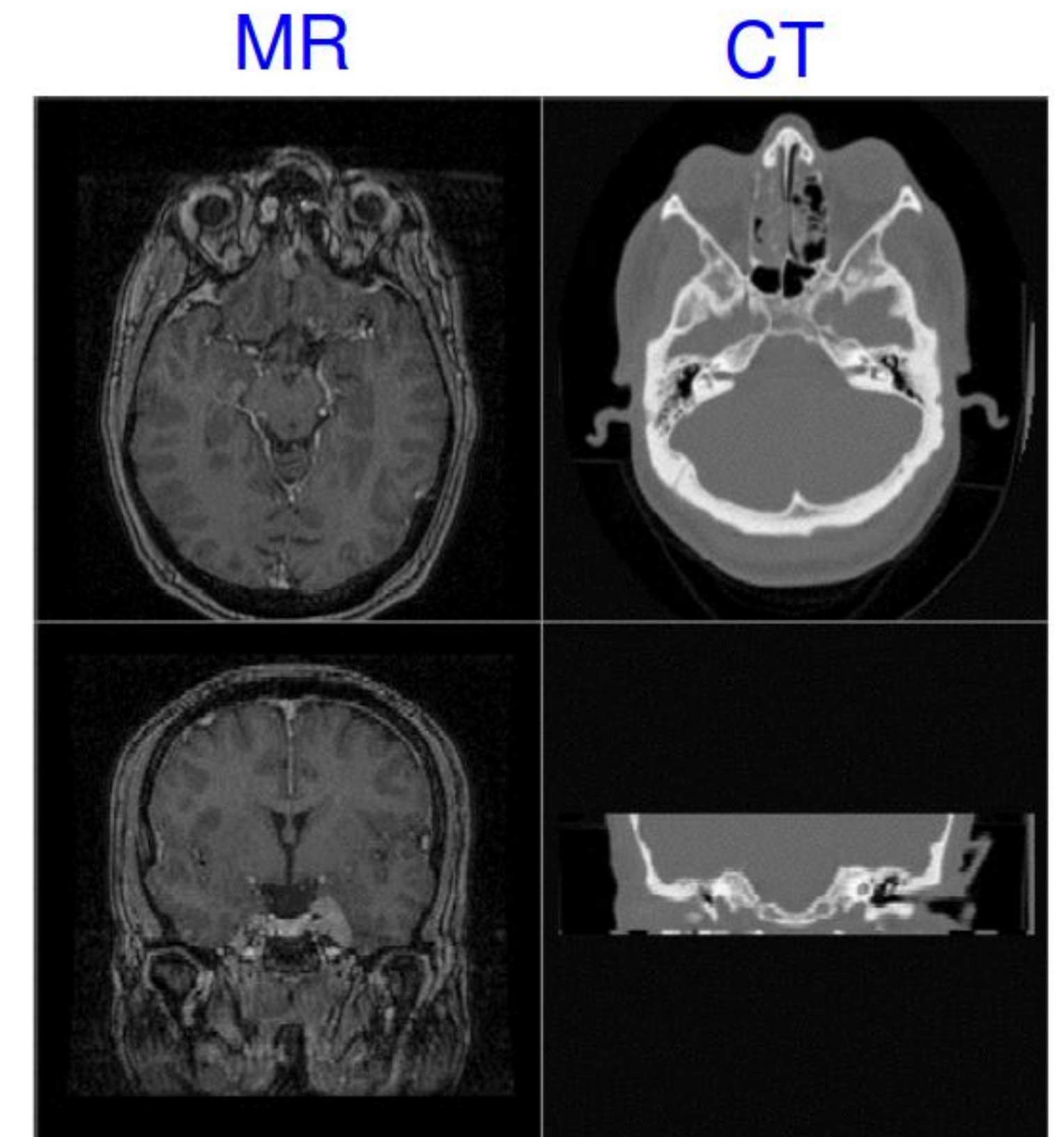
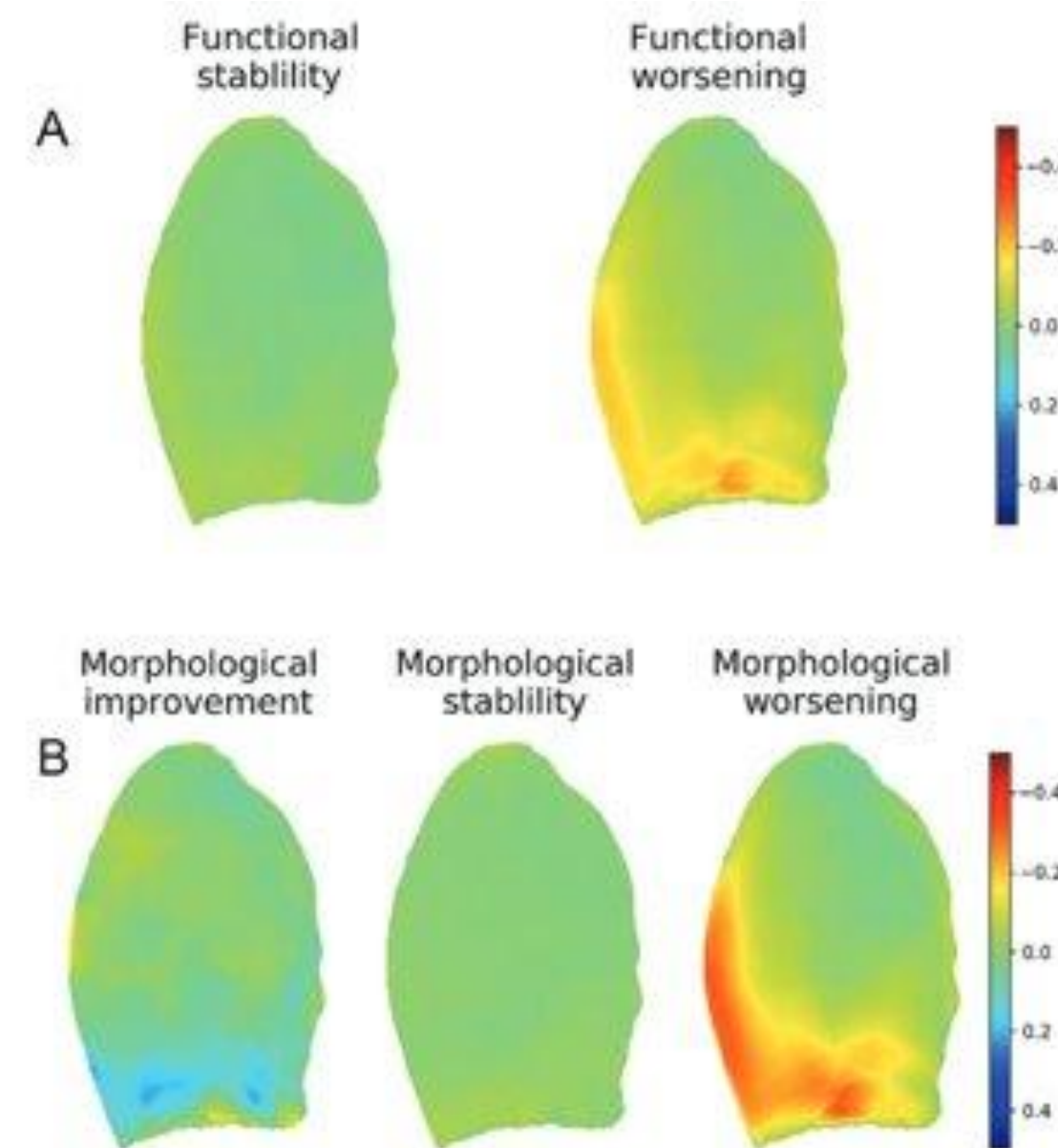


Image Registration

- **Christodoulidis et al.** “Linear and Deformable Image Registration with 3D Convolutional Neural Networks” **In: RAMBO MICCAI 2018**
 - Non-rigid, deformable procedure, the observed signals are associated through a non-linear dense transformation, or a spatially varying model.
- Three main components
 - 3D Spatial Transformer
 - 3D CNN Architecture
 - Loss Function

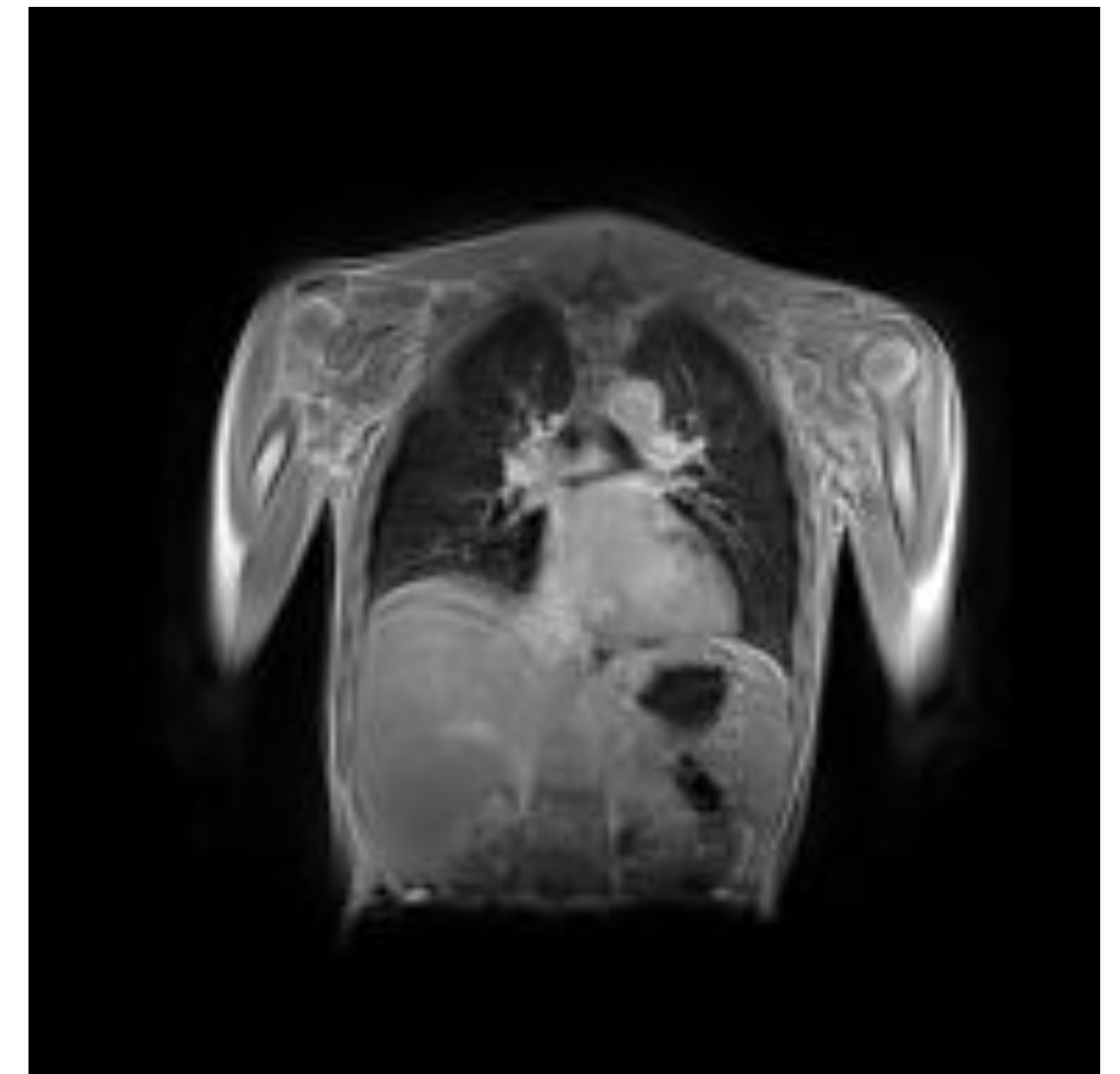


Image Registration

- **Christodoulidis et al.** “Linear and Deformable Image Registration with 3D Convolutional Neural Networks” **In: RAMBO MICCAI 2018**

- 3D Spatial transformer
 - A differentiable module which applies a spatial transformation to an image. Backward trilinear interpolation sampling

$$D(\mathbf{p}) = \sum_{\mathbf{q}} S(\mathbf{q}) \prod_d \max(0, 1 - |[G(\mathbf{p})]_d - \mathbf{q}_d|),$$

- 3D CNN Architecture
 - Based on an encoder-decoder framework, using dilated convolutional kernels. Output of the decoder is the spatial gradient along each axis ∇G . The grid is calculated by:

$$G(\mathbf{p}) = \int_{-\infty}^{\mathbf{p}} \nabla G \, d\mathbf{p}$$

- Loss Function

$$\text{Loss} = \|R - \mathcal{W}_G(S)\|^2 + \alpha \|A - A_I\|_1 + \beta \|\Phi - \Phi_I\|_1,$$

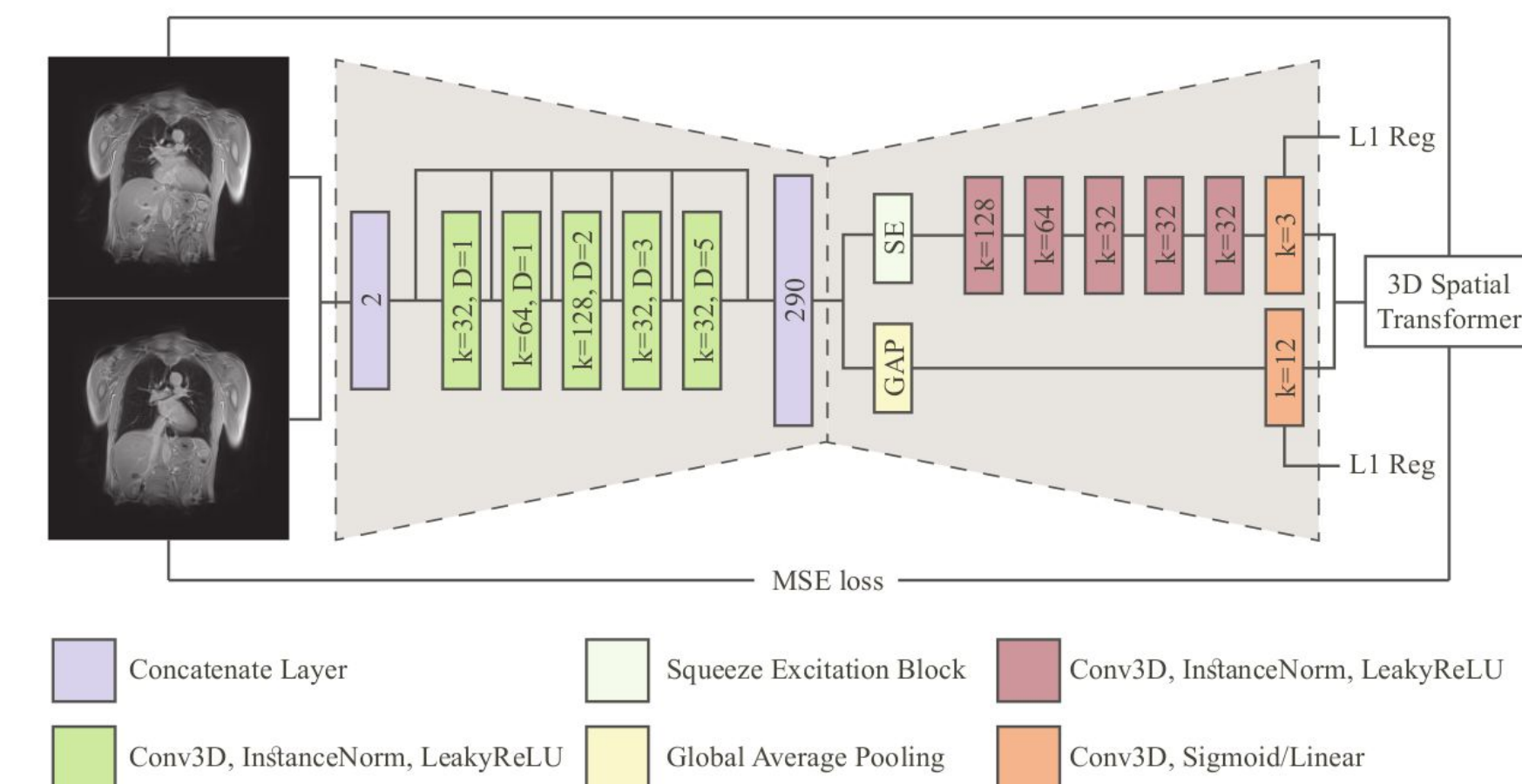
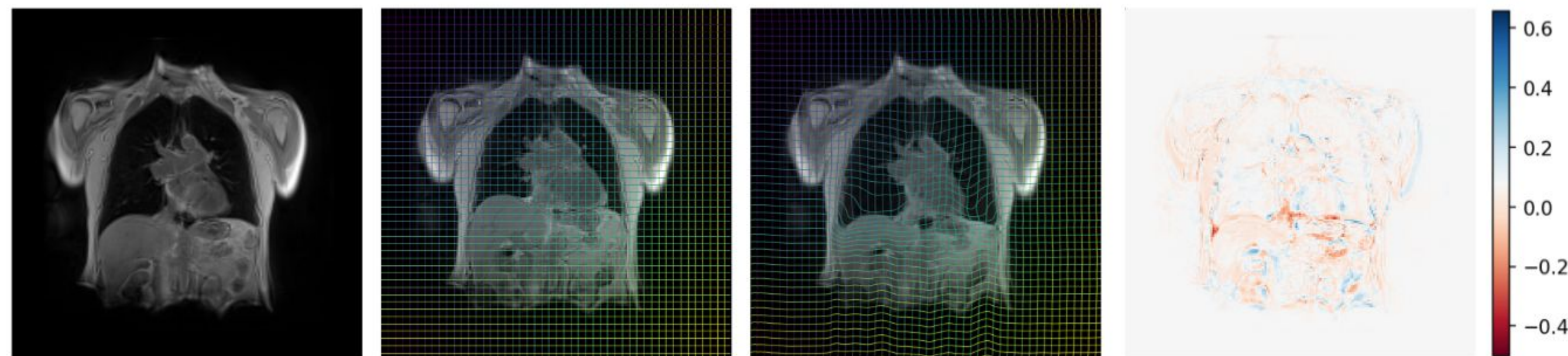


Image Registration

- **Christodoulidis et al.** “Linear and Deformable Image Registration with 3D Convolutional Neural Networks” **In: RAMBO MICCAI 2018**

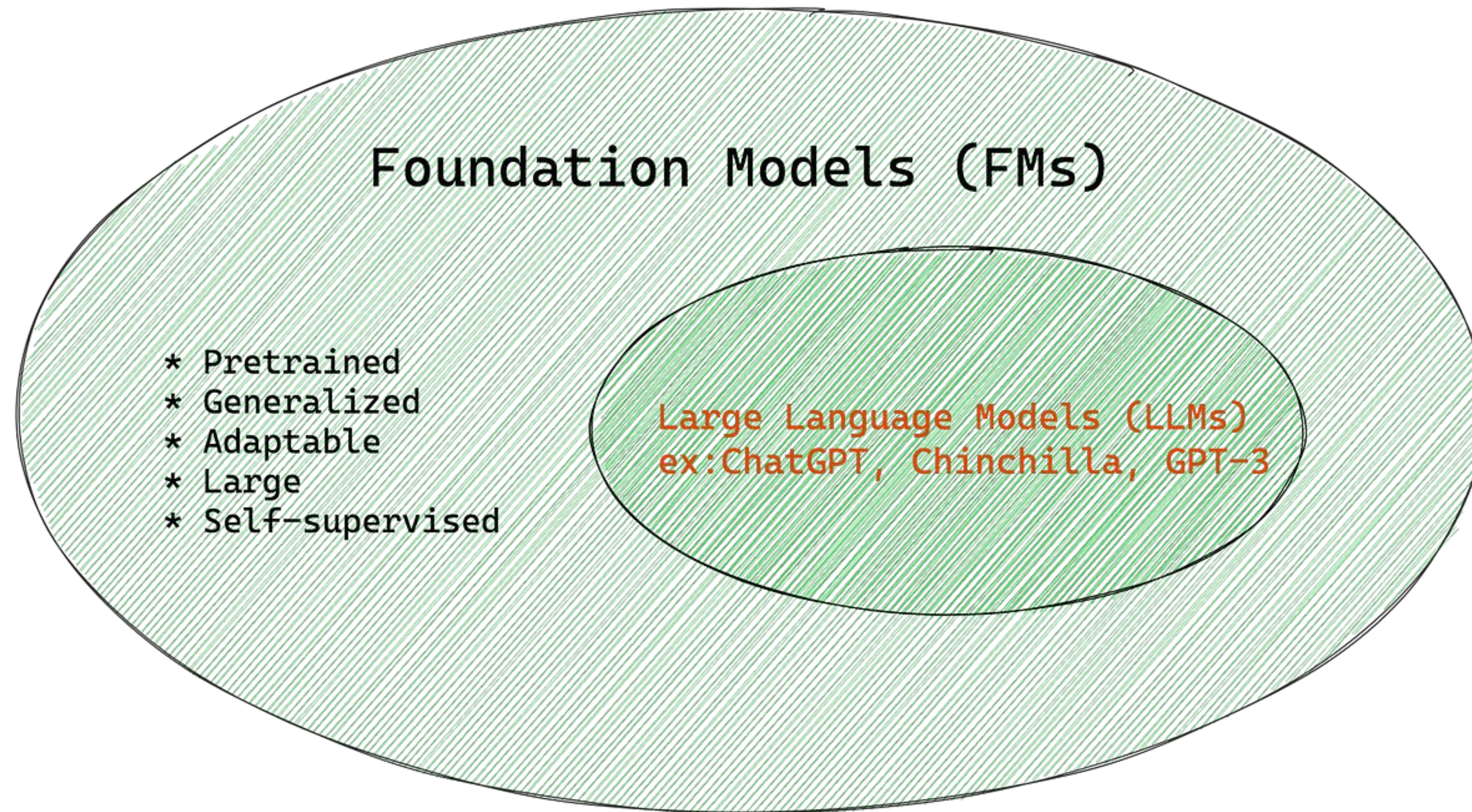
Method	dx	dy	dz	ds
Inter-observer	1.664	2.545	1.555	3.905
Deformable with NCC, DWM, and MI [7]	1.855	3.169	2.229	4.699
Proposed w/o Affine	2.014	2.947	1.858	4.569
Proposed	1.793	2.904	1.822	4.358

Method	Inhale-Exhale	All Combinations	Time/subject (s)
Unregistered	75.62±10.89	57.22±12.90	—
Deformable with NCC [7]	84.25±6.89	76.10±7.92	~1 (GPU)
Deformable with DWM [7]	88.63±4.67	75.92±8.81	~2 (GPU)
Deformable with MI [7]	88.86±5.13	76.33±8.74	~2 (GPU)
Deformable with all above [7]	88.81±5.85	78.71±8.56	~2 (GPU)
SyN [2]	83.86±6.04	—	~2500 (CPU)
Proposed w/o Affine	91.28±2.47	81.75±7.88	~0.5 (GPU)
Proposed	91.48±2.33	82.34±7.68	~0.5 (GPU)



(a) Reference image (b) Moving image (c) Deformed image (d) Difference

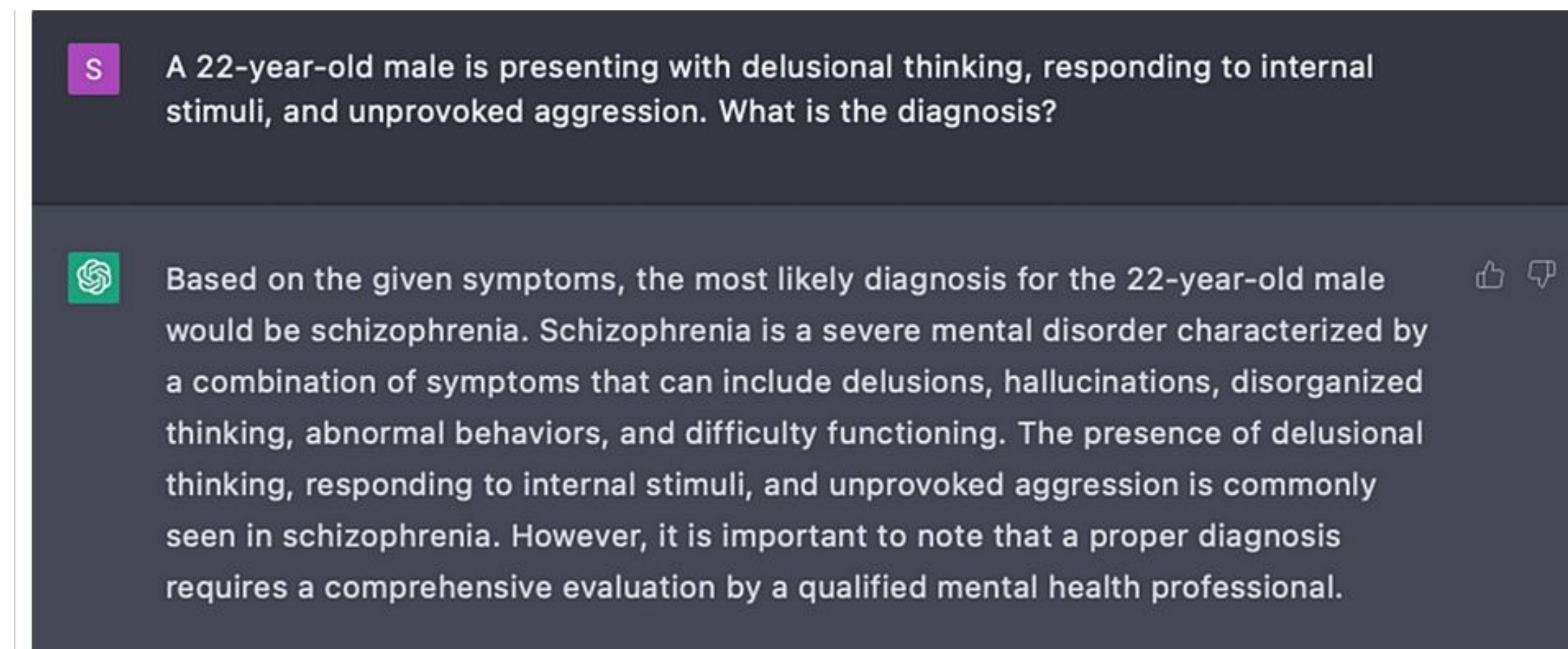
Foundation Models and Medical Imaging



FMs are models trained on broad data (using self-supervision at scale)
that can be adapted to a wide range of downstream tasks.
<https://hai.stanford.edu/news/reflections-foundation-models>

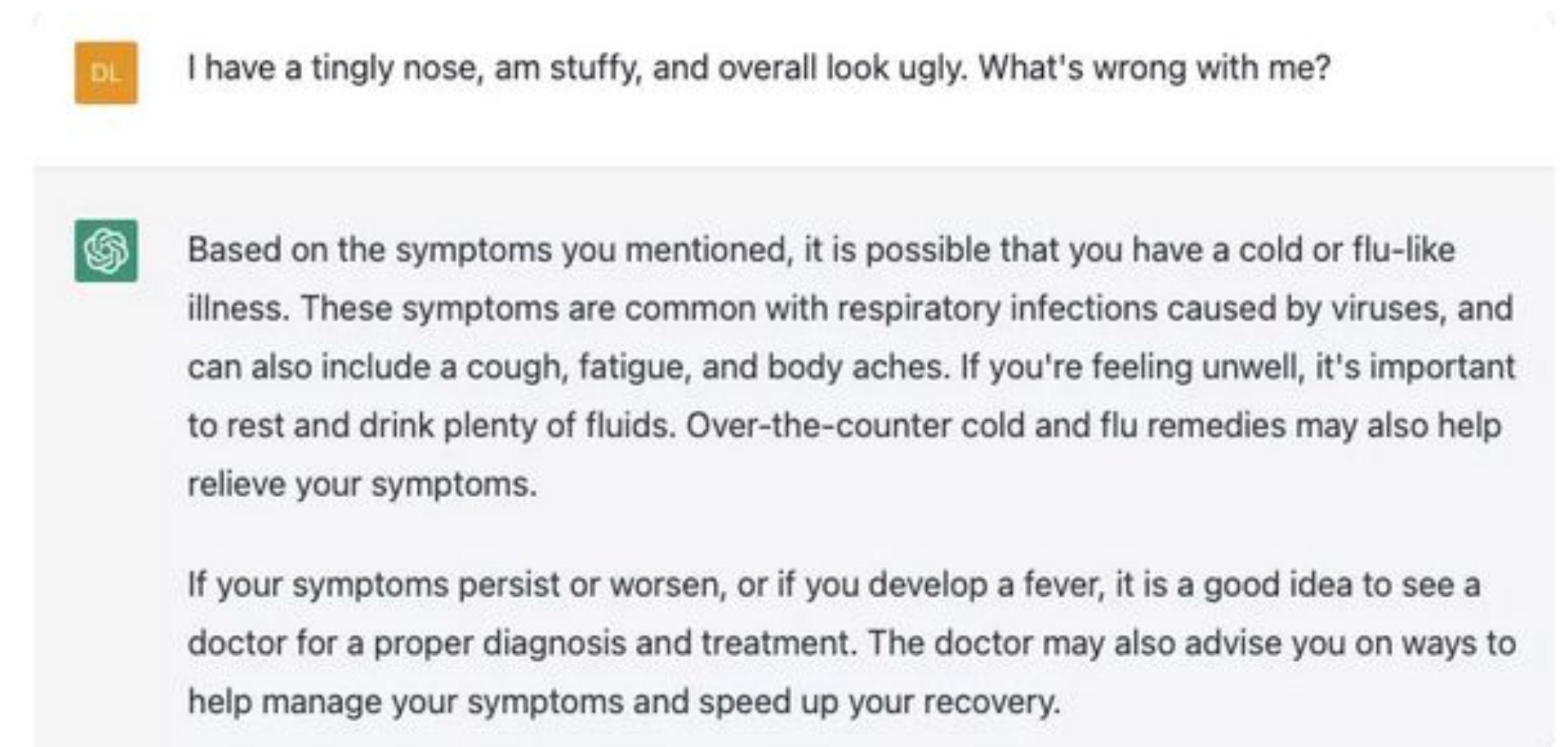
Towards Large Deep Learning Models

- The community is working on building huge and powerful models for different tasks with currently impressive results on:
 - Large Language Models inferring relationships between words within the text
 - ChatGPT: 176-billion parameters trained in 570GB data from text, books, websites, and more



S A 22-year-old male is presenting with delusional thinking, responding to internal stimuli, and unprovoked aggression. What is the diagnosis?

GPT Based on the given symptoms, the most likely diagnosis for the 22-year-old male would be schizophrenia. Schizophrenia is a severe mental disorder characterized by a combination of symptoms that can include delusions, hallucinations, disorganized thinking, abnormal behaviors, and difficulty functioning. The presence of delusional thinking, responding to internal stimuli, and unprovoked aggression is commonly seen in schizophrenia. However, it is important to note that a proper diagnosis requires a comprehensive evaluation by a qualified mental health professional.



DL I have a tingly nose, am stuffy, and overall look ugly. What's wrong with me?

GPT Based on the symptoms you mentioned, it is possible that you have a cold or flu-like illness. These symptoms are common with respiratory infections caused by viruses, and can also include a cough, fatigue, and body aches. If you're feeling unwell, it's important to rest and drink plenty of fluids. Over-the-counter cold and flu remedies may also help relieve your symptoms.

If your symptoms persist or worsen, or if you develop a fever, it is a good idea to see a doctor for a proper diagnosis and treatment. The doctor may also advise you on ways to help manage your symptoms and speed up your recovery.

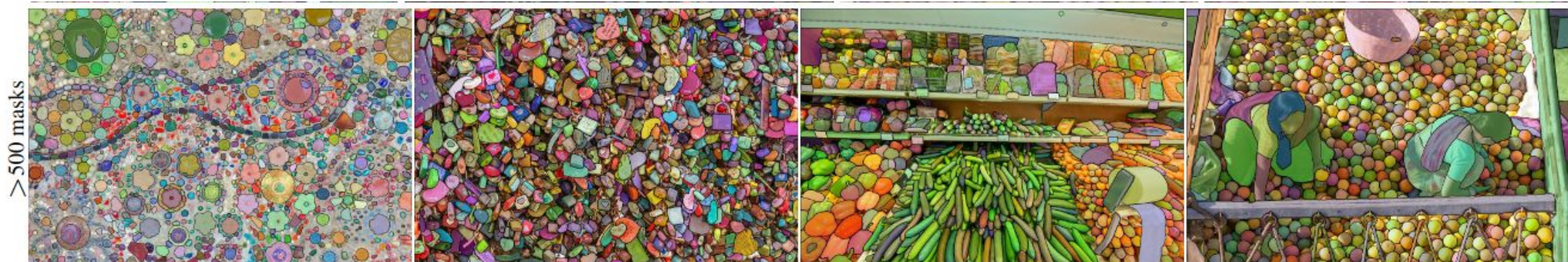
Galido P, Butala S, Chakerian M, et al. (April 26, 2023) A Case Study Demonstrating Applications of ChatGPT in the Clinical Management of Treatment-Resistant Schizophrenia . Cureus 15(4): e38166. doi:10.7759/cureus.38166

<https://www.verywellhealth.com/chatgpt-in-healthcare-7107800>

<https://openai.com/blog/chatgpt>

Towards Large Deep Learning Models

- The community is working on building huge and powerful models for different tasks with currently impressive results on:
 - Imaging tasks and image segmentation
 - SAM: trained on 1B masks and from 11M licensed and privacy-preserving image



Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y. and Dollár, P., 2023. Segment anything. arXiv preprint arXiv:2304.02643.

Towards Large Deep Learning Models

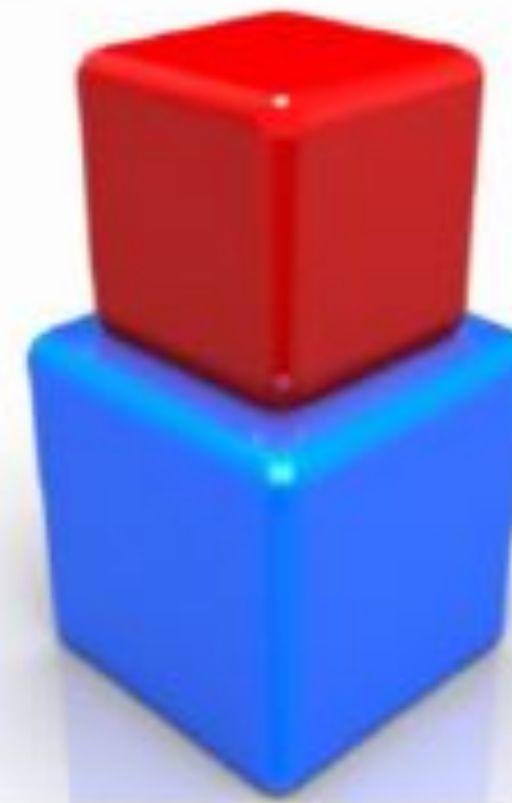
- The community is working on building huge and powerful models for different tasks with currently impressive results on:
 - Image generation from text
 - DALL-E: 12-billion parameters trained in 250 million text-images pairs
 - GLIDE: 3.5-billion parameter text-conditional diffusion, and another 1.5 billion parameter text-conditional upsampling diffusion model.



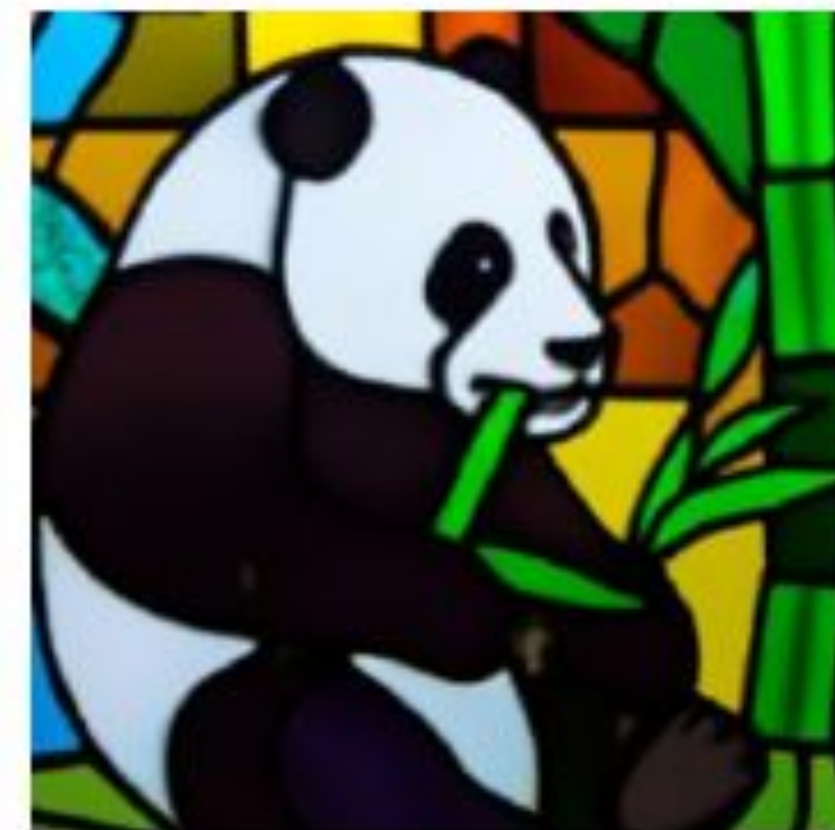
"a boat in the canals of venice"



"a painting of a fox in the style of starry night"



"a red cube on top of a blue cube"

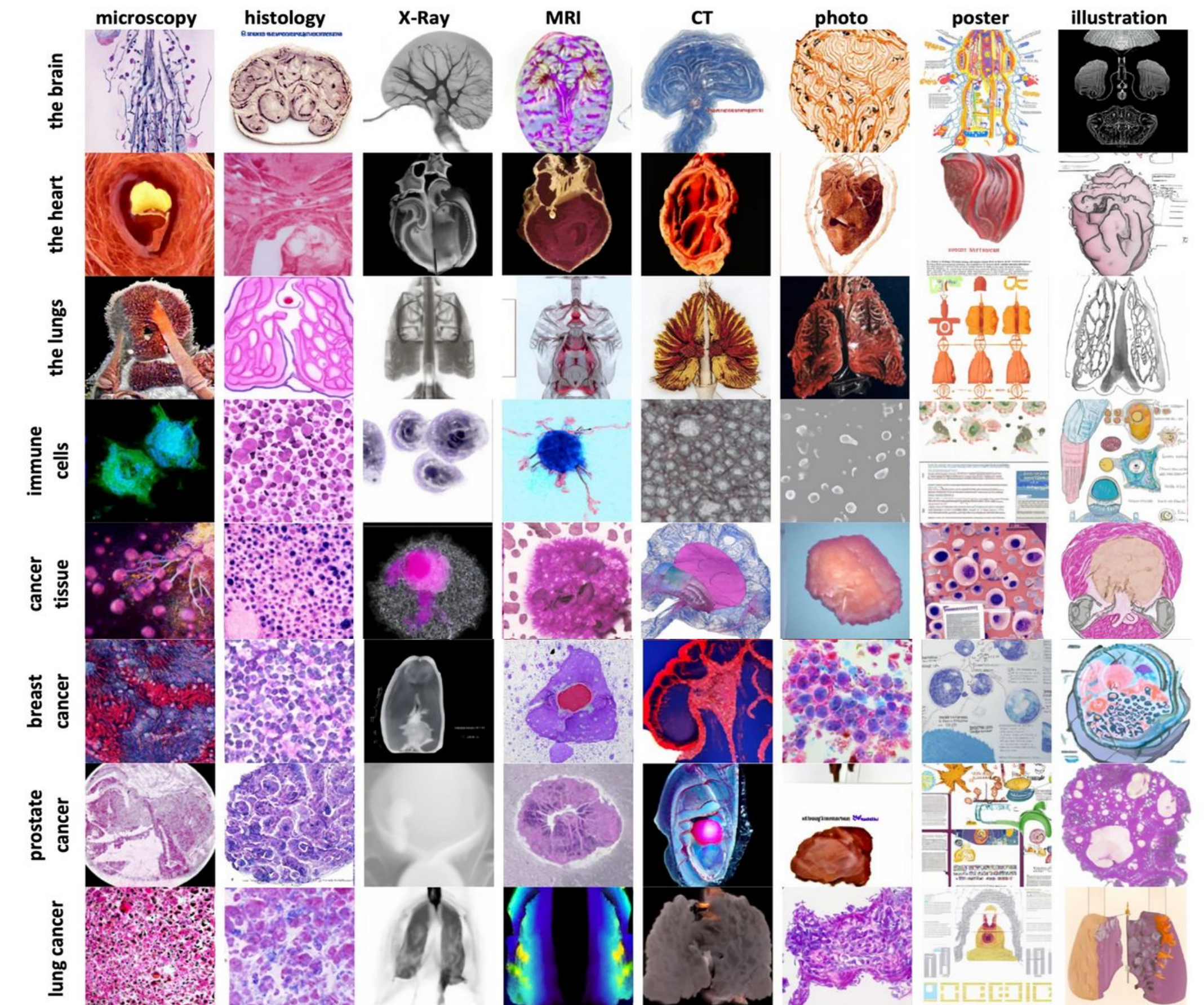


"a stained glass window of a panda eating bamboo"

Nichol, A., Dhariwal, P., Ramesh, A., Shyam, P., Mishkin, P., McGrew, B., Sutskever, I. and Chen, M., 2021. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. arXiv preprint arXiv:2112.10741.

Challenges in the Medical Domain

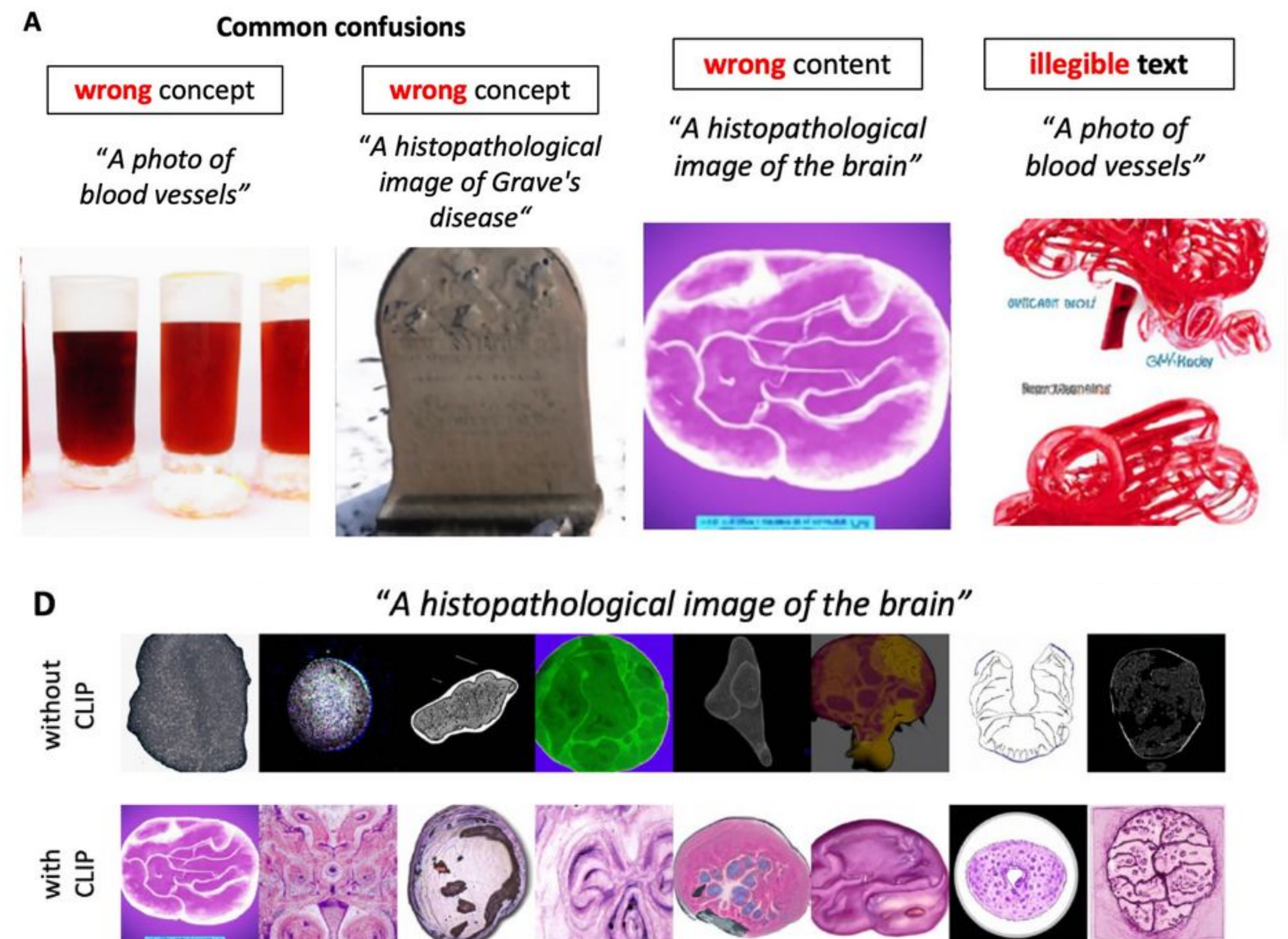
- GLIDE model: a diffusion model for the problem of text-conditional synthesis and compare two different guidance strategies: CLIP guidance and classifier-free guidance. [December 2021]
- Dataset: 250 million text-image from the internet incorporating Conceptual Captions, the text-image pairs from Wikipedia and a subset of YFCC100M.



Kather, J.N., Ghaffari Laleh, N., Foersch, S. *et al.* Medical domain knowledge in domain-agnostic generative AI. *npj Digit. Med.* **5**, 90 (2022)

Challenges in their use on the Medical Domain

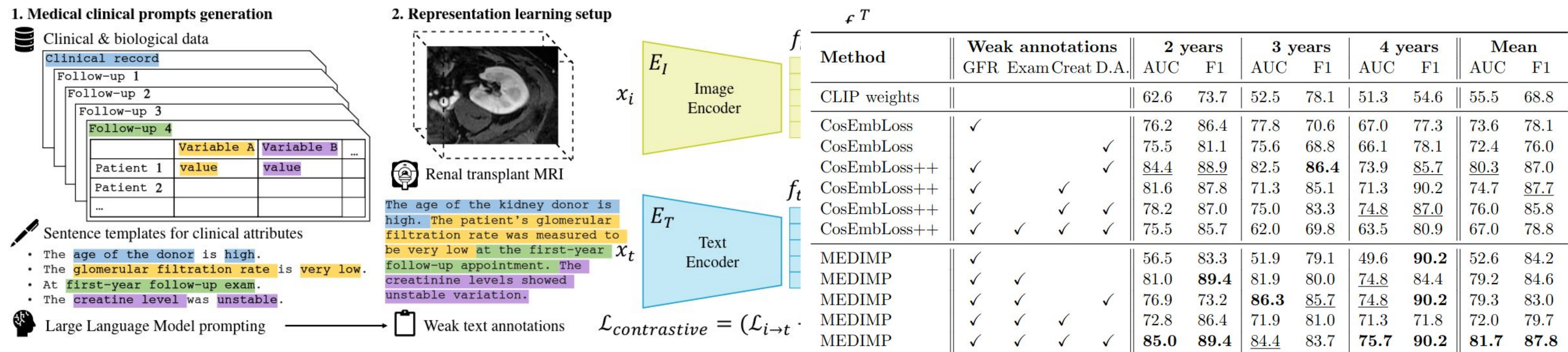
- These models **do not generalize** well on medical data and particular clinical questions
 - GLIDE performance evaluation:
 - high score for: histopathology, scientific posters and scientific illustration
 - poor score for: X-rays, MRI and CT
 - Similar models could be used for exploring biological mechanisms:
 - *A histology image of a patient who benefits from immunotherapy*
 - *An MRI image of a patient who should be treated with a statin*



Kather, J.N., Ghaffari Laleh, N., Foersch, S. *et al.* Medical domain knowledge in domain-agnostic generative AI. *npj Digit. Med.* **5**, 90 (2022)

Integrate LLMs on medical applications

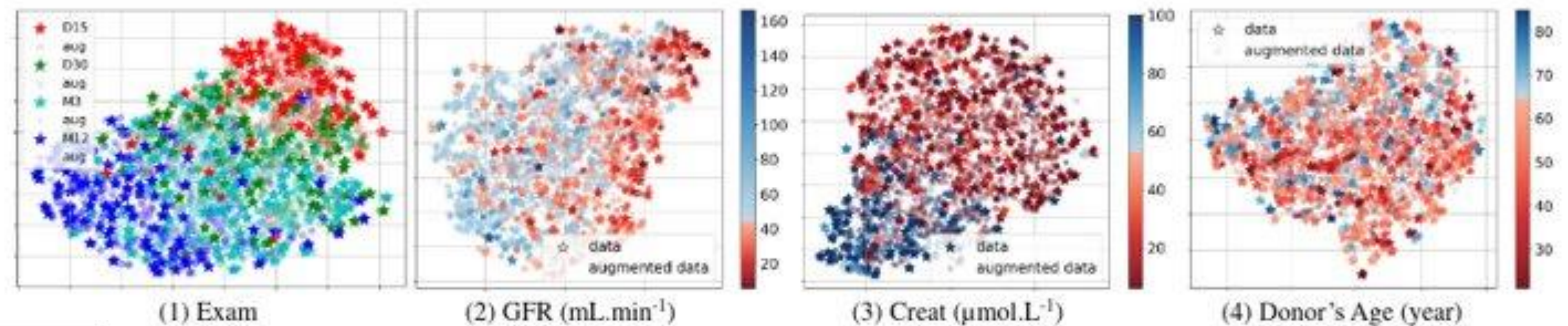
- Explore pretrained models and their use on multimodal learning and integration of clinical attributes into text.
- Use contrastive learning to match medical imaging with text, enhancing explainability and boosting performance on prognosis of graft kidney transplant.



Milecki, L., Kalogeiton, V., Bodard, S., Anglicheau, D., Correas, J. M., Timsit, M. O., & Vakalopoulou, M. (2023). MEDIMP: Medical Images and Prompts for renal transplant representation learning. MIDL 2023

Integrate LLMs on medical applications

- This approach provides an elegant way to incorporate clinical or biological information into the learning process of feature extraction of medical imaging data.



Method	Weak annotations				2 years		3 years		4 years		Mean	
	GFR	Exam	Creat	D.A.	AUC	F1	AUC	F1	AUC	F1	AUC	F1
CLIP weights					62.6	73.7	52.5	78.1	51.3	54.6	55.5	68.8
CosEmbLoss	✓				76.2	86.4	77.8	70.6	67.0	77.3	73.6	78.1
CosEmbLoss				✓	75.5	81.1	75.6	68.8	66.1	78.1	72.4	76.0
CosEmbLoss++	✓			✓	<u>84.4</u>	<u>88.9</u>	82.5	86.4	73.9	<u>85.7</u>	<u>80.3</u>	87.0
CosEmbLoss++	✓		✓		81.6	87.8	71.3	85.1	71.3	90.2	74.7	<u>87.7</u>
CosEmbLoss++	✓		✓	✓	78.2	87.0	75.0	83.3	<u>74.8</u>	<u>87.0</u>	76.0	85.8
CosEmbLoss++	✓	✓	✓	✓	75.5	85.7	62.0	69.8	63.5	80.9	67.0	78.8
MEDIMP	✓				56.5	83.3	51.9	79.1	49.6	90.2	52.6	84.2
MEDIMP	✓	✓			81.0	89.4	81.9	80.0	<u>74.8</u>	84.4	79.2	84.6
MEDIMP	✓	✓		✓	76.9	73.2	86.3	<u>85.7</u>	<u>74.8</u>	90.2	79.3	83.0
MEDIMP	✓	✓	✓		72.8	86.4	71.9	81.0	71.3	71.8	72.0	79.7
MEDIMP	✓	✓	✓	✓	85.0	89.4	<u>84.4</u>	83.7	75.7	90.2	81.7	87.8

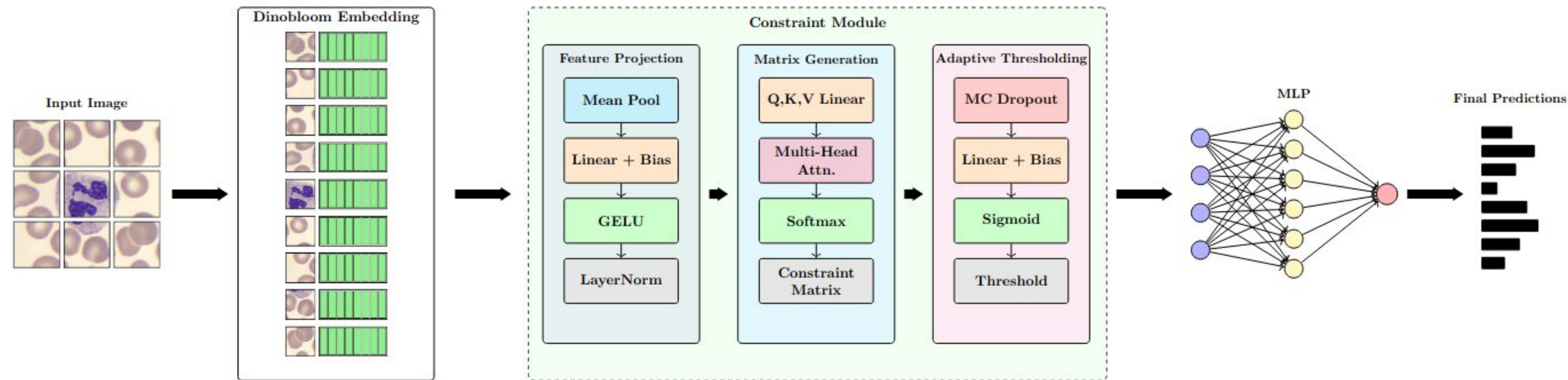
- MEDIMP augmentation perform better than other sentence augmentations

Method	2 years		3 years		4 years		Mean	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1
MEDIMP	85.0	89.4	84.4	83.7	<u>75.7</u>	90.2	81.7	87.8
Manual	74.2	76.2	<u>80.6</u>	62.1	80.0	76.9	<u>78.3</u>	71.7
T5 WebNLG	<u>74.8</u>	<u>85.7</u>	78.8	<u>83.3</u>	74.8	<u>85.7</u>	76.1	<u>84.9</u>

Milecki, L., Kalogeiton, V., Bodard, S., Anglicheau, D., Correias, J. M., Timsit, M. O., & Vakalopoulou, M. (2023). MEDIMP: Medical Images and Prompts for renal transplant representation learning. MIDL 2023

Biologically-Constrained Foundation model

- Using DinoBloom foundation model for hematological analysis, we add **more constraints** with respect to the different classes that are available to integrate information about the different classes and abnormalities available in the dataset.



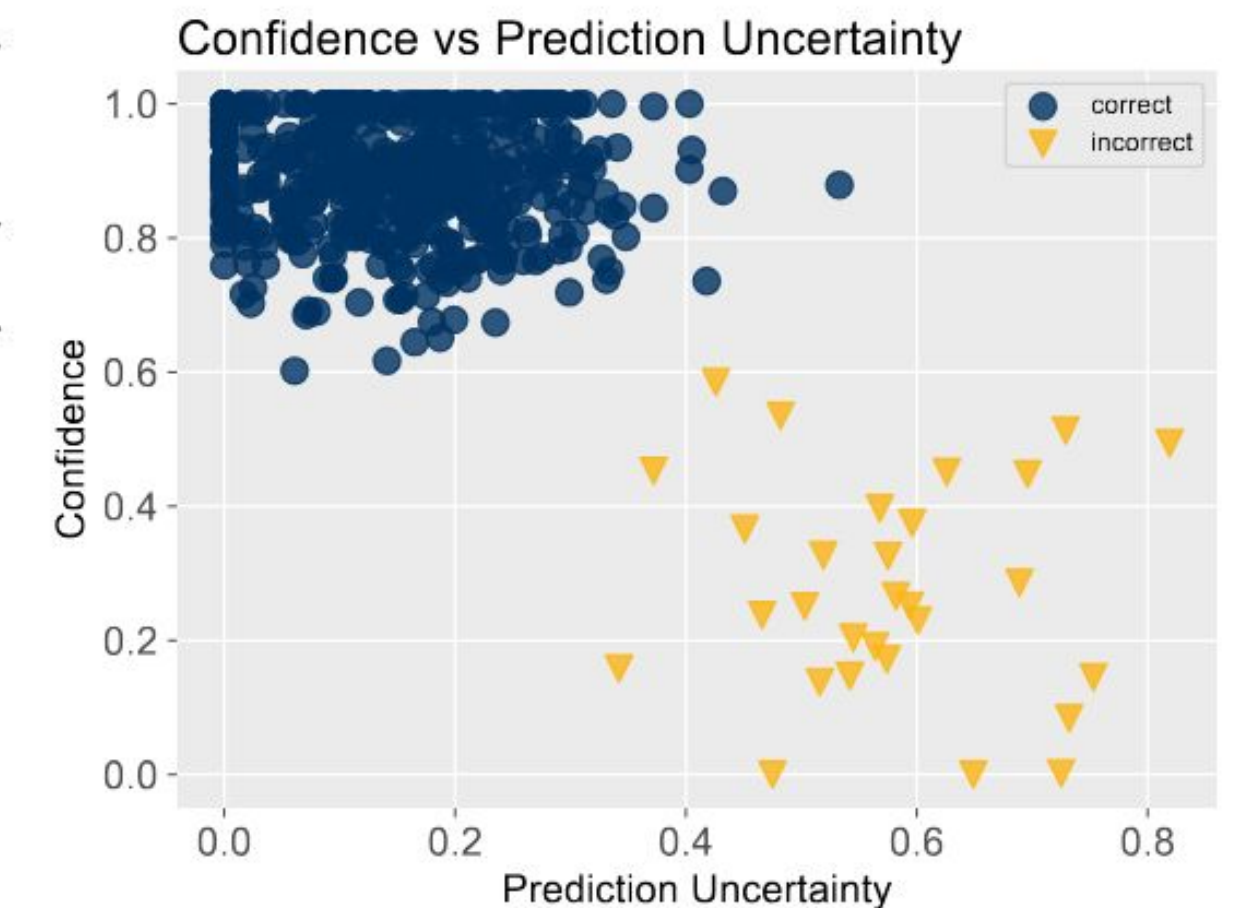
- Our dataset comprises **1,934 high-resolution microscopy images of neutrophils**, including both normal cells (878 images) and various abnormalities (582 images) without any patient level information.

Nabil Mouadden, Véronique Vergé, Ahmadreza Arbab, Jean-Baptiste Micol, Elsa Bernard, Aline Renneville, Stergios Christodoulidis, and Maria Vakalopoulou. "Biologically-Constrained Multi-Label Classification with Learnable Domain Knowledge." In Medical Imaging with Deep Learning. 2025.

Biologically-Constrained Foundation model

- Very good and robust performance

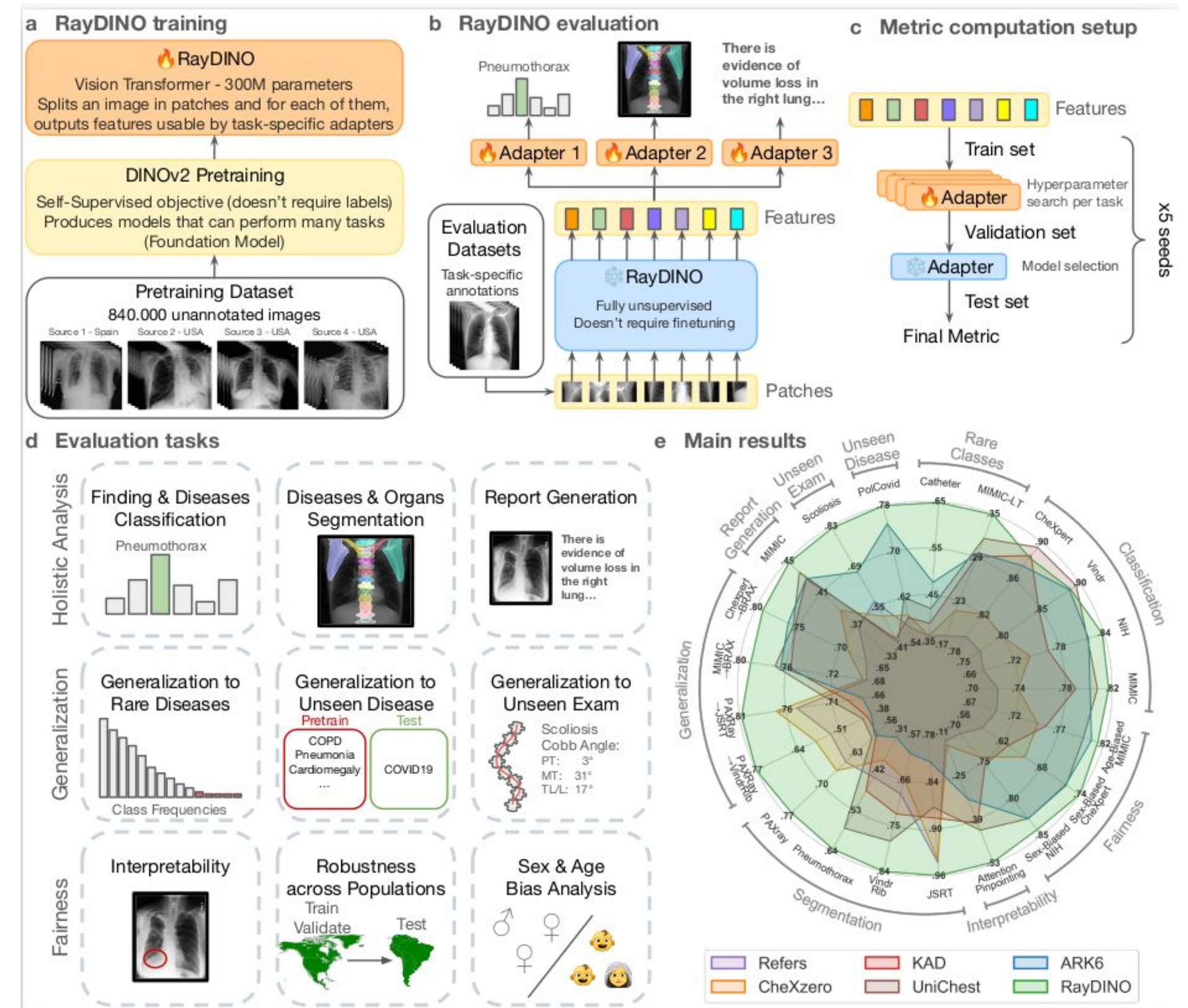
Method	Normal		Chromatin		Dohle		Hypergr.		Hyperseg.		Hypogr.		Hyposeg.		Overall	
	wF1	bAcc	wF1	bAcc	wF1	bAcc	wF1	bAcc	wF1	bAcc	wF1	bAcc	wF1	bAcc	wF1	bAcc
DINOv2 ViT-S/14	0.72	0.70	0.43	0.41	0.76	0.74	0.94	0.92	0.79	0.77	0.80	0.78	0.84	0.82	0.68	0.66
DINOv2 ViT-B/14	0.71	0.69	0.48	0.46	0.77	0.75	0.92	0.90	0.81	0.79	0.82	0.80	0.83	0.81	0.70	0.68
DINOv2 ViT-L/14	0.72	0.70	0.49	0.47	0.77	0.75	0.89	0.87	0.83	0.81	0.81	0.79	0.84	0.82	0.71	0.69
DinoBloom-S	0.86	0.84	0.55	0.53	0.86	0.84	0.94	0.92	0.89	0.87	0.90	0.88	0.84	0.82	0.85	0.83
DinoBloom-B	0.87	0.85	0.61	0.59	0.87	0.85	0.92	0.90	0.91	0.89	<u>0.91</u>	<u>0.89</u>	0.83	0.81	<u>0.86</u>	<u>0.84</u>
DinoBloom-L	<u>0.88</u>	<u>0.86</u>	<u>0.63</u>	<u>0.61</u>	<u>0.87</u>	<u>0.85</u>	<u>0.91</u>	<u>0.89</u>	<u>0.91</u>	<u>0.89</u>	0.90	0.88	<u>0.84</u>	<u>0.82</u>	<u>0.86</u>	<u>0.84</u>
CTransPath	0.80	0.78	0.52	0.50	0.83	0.81	0.88	0.86	0.80	0.78	0.82	0.80	0.83	0.81	0.74	0.72
Phikon ViT-B	0.83	0.81	0.54	0.52	0.85	0.83	0.88	0.86	0.82	0.80	0.85	0.83	0.83	0.81	0.76	0.74
Ours	0.99	0.97	0.87	0.85	0.95	0.93	1.00	1.00	0.97	0.95	0.90	0.88	0.90	0.88	0.94	0.92



Nabil Mouadden, Véronique Vergé, Ahmadreza Arbab, Jean-Baptiste Micol, Elsa Bernard, Aline Renneville, Stergios Christodoulidis, and Maria Vakalopoulou. "Biologically-Constrained Multi-Label Classification with Learnable Domain Knowledge." In Medical Imaging with Deep Learning. 2025.

RayDINO foundation model for X-Rays

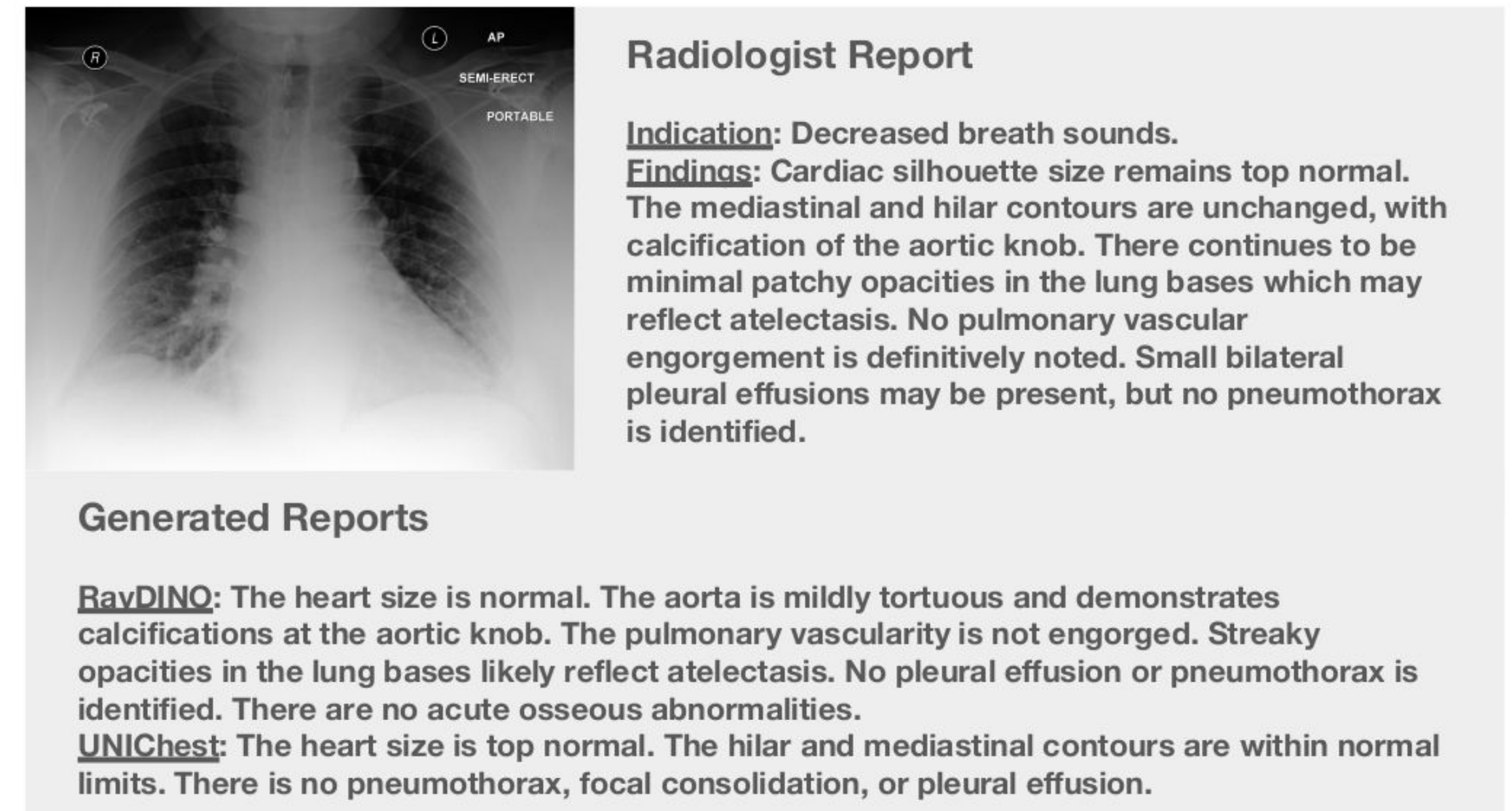
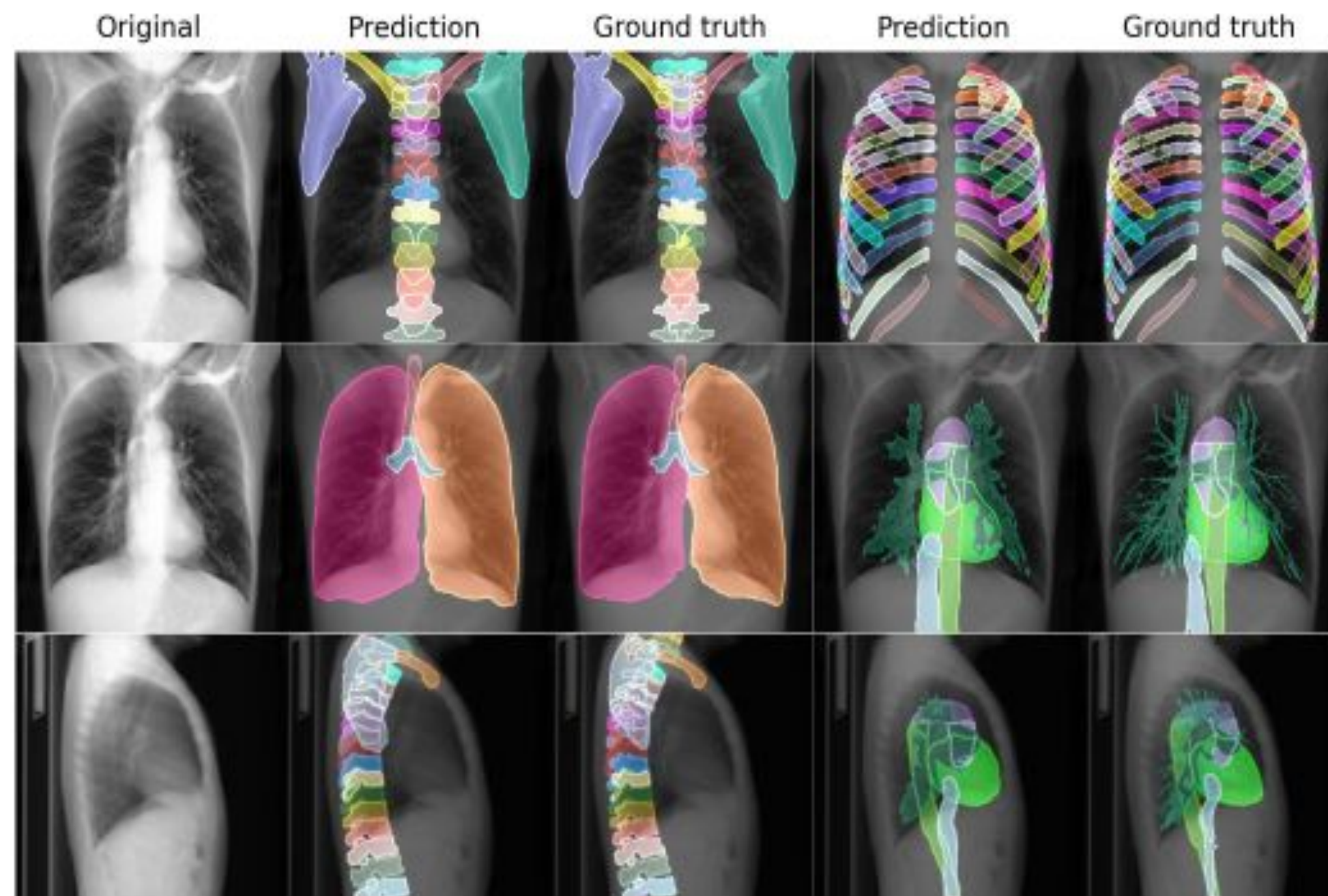
- Bigger and bigger models are available currently with a lot of capabilities.
- RayDINO the largest vision transformer (ViT) model that is currently available of 307M parameters that can generate holistic representations of chest X-rays using the self-supervised DINOv2 method
- Trained on public data and a collection of 873k chest X-rays and evaluated on 9 different tasks.



T. Moutakanni P. Bojanowski, G. Chassagon, C. Hudelot, A. Joulin, Y. LeCun, M. Muskley, M. Oquab, M.P. Revel and M. Vakalopoulou. "Advancing patient-centric AI for robust X-ray analysis through holistic self-supervised learning". In: Under Submission, 2024

RayDINO foundation model for X-Rays

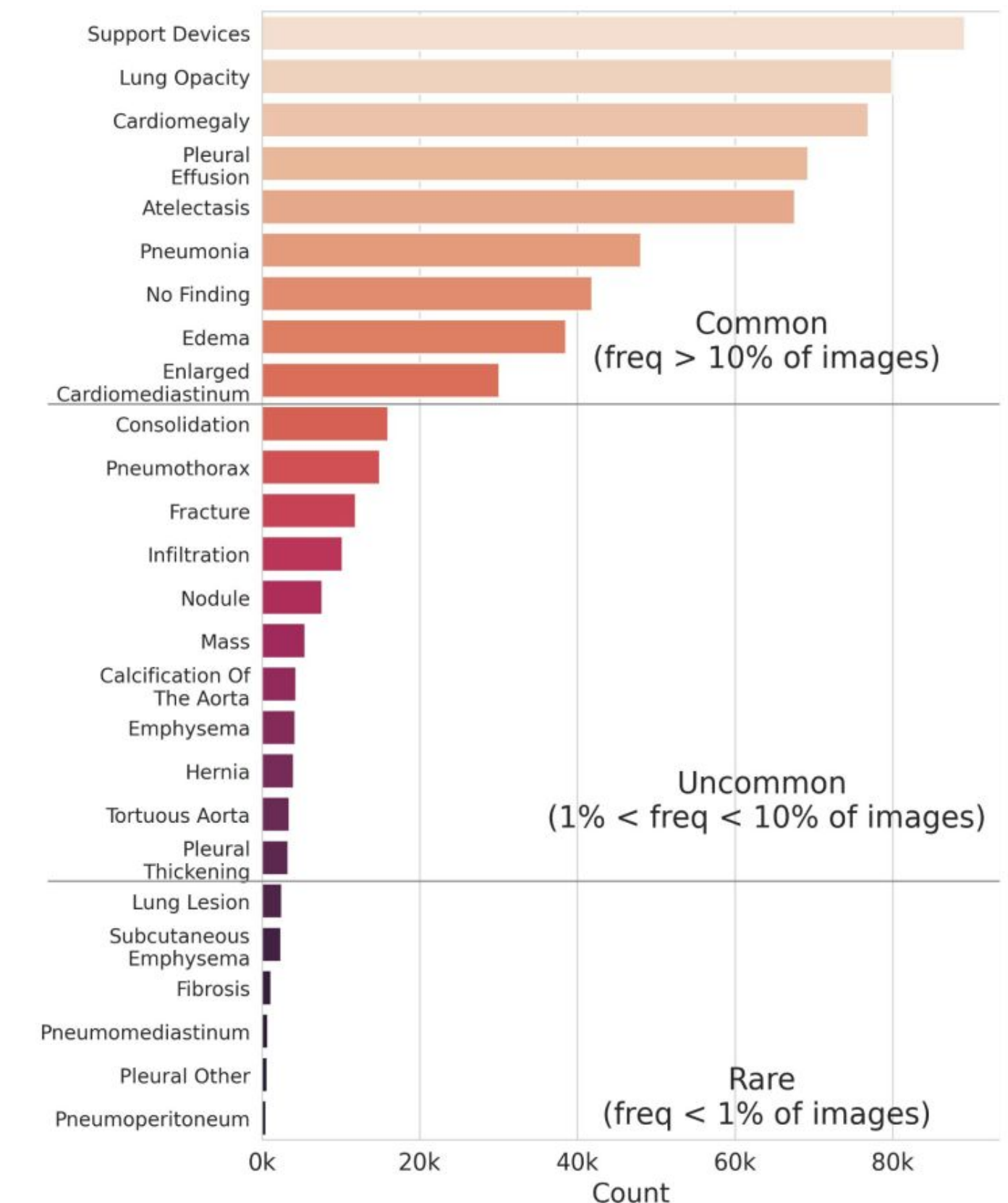
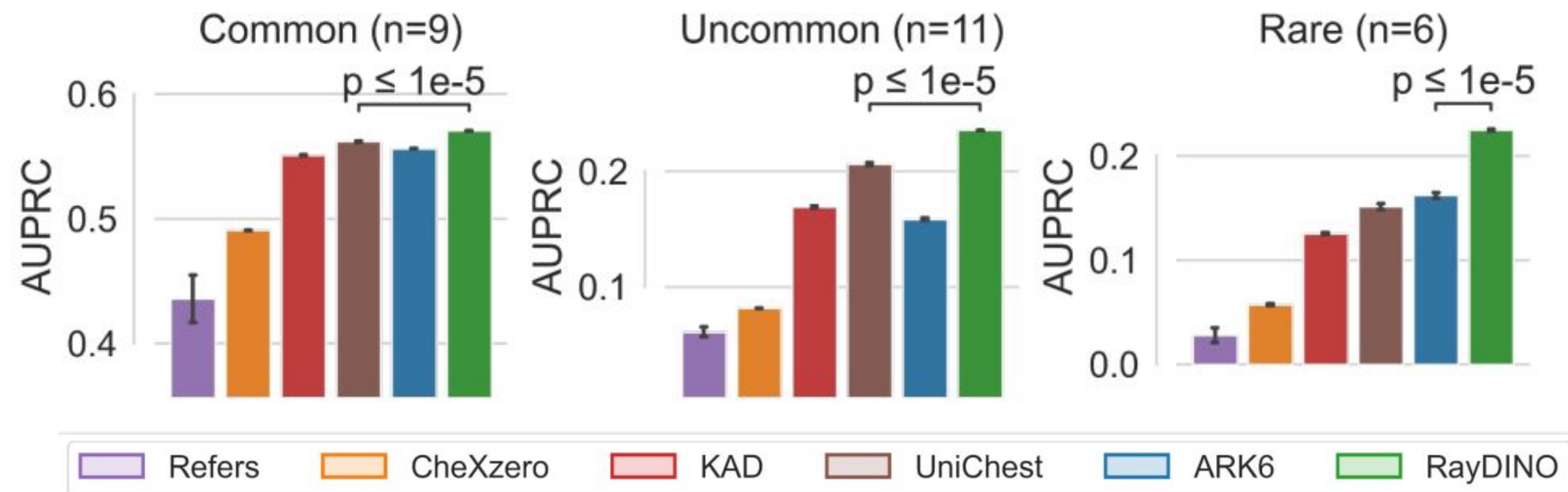
- Semantic segmentation evaluation for organs and diseases
- Report generation evaluation for the automatic report generation



T. Moutakanni P. Bojanowski, G. Chassagon, C. Hudelot, A. Joulin, Y. LeCun, M. Muskley, M. Oquab, M.P. Revel and M. Vakalopoulou. “Advancing patient-centric AI for robust X-ray analysis through holistic self-supervised learning”. In: Under Submission, 2024

RayDINO foundation model for X-Rays

- Long tail classification evaluation with under representative classes

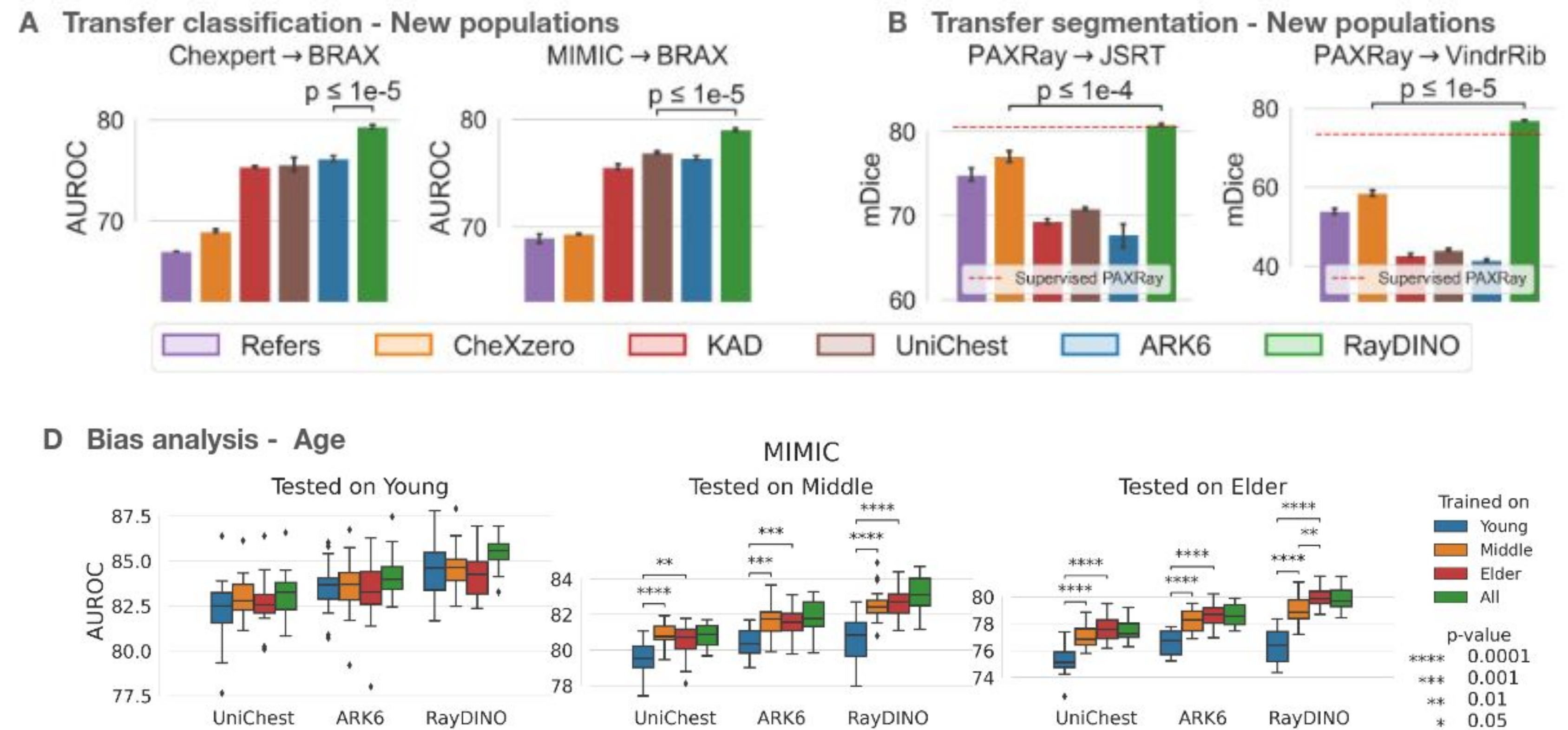


T. Moutakanni P. Bojanowski, G. Chassagon, C. Hudelot, A. Joulin, Y. LeCun, M. Muskley, M. Oquab, M.P. Revel and M. Vakalopoulou. “Advancing patient-centric AI for robust X-ray analysis through holistic self-supervised learning”. In: Under Submission, 2024

RayDINO foundation model for X-Rays

- Transferability, Bias analysis and Interpretability

- A) Transferability from MIMIC (USA) -> BRAX (Brazil)
- B) Transferability from 2D CT-scan projections (USA, China), and testing it on real world datasets JSRT (Japan) and Vindr Rib (Vietnam)
- D) Fairness comparisons by training on specific group and testing on different age groups.



T. Moutakanni P. Bojanowski, G. Chassagon, C. Hudelot, A. Joulin, Y. LeCun, M. Muskley, M. Oquab, M.P. Revel and M. Vakalopoulou. “Advancing patient-centric AI for robust X-ray analysis through holistic self-supervised learning”. In: Under Submission, 2024

Towards Large Deep Learning Models

- The community is working on building huge and powerful models for different tasks with currently impressive results on:
 - Large Language Models and Vision Language Models
 - Llama [Meta]
 - LLAVA [Microsoft]
 - ChatGPT [OpenAI]
 - GEMINI [Google]
 - DeepSeek-VL2 [DeepSeek]



Explain this meme to me.



This image depicts a humorous situation involving a young child and a cake. The child, appearing determined and mischievous, is depicted biting into a large chocolate cake on a table. The text at the top of the image reads: "Mom told me not to touch it."

Analysis and Meaning:

1. ****Disobedience theme****: The text implies that the child has intentionally ignored the instruction or warning given by their mom. This reflects the common tendency of children to follow rules reluctantly and test boundaries when told otherwise.
2. ****Humor in contrast****: The exaggerated act of biting the cake makes the situation absurd, adding to the humor. The over-the-top behavior contrasts with the seemingly simple instruction, creating an unexpected and funny outcome.
3. ****Relatable scenario****: This image humorously captures a widely understood, everyday behavior—children wanting things they are told they cannot have, leading to playful and sometimes rebellious actions.
4. ****Visual cue****: The child's action of taking a bite adds a dynamic and engaging element to the humor, amplifying the sense of mischief and delight in the moment.

In short, this image humorously captures the playful defiance of childhood, using the visual of a child taking a bite of an untouchable cake to amplify the humor and relatability.

Wu, Zhiyu, et al. "DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding." *arXiv preprint arXiv:2412.10302* (2024).

Towards Large Deep Learning Models

Initial Reasoning: Based on the patient information provided, the patient is a white female with a summary of erythromycin ophthalmic ointment being used for treatment. There is no mention of glaucoma in the patient's medical note or summary. Therefore, the patient does not have glaucoma.

Initial Answer: No

Self-Reflection: Reasoning: Based on the patient information provided, the patient is a white female with a summary of erythromycin ophthalmic ointment being used for treatment. There is no mention of glaucoma in the patient's medical note or summary. Therefore, the patient does not have glaucoma.

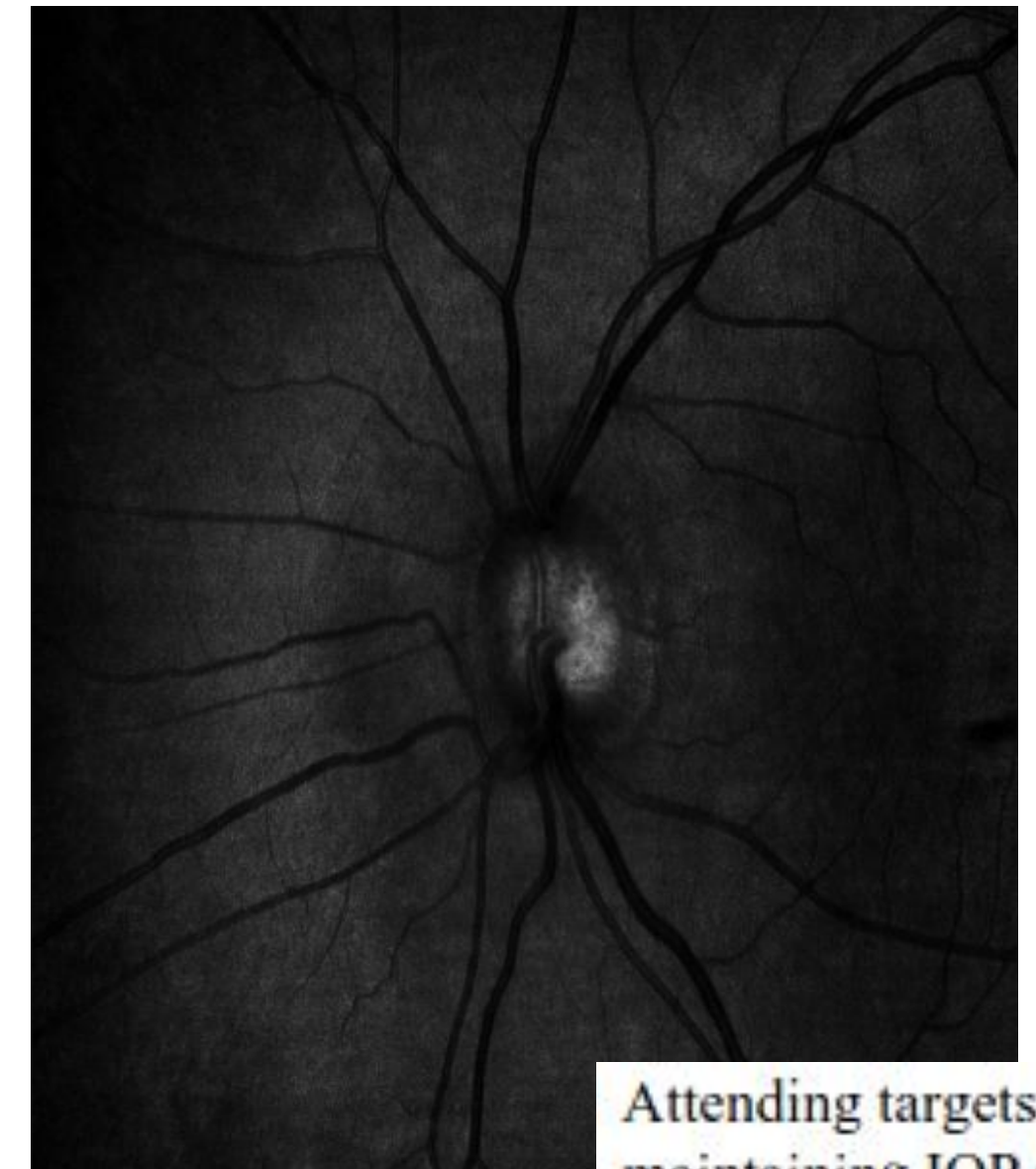
Prediction: No

Flaws: The patient's summary does not mention glaucoma, which is a medical condition that can be diagnosed based on visual content of the image.

Assumptions: The patient's summary does not mention glaucoma, which is a medical condition that can be diagnosed based on visual content of the image.

Final Reasoning: Final Based on the patient information provided, the patient is a white female with a summary of erythromycin ophthalmic ointment being used for treatment. There is no mention of glaucoma in the patient's medical note or summary. Therefore, the patient does not have glaucoma.

Final Answer: Final No

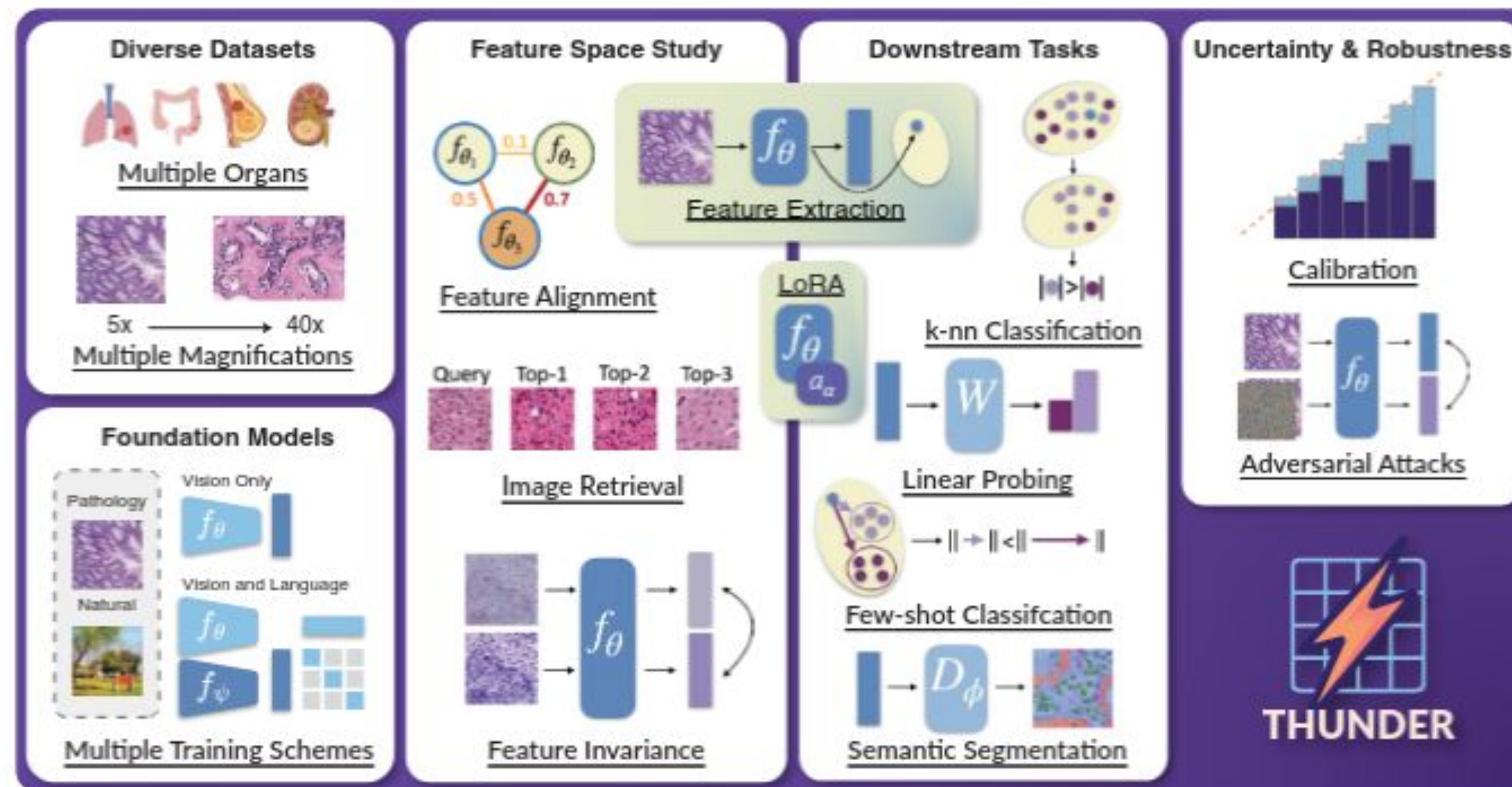


Attending targets maintaining IOP ≤ 18 mmHg in both eyes without glaucoma meds, prescribes artificial tears, schedules follow-up for comprehensive eye evaluation, and confirms the accuracy of the scribe's notes.

Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., ... & Lin, J. (2024). Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*.

How to compare/ evaluate foundation models?

- We proposed a new **open source** benchmark for the comparison and evaluation of foundation models in **digital pathology**. Evaluation on variety of aspects



Pierre Marza, Leo Fillioux, Sofiène Boutaj, Kunal Mahatha, Christian Desrosiers, Pablo Piantanida, Jose Dolz, Stergios Christodoulidis, Maria Vakalopoulou
“THUNDER: Tile-level Histopathology image UNDERstanding benchmark”. In: Under Submission, 2025

How to compare/ evaluate foundation models?

- Variety of models (#23) and datasets (#16) are included already on the benchmark with the addition to be used for any other user defined model.

Name	Short name	Labels	Nb. cls.	Organ(s)	Im. size	Magnif.	Nb. im.
BACH [4]	bach	Classif.	4	Breast	1,536 × 2,048	20×	408
BRACS [6]	bracs	Classif.	7	Breast	Variable	40×	4,539
BreakHis [53]	break-h	Classif.	8	Breast	700 × 460	40×	1,995
Camelyon17 WILDS [35]	wilds	Classif.	2	Breast	96 × 96	10×	302,436
Patch Camelyon [58]	pcam	Classif.	2	Breast	96 × 96	10×	327,680
CRC-100k [34]	crc	Classif.	9	CRC	224 × 224	20×	107,180
MHIST [62]	mhist	Classif.	2	CRC	224 × 224	5×	3,152
TCGA CRC-MSI [33]	tcga-crc	Classif.	2	CRC	512 × 512	20×	51,918
CCRCC [8]	crrcc	Classif.	3	Renal	300 × 300	40×	52,713
ESCA [55]	esca	Classif.	11	Oeso.	256 × 256	10×	367,229
TCGA TILS [38]	tcga-tils	Classif.	2	Multi	100 × 100	20×	304,097
TCGA Uniform [30]	tcga-unif	Classif.	32	Multi	256 × 256	20×	271,170
Ocelot [49]	ocelot	Segm.	2	Multi	256 × 256	40×	10,608
PanNuke [19,20]	pannuke	Segm.	6	Multi	256 × 256	40×	7,901
SegPath Epithelial [36]	segp-ep	Segm.	2	Multi	256 × 256	40×	238,581
SegPath Lymphocytes [37]	segp-ly	Segm.	2	Multi	256 × 256	40×	110,457

Name	Short name	Vision arch.
HIBOU-B [12]	hiboub	ViT-B/14
HIBOU-L [12]	hiboul	ViT-L/14
H-OPTIMUS-0 [15]	hopt0	ViT-G/14
H-OPTIMUS-1	hopt1	ViT-G/14
MIDNIGHT [9]	midnight	ViT-G/14
PHIKON [5]	phikon	ViT-B/16
PHIKON2 [6]	phikon2	ViT-L/16
UNI [1]	uni	ViT-L/16
UNI2-H	uni2h	ViT-H/14
VIRCHOW [16]	virchow	ViT-H/14
VIRCHOW2 [19]	virchow2	ViT-H/14
CONCH [10]	conch	ViT-B/16
CONCH 1.5 [2]	titan	
KEEP [18]	keep	ViT-L/16
MUSK [17]	musk	V-FFN
PLIP [7]	plip	ViT-B/32
QUILTNET [8]	quilt	ViT-B/32
DINOv2-B [13]	dinob	ViT-B/14
DINOv2-L [13]	dinol	ViT-L/14
ViT-B/16 [3]	vitb	ViT-B/16
ViT-L/16 [3]	vitl	ViT-L/16
CLIP-B/32 [14]	clipb	ViT-B/32
CLIP-L/14 [14]	clipl	ViT-L/14

Pierre Marza, Leo Fillioux, Sofiène Boutaj, Kunal Mahatha, Christian Desrosiers, Pablo Piantanida, Jose Dolz, Stergios Christodoulidis, Maria Vakalopoulou
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How to compare/ evaluate foundation models?

- Downstream tasks

Task	Histopathology models												Natural-image models										
	hiboub	hiboul	hopt0	hopt1	midnight	phikon	phikon2	uni	uni2h	virchow	virchow2	conch	titan	keep	musk	plip	quilt	dinob	dinol	vitb	vitl	clipb	clipl
knn	75.8 (10)	75.2 (12)	79.2 (5)	80.5 (3)	78.2 (8)	72.8 (14)	70.1 (15)	78.8 (6)	81.7 (1)	74.2 (13)	81.2 (2)	77.3 (9)	78.6 (7)	79.7 (4)	75.6 (11)	67.8 (19)	68.3 (17)	67.9 (18)	69.5 (16)	64.4 (21)	67.5 (20)	61.9 (23)	64.2 (22)
Lin. prob.	78.0 (14)	81.2 (7)	81.4 (5)	83.3 (2)	82.9 (3)	78.5 (13)	76.5 (15)	81.3 (6)	83.9 (1)	80.2 (11)	82.7 (4)	80.2 (10)	80.8 (9)	81.1 (8)	79.0 (12)	71.0 (21)	71.0 (22)	74.8 (17)	75.3 (16)	71.9 (19)	72.8 (18)	65.8 (23)	71.3 (20)
Few-shot	74.2 (6)	70.4 (12)	73.4 (7)	74.8 (4)	70.6 (11)	72.2 (10)	70.1 (13)	76.4 (2)	78.4 (1)	68.5 (15)	72.5 (9)	73.1 (8)	74.6 (5)	75.8 (3)	70.0 (14)	63.4 (17)	65.7 (16)	61.0 (18)	59.2 (19)	57.8 (21)	56.5 (22)	53.3 (23)	58.2 (20)
Calib.	3.7 (2)	5.5 (21)	4.7 (12)	4.1 (4)	3.2 (1)	6.4 (22)	4.6 (10)	4.3 (7)	4.5 (8)	5.5 (19)	4.6 (11)	4.3 (6)	4.9 (15)	4.7 (13)	4.5 (9)	4.9 (14)	7.0 (23)	5.5 (20)	5.3 (17)	3.9 (3)	5.0 (16)	5.5 (18)	4.2 (5)
Adv. attack	52.8 (13)	40.0 (5)	44.1 (9)	58.0 (17)	36.3 (4)	34.4 (3)	45.6 (11)	42.8 (7)	34.3 (2)	41.0 (6)	33.7 (1)	55.0 (15)	75.3 (23)	44.7 (10)	69.1 (22)	56.9 (16)	53.0 (14)	65.8 (20)	64.5 (19)	46.8 (12)	44.0 (8)	60.4 (18)	67.8 (21)
Rank sum	45 (8)	57 (10)	38 (7)	30 (5)	27 (2)	62 (12)	64 (14)	28 (4)	13 (1)	64 (13)	27 (3)	48 (9)	59 (11)	38 (6)	68 (15)	87 (18)	92 (21)	93 (22)	87 (19)	76 (16)	84 (17)	105 (23)	88 (20)

Table 4: **Classification-related downstream tasks:** Gain in linear probing F1 score from LoRA.

Dataset	uni	uni2h	virchow	virchow2	keep	conch
mhst	+1.8	+7.7	+2.3	+5.8	+6.6	+3.4
bracs	+1.0	-2.2	+3.2	+2.7	+0.9	+2.1

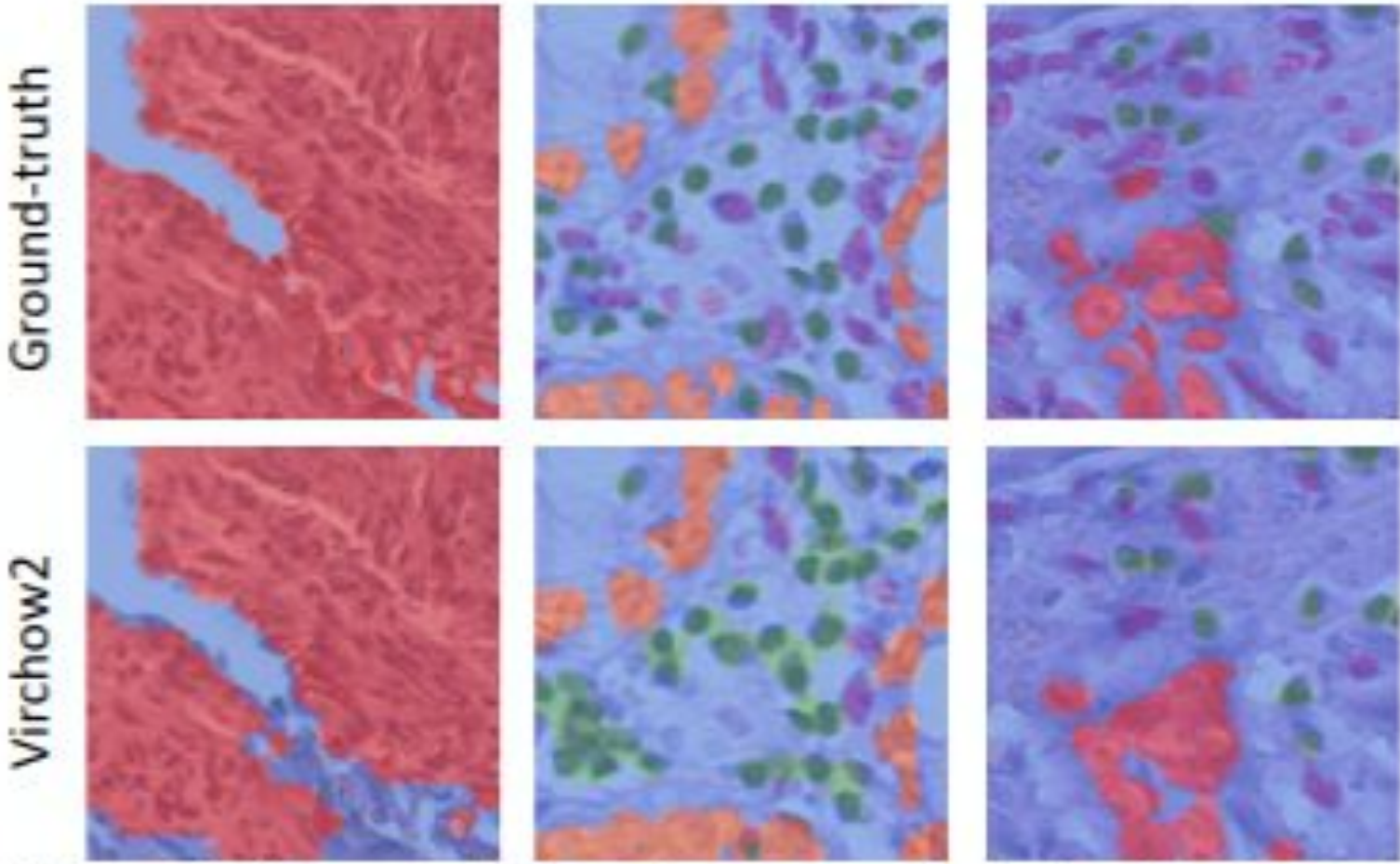


Figure 3: **Segmentation downstream task:** Qualitative semantic maps.

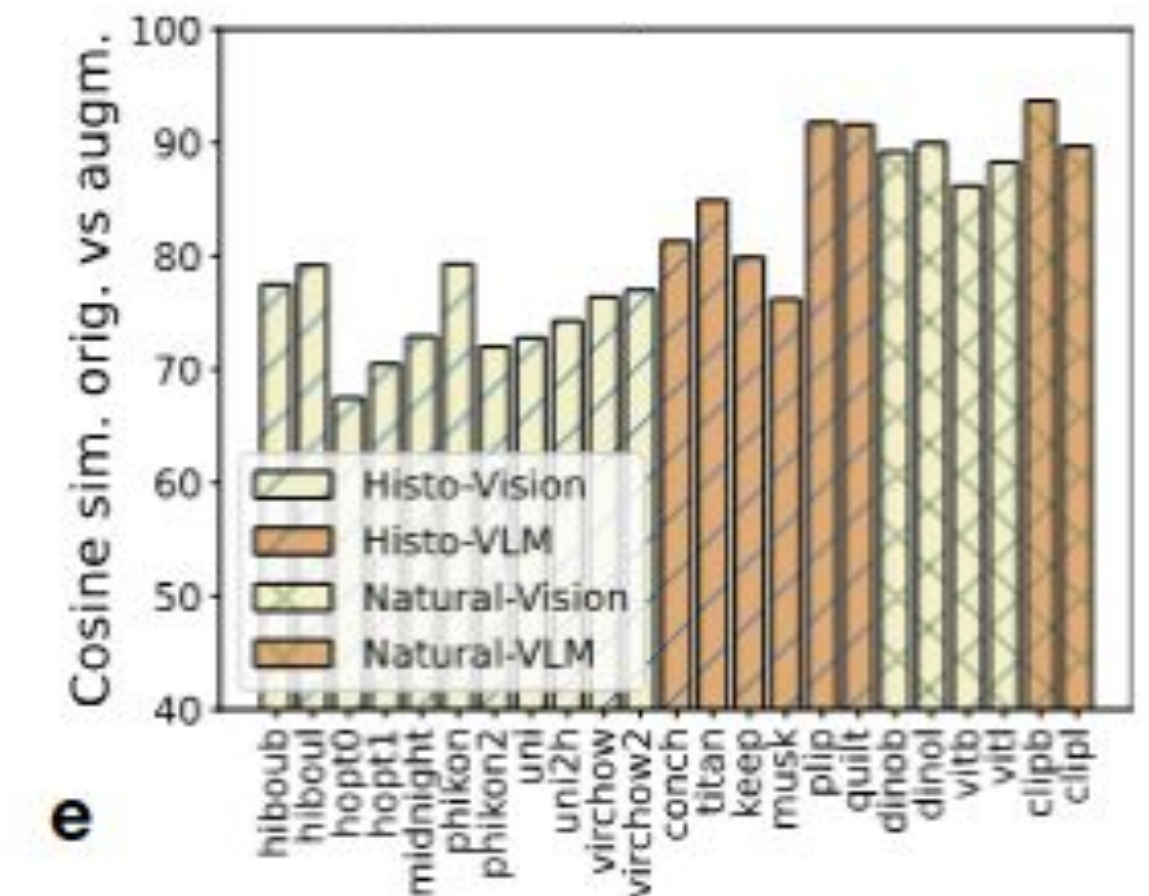
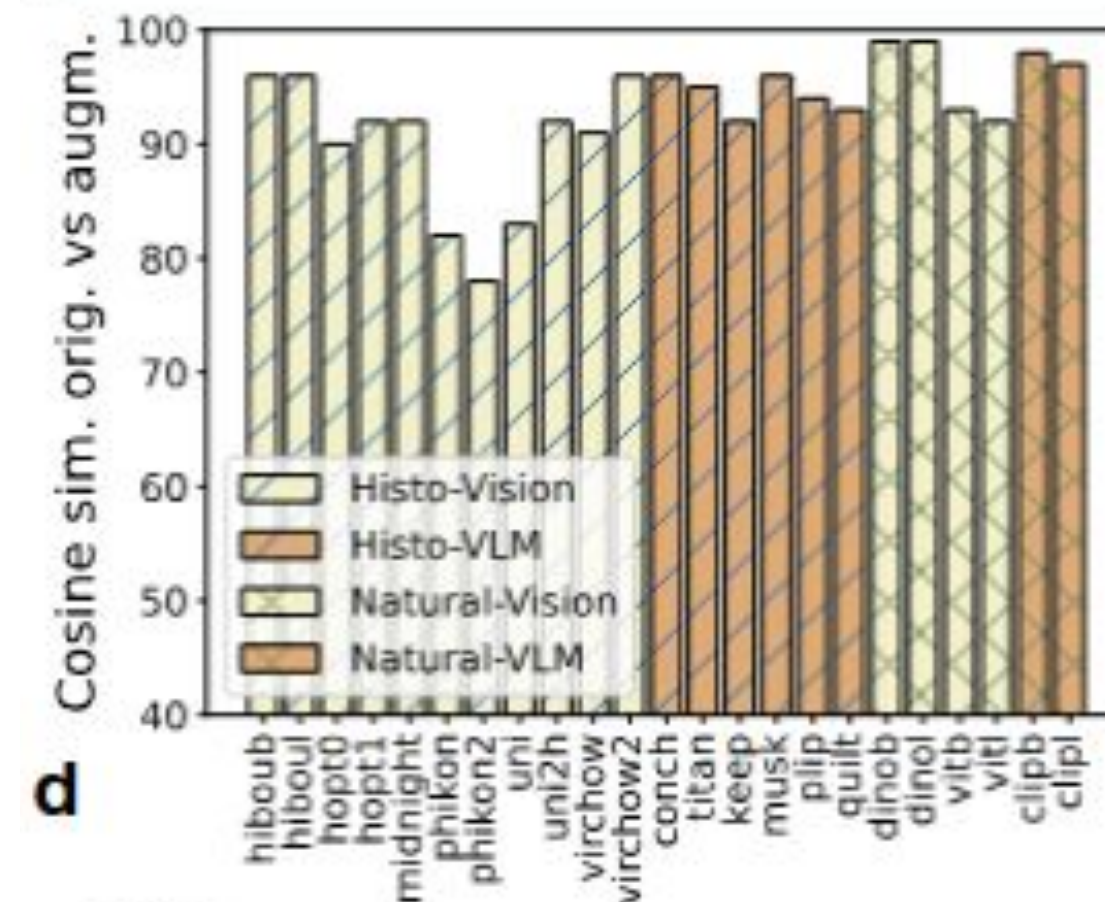
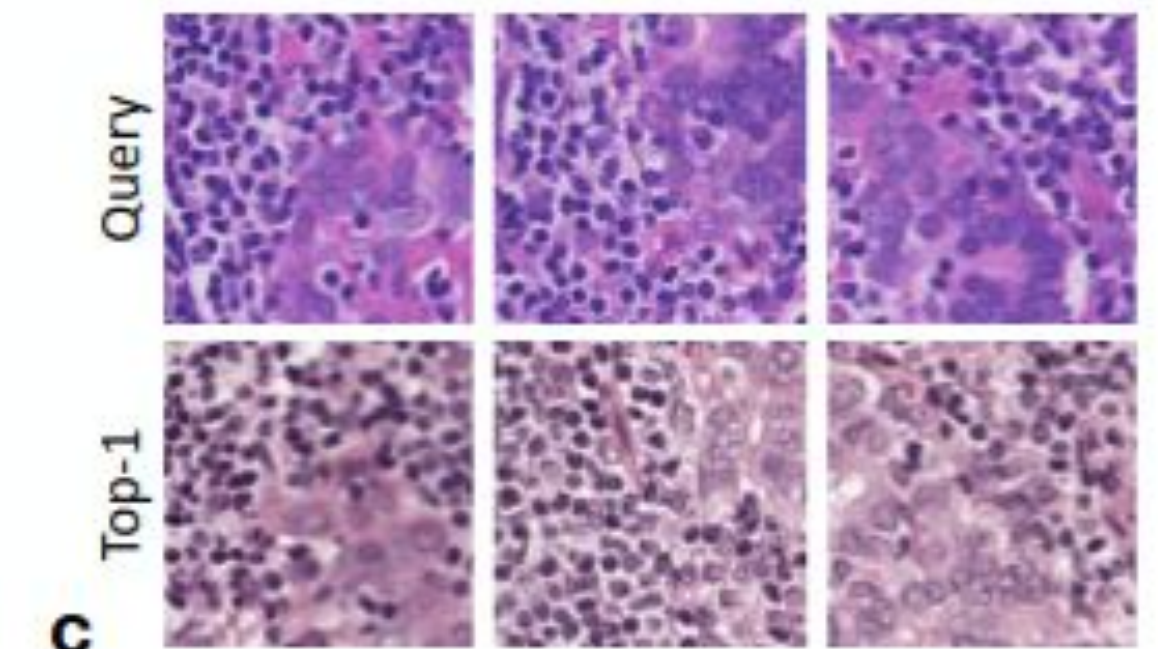
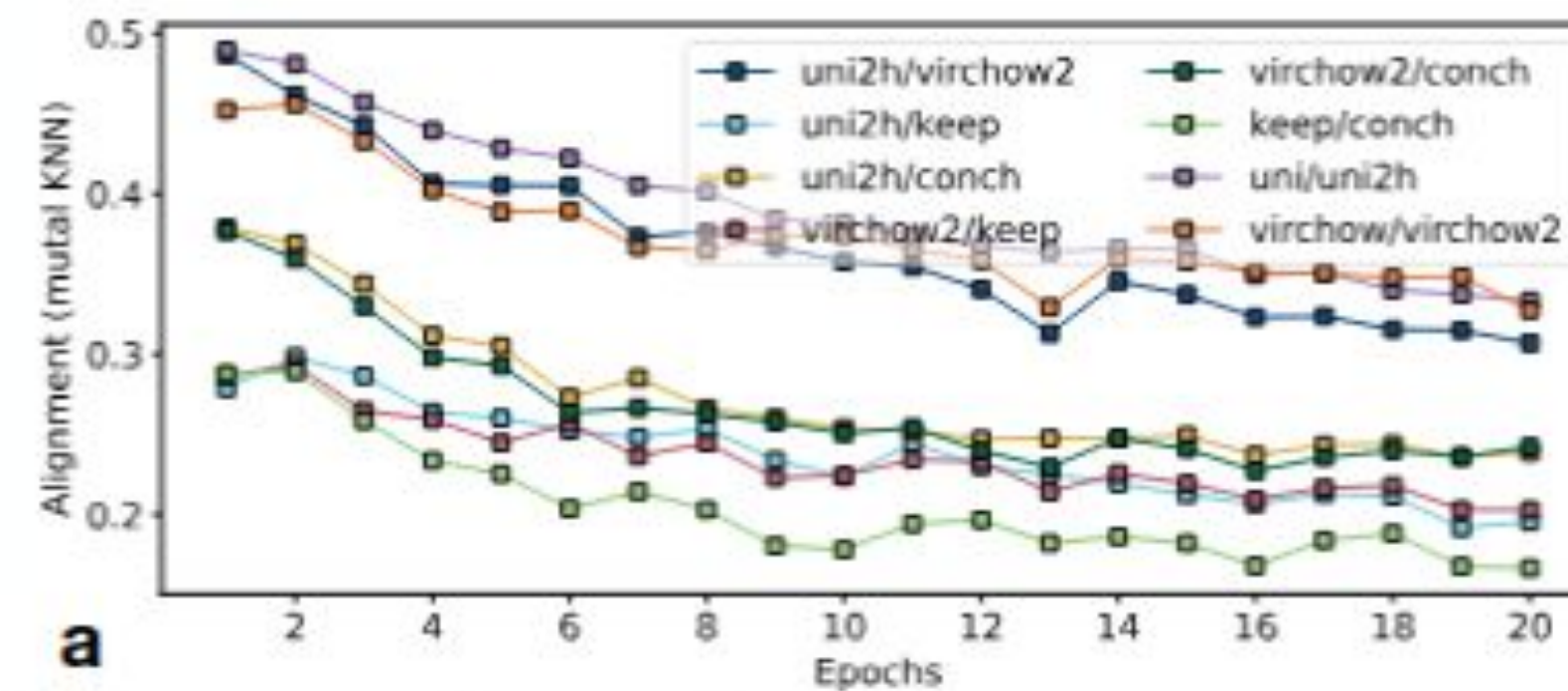
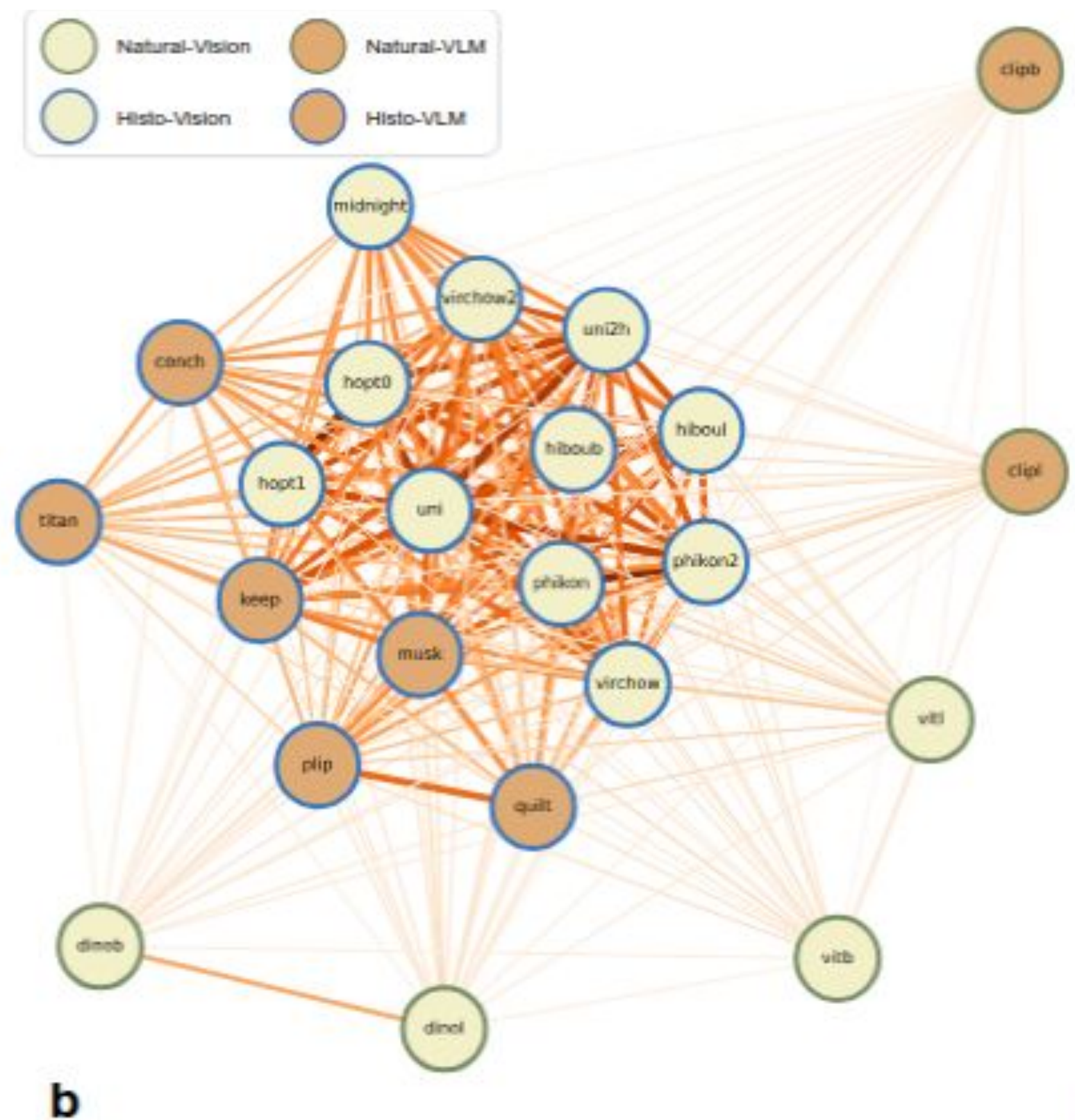
Table 3: **Segmentation downstream task:** Average dice score per dataset.

Model	pannuke	ocelot	segp-ep	segp-ly
conch	31.0	83.1	77.8	60.6
hopt1	51.9	89.5	86.9	64.2
keep	59.8	92.5	86.7	69.2
uni2h	60.5	92.4	88.0	69.1
virchow2	63.0	92.7	88.0	70.7

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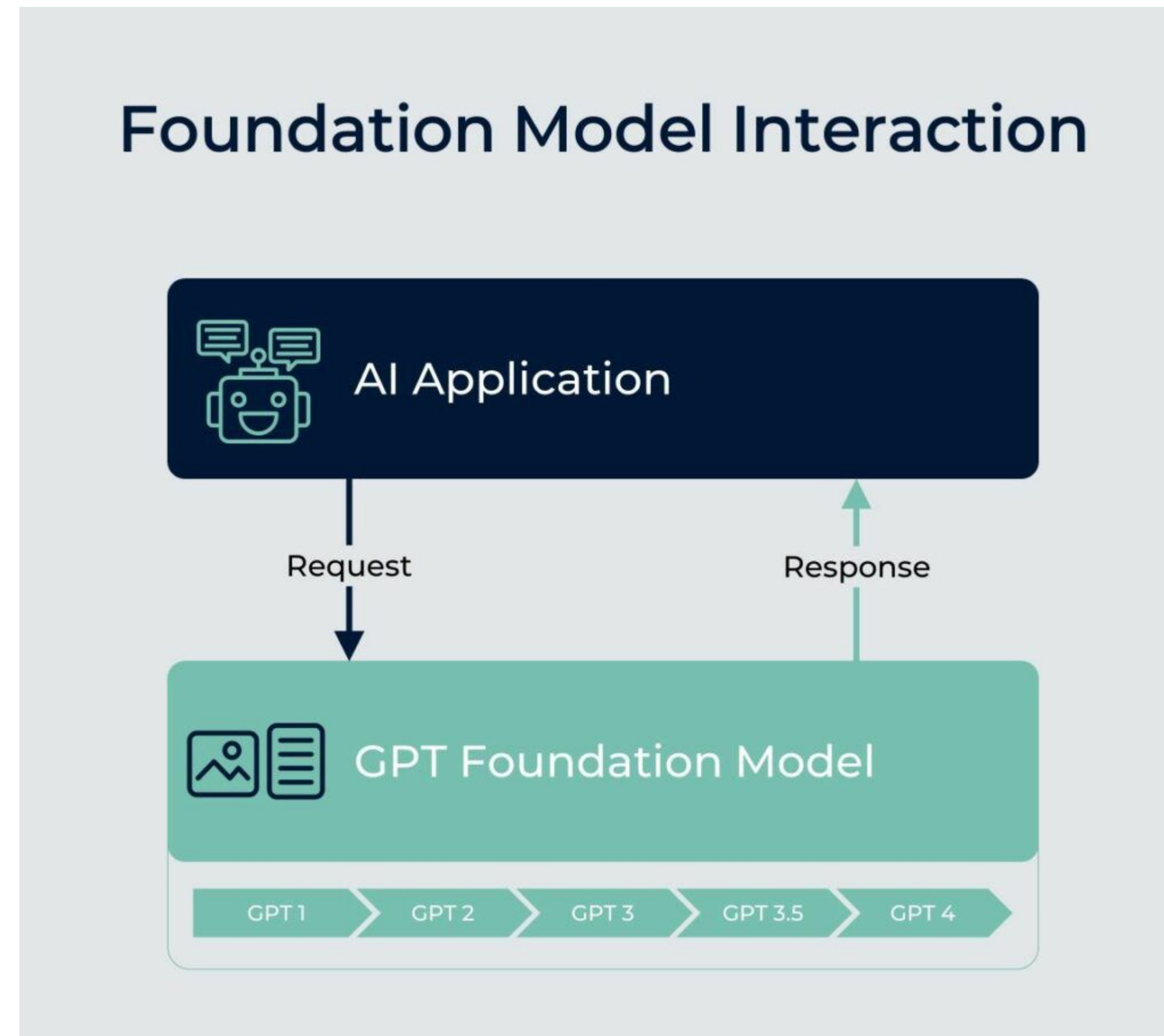
How to compare/ evaluate foundation models?

- Representation Space



Pierre Marza, Leo Fillioux, Sofiène Boutaj, Kunal Mahatha, Christian Desrosiers, Pablo Piantanida, Jose Dolz, Stergios Christodoulidis, Maria Vakalopoulou
“THUNDER: Tile-level Histopathology image UNDERstanding benchmark”. In: Under Submission, 2025

Adaptation of Foundation Models



Case study in Digital Pathology

Many many more foundation models

- **Phikon [1]**: 40 million pan-cancer tiles extracted from TCGA trained with iBOT method.
- **Virchow [2]**: 1.5 million WSIs using DINOv2.
- **Pathoduet [3]**: based on MoCo v3 using cross scale positioning and cross stain transfer between registered pairs of IHC and H&E.
- **PLUTO [4]**: 158K slides with more than 100 stains and more than 12 scanners. Trained with DINOv2 loss by adding a Masked Autoencoder (MAE) objective and a Fourier-loss-based term to modulate the preservation of low- and high-frequency components.
- And many many more!!!! ***How to adapt them?***

[1] Filiot, Alexandre, et al. "Scaling self-supervised learning for histopathology with masked image modeling." medRxiv (2023): 2023-07.

[2] Vorontsov, Eugene, et al. "Virchow: A million-slide digital pathology foundation model." arXiv preprint arXiv:2309.07778 (2023).

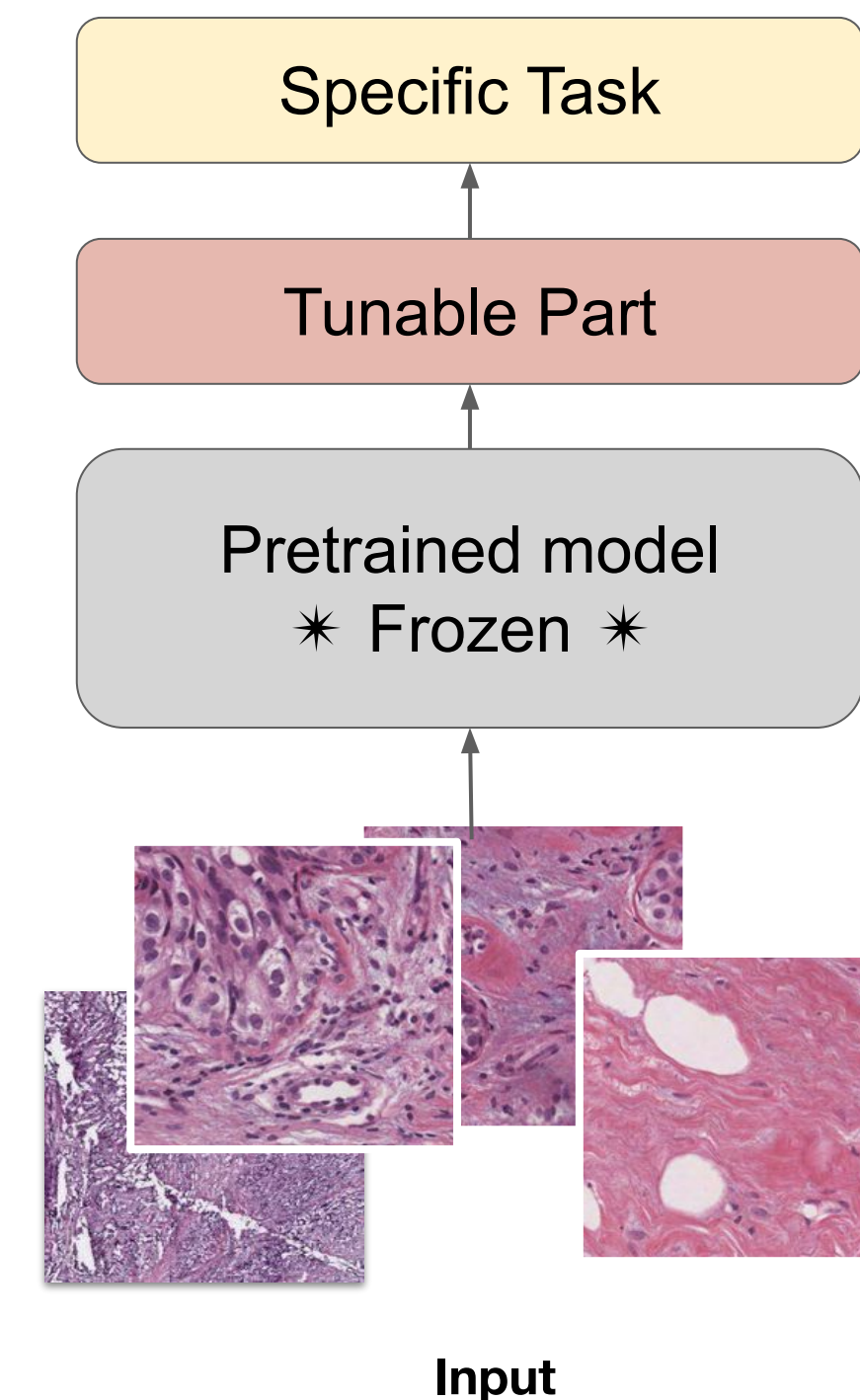
[3] Hua, Shengyi, et al. "Pathoduet: Foundation models for pathological slide analysis of h&e and ihc stains." arXiv preprint arXiv:2312.09894 (2023).

[4] Juyal, Dinkar, et al. "PLUTO: Pathology-Universal Transformer." arXiv preprint arXiv:2405.07905 (2024).

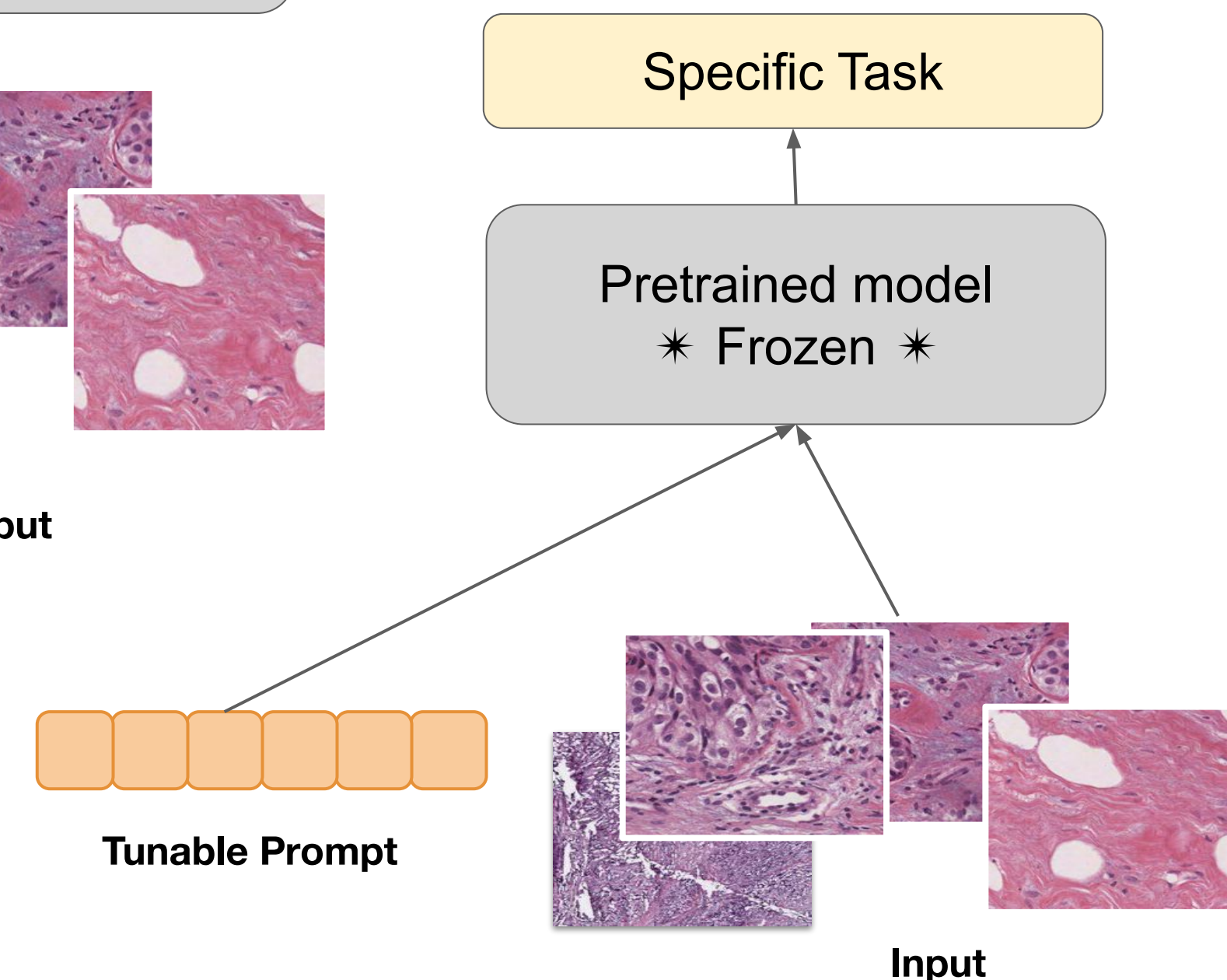
Prompt-based learning

- Instead of traditional approaches that use pre-trained models and fine-tune them on specific tasks, prompts can be used.
- Prompt-based learning or prompting, started from the NLP community one year ago as an instruction by the user for the model to execute or complete.
- Several advantages with respect to traditional methods
 - Data-efficient *[also achieved with few-shot and zero-shot learning]*
 - Parameter-efficient *[also achieved with tuning]*

Classical Fine-Tuning Learning

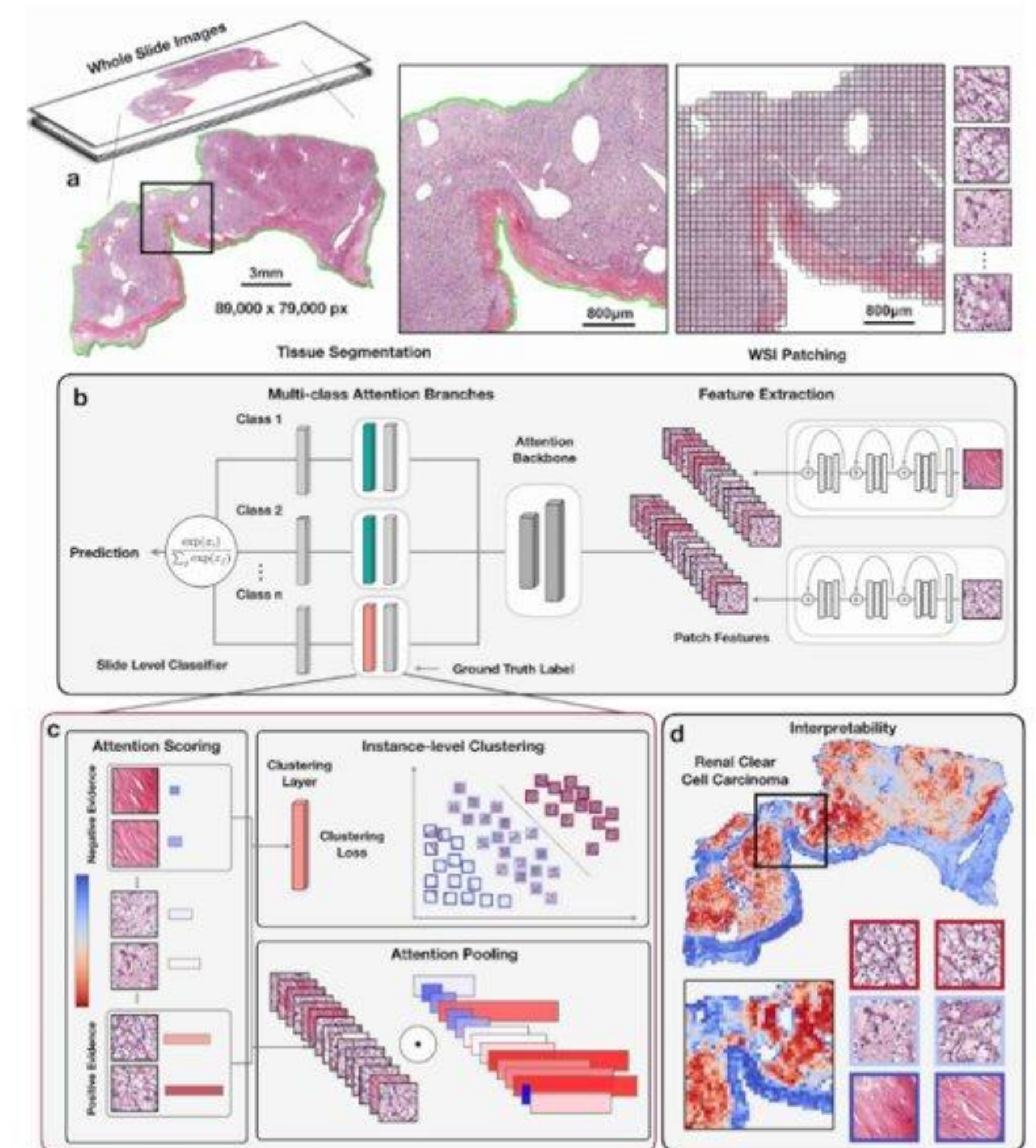


Prompt-based Learning



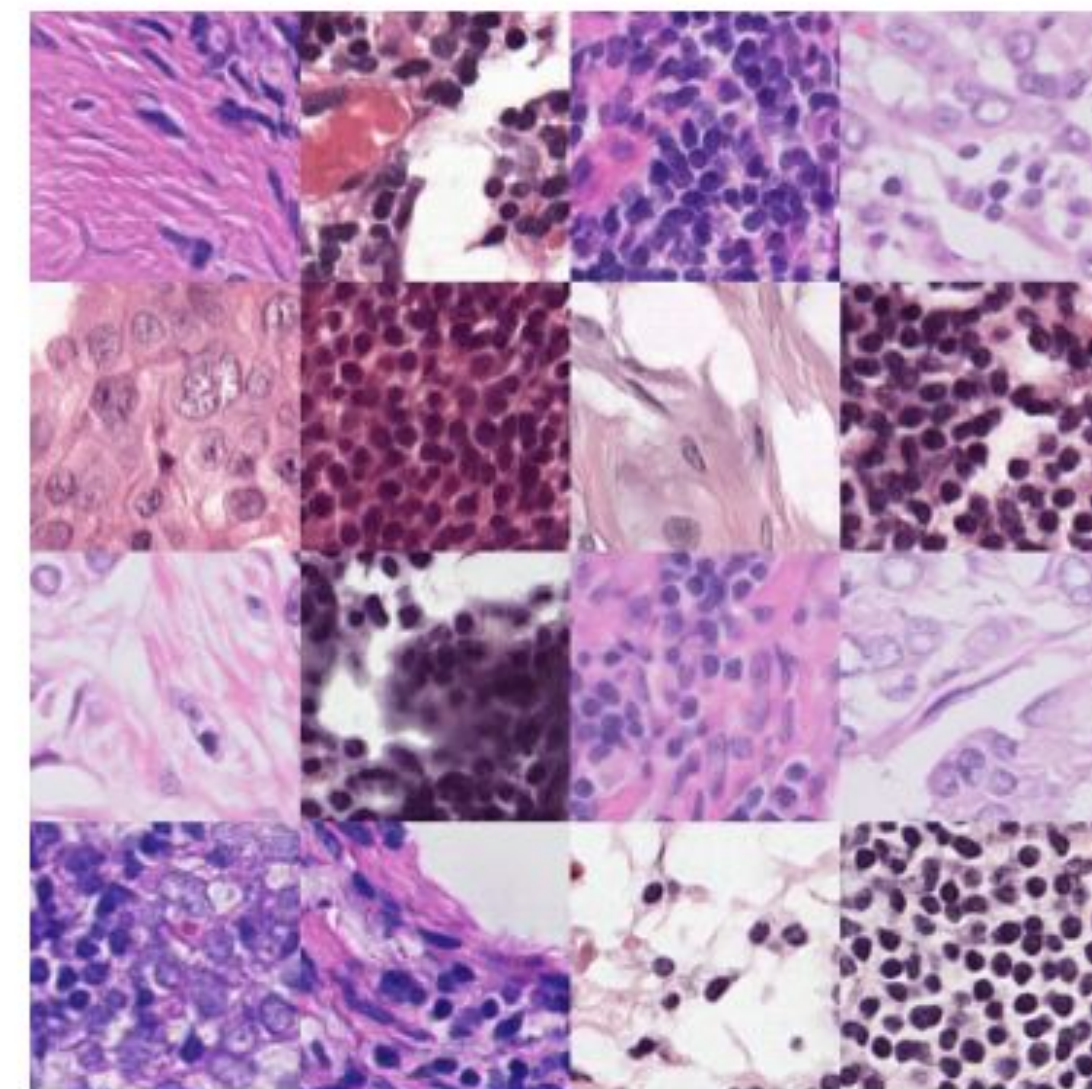
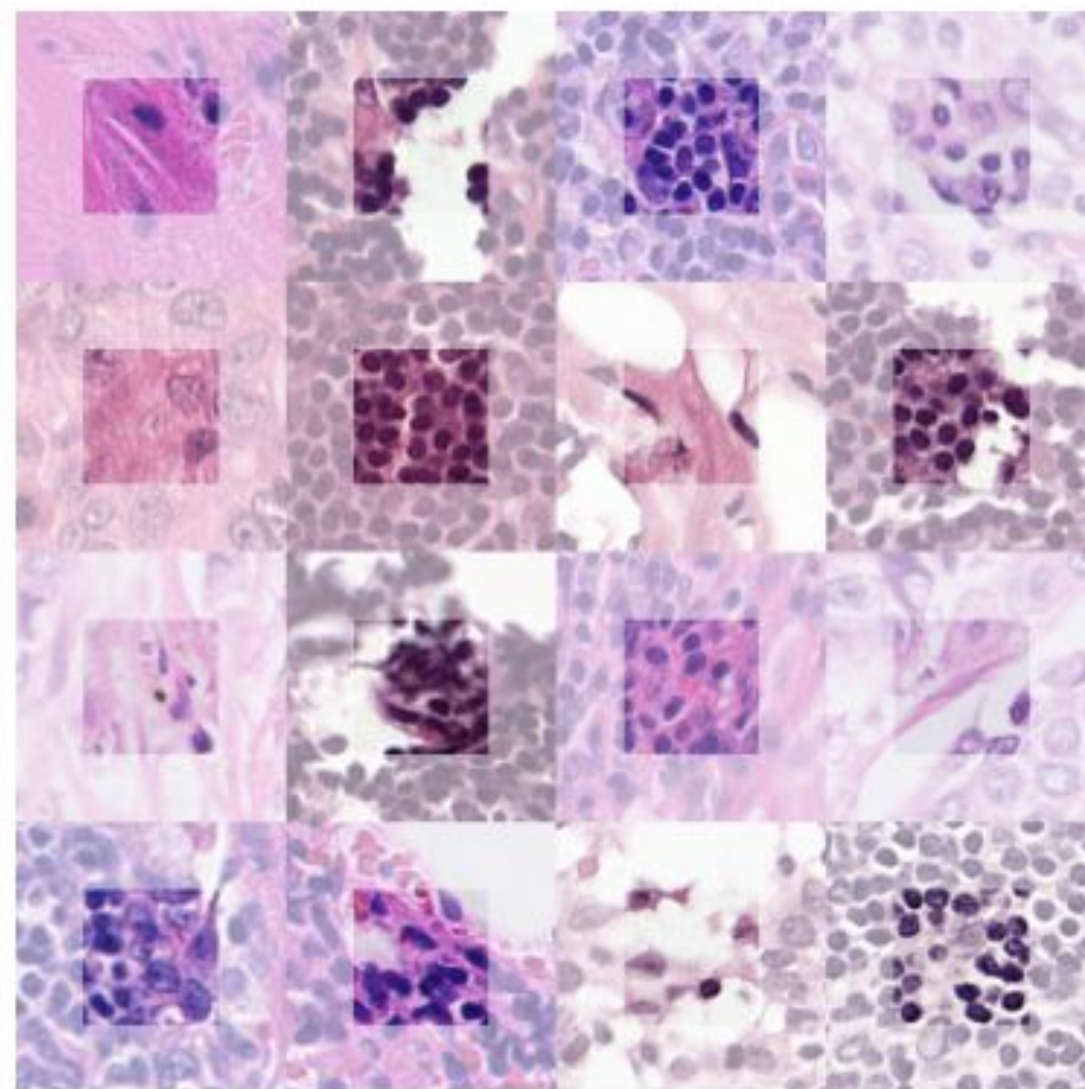
Classical WSI-level MIL methods

- Whole slide image (WSI) classification is a critical task in computational pathology.
- Multi-instance learning (MIL) methods is currently widely used in computationally pathology
- Clustering-constrained attention multiple instance learning (CLAM)
 - Use of a pre-trained ResNet50 network per patch.
- However, the representation of the patch are *task and data agnostic*

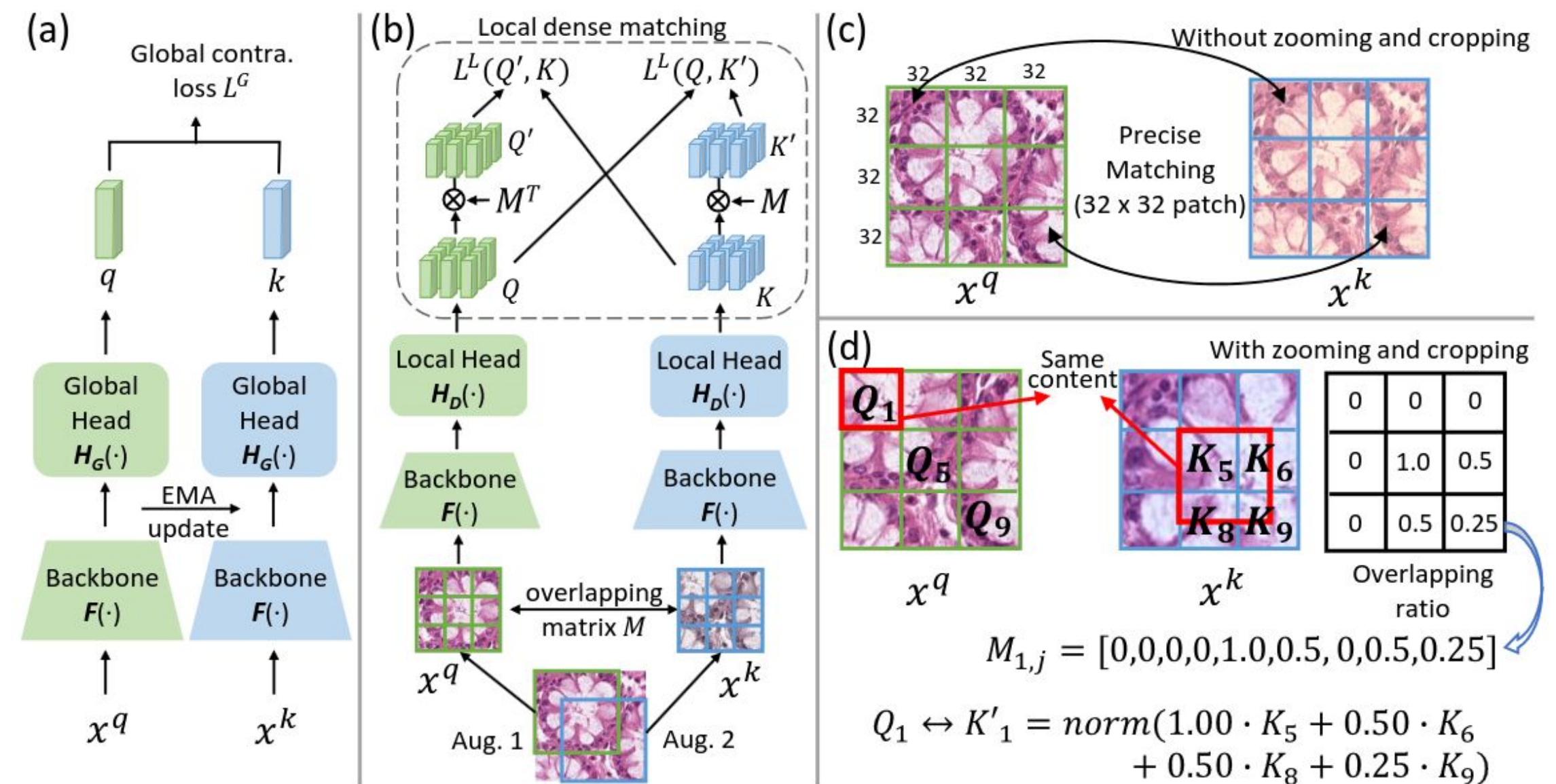


Data-Specific Representations

- Existing MIL methods do not fine-tune their features together with the classification task due to GPU memory limitations.
- Huge large models are not easy to be trained on histopathology data, due to model complexity and limited data - *current works use mainly self-supervised tasks for the pretraining of such models* -



Boyd, J., Liashuha, M., Deutsch, E., Paragios, N., Christodoulidis, S., & Vakalopoulou, M. (2021). Self-supervised representation learning using visual field expansion on digital pathology. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 639-647).



Zhang, J., Kapse, S., Ma, K., Prasanna, P., Vakalopoulou, M., Saltz, J., & Samaras, D. (2023). Precise Location Matching Improves Dense Contrastive Learning in Digital Pathology. IPMI 2023.

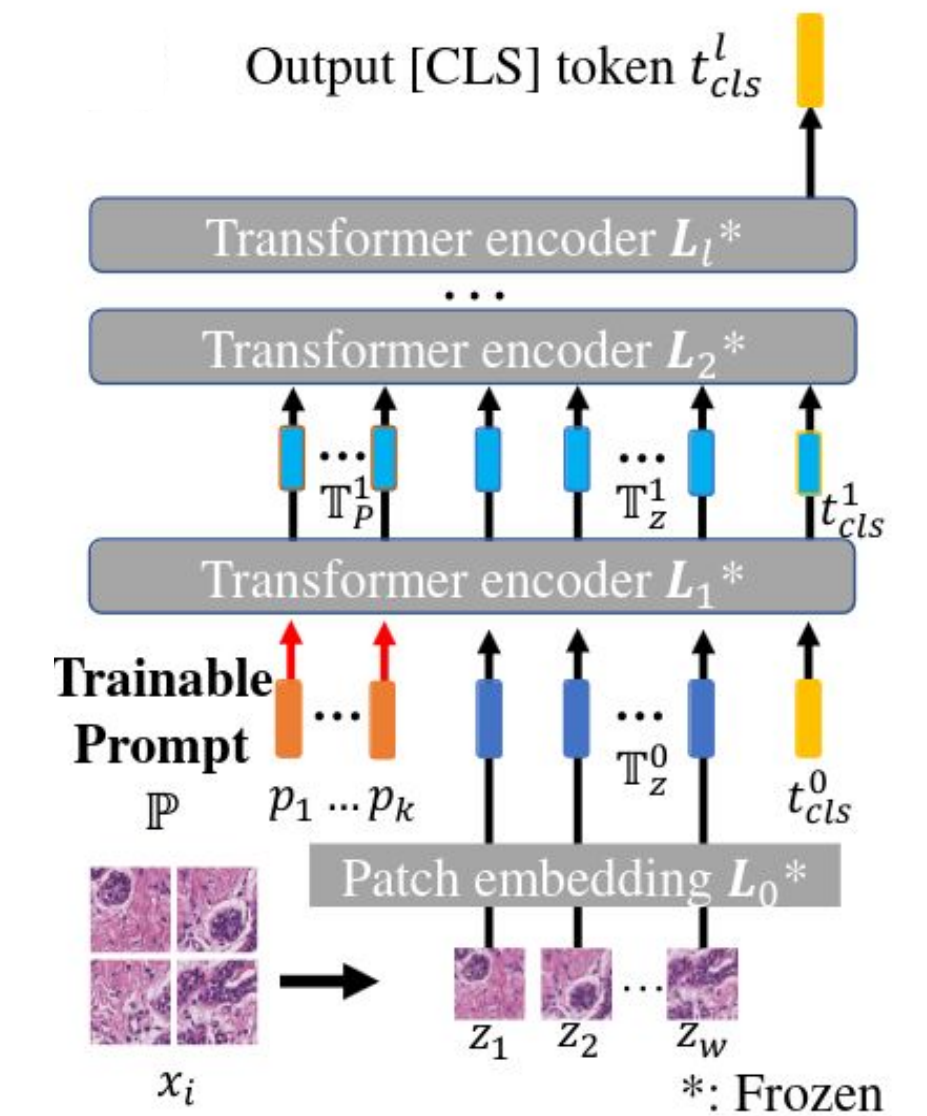
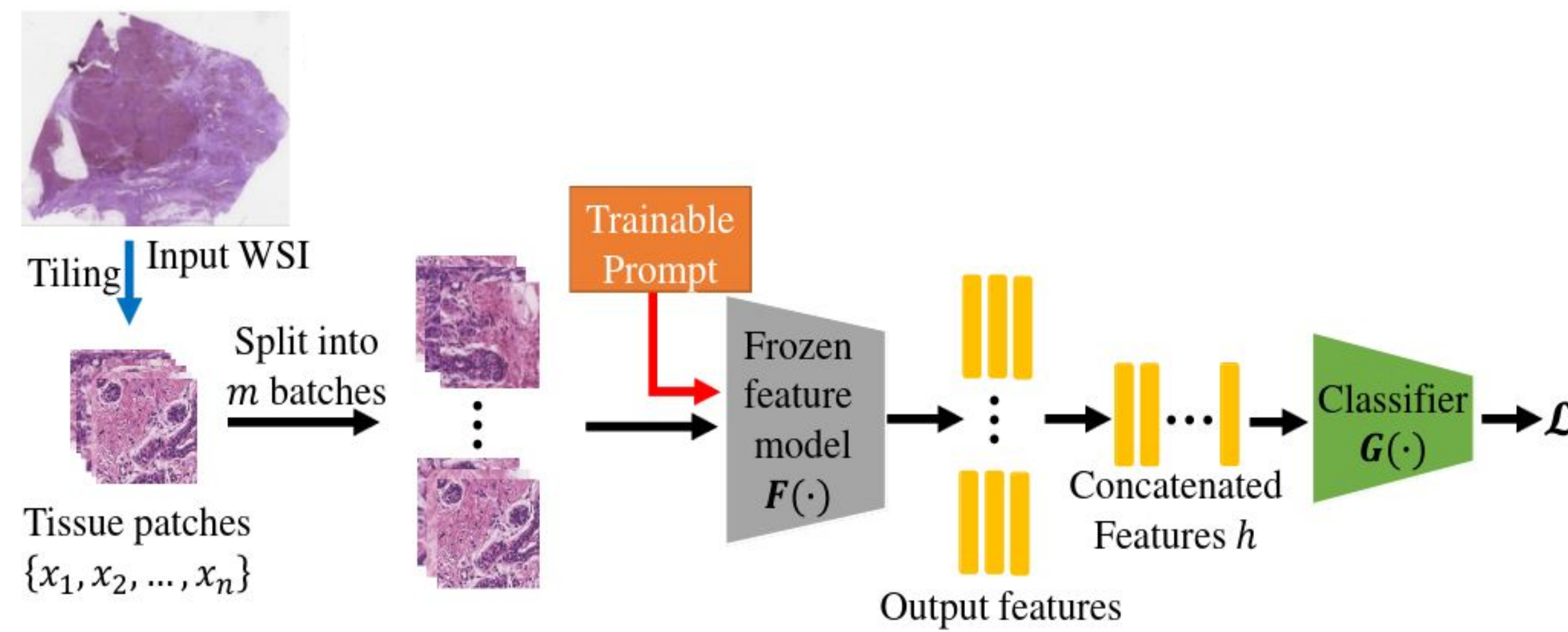
Task-specific Prompt for MIL Schemes

- Existing MIL methods do not fine-tune their features together with the classification task due to GPU memory limitations.
- Huge large models are not easy to be trained on histopathology data, due to model complexity and limited data - *current works use mainly self-supervised tasks for the pretraining of such models* -
- Prompt-MIL: Calibration of the pre-trained on task-specific information using only a small fraction of tunable parameters rather than conventional fine-tuning.
 - Train only the prompt and downstream network without re-training the large backbone
 - Better performances on limited labeled data - ideal for computational pathology

Zhang, J., Kapse, S., Ma, K., Prasanna, P., Saltz, J., Vakalopoulou, M., & Samaras, D. (2023). Prompt-MIL: Boosting Multi-Instance Learning Schemes via Task-specific Prompt Tuning. arXiv preprint arXiv:2303.12214.

Task-specific Prompt for MIL Schemes

- Prompt-MIL framework
 - a frozen feature model $F(\cdot)$
 - a classifier to perform the task ($G(\cdot)$)
 - a trainable prompt $P(\cdot)$
- The number of the parameters is negligible with respect to the total amount of parameters of the architecture (192, less than 0.3% than the total parameters).



Zhang, J., Kapse, S., Ma, K., Prasanna, P., Saltz, J., Vakalopoulou, M., & Samaras, D. (2023). Prompt-MIL: Boosting Multi-Instance Learning Schemes via Task-specific Prompt Tuning. arXiv preprint arXiv:2303.12214.

Experimental Setting - Datasets

- Experiments in 3 public available WSI H&E datasets
 - Subtyping in TCGA-BRCA: 1034 diagnostic digital slides of two breast cancer subtyping: invasive ductal carcinoma (IDC) and invasive lobular carcinoma (ILC)
 - Classification in TCGA-CRC: 430 diagnostic digital slides of colorectal cancer for classification in chromosomal instability (CIN) or genome stable (GS)
 - Classification in BRIGHT: 503 diagnostic slides for breast tissue for classification in non-cancerous, pre-cancerous and cancerous sides
- Splitting of the datasets in training/ validation/ testing using either the official splitting or using the schemes presented in bibliography

Zhang, J., Kapse, S., Ma, K., Prasanna, P., Saltz, J., Vakalopoulou, M., & Samaras, D. (2023). Prompt-MIL: Boosting Multi-Instance Learning Schemes via Task-specific Prompt Tuning. arXiv preprint arXiv:2303.12214.

Experimental Setting - Quantitative

- Our prompt-MIL outperform in all our experiments conventional MIL approaches as well as full fine-tuning using also much less parameters and GPU memory

Dataset Metric	TCGA-BRCA		TCGA-CRC		BRIGHT		Num. of Parameters
	Accuracy	AUROC	Accuracy	AUROC	Accuracy	AUROC	
Conventional MIL	92.10	96.65	73.02	69.24	62.08	80.96	70k
Full fine-tuning	88.14	93.78	74.53	56.63	56.13	75.87	5.6M
Prompt-MIL (ours)	93.47	96.89	75.47	75.45	64.58	81.31	70k+192

- Increase performance when using SSL-data specific representations trained on TCGA Pan-cancer dataset

Dataset Metric	TCGA-BRCA		BRIGHT	
	Accuracy	AUROC	Accuracy	AUROC
ViT-small [27]	91.75	97.03	54.17	76.76
ViT-small w/ Prompt-MIL	92.78	97.53	57.50	78.29

Zhang, J., Kapse, S., Ma, K., Prasanna, P., Saltz, J., Vakalopoulou, M., & Samaras, D. (2023). Prompt-MIL: Boosting Multi-Instance Learning Schemes via Task-specific Prompt Tuning. arXiv preprint arXiv:2303.12214.

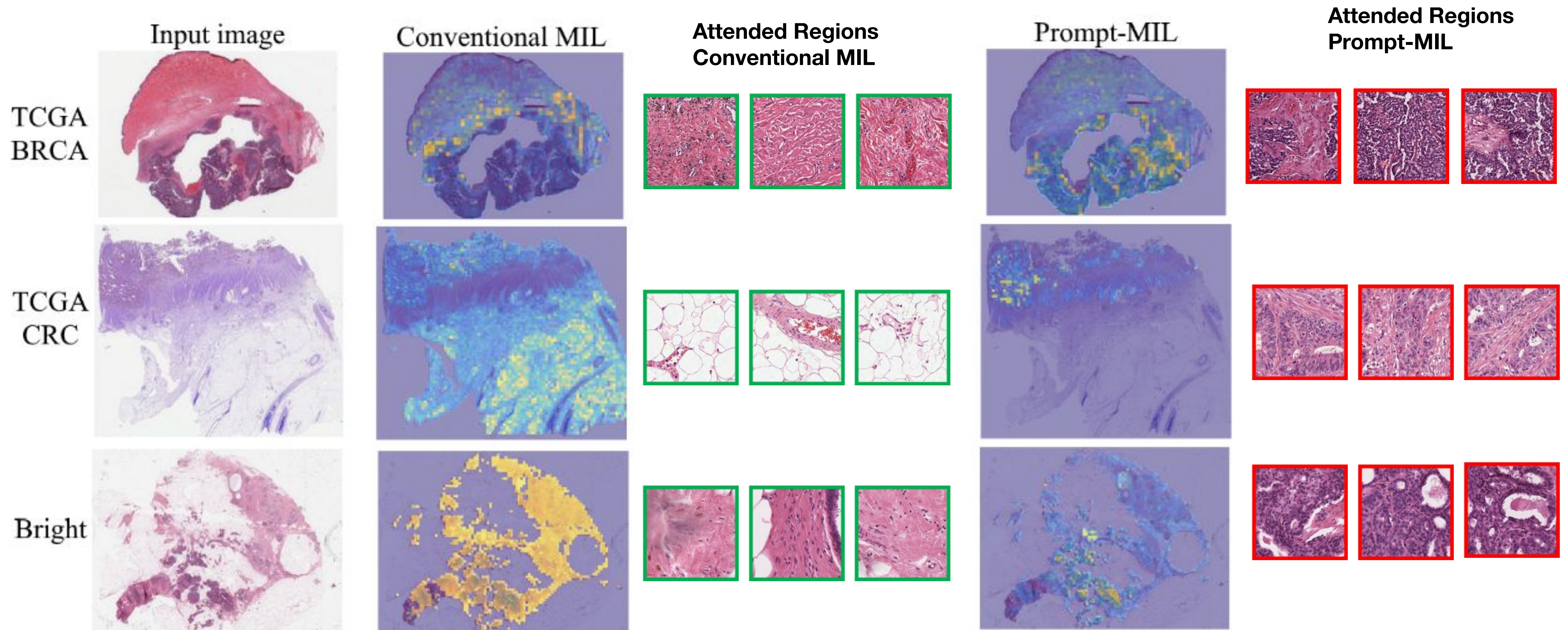
Experimental Setting - Quantitative

- Increase on performance when adapted on different MIL schemes

MIL scheme	Method	TCGA-CRC		BRIGHT		Num. of Parameters
		Accuracy	AUROC	Accuracy	AUROC	
DSMIL	Conventional MIL	73.02	69.24	62.08	80.96	64k
	Fine-tuning L_{12}	69.81	70.72	61.25	80.10	509k
	Prompt-MIL (ours)	75.47	75.45	64.58	81.31	64k+192
ABMIL	Conventional MIL	74.10	68.56	61.25	80.35	25k
	Fine-tuning L_{12}	70.37	70.78	62.50	78.92	470k
	Prompt-MIL (ours)	75.87	70.10	62.50	79.30	25k+192
CLAM	Conventional MIL	75.87	77.50	62.08	82.97	59k
	Fine-tuning L_{12}	74.07	71.40	63.33	81.32	504k
	Prompt-MIL (ours)	76.19	80.84	64.17	84.31	59k+192

Zhang, J., Kapse, S., Ma, K., Prasanna, P., Saltz, J., Vakalopoulou, M., & Samaras, D. (2023). Prompt-MIL: Boosting Multi-Instance Learning Schemes via Task-specific Prompt Tuning. arXiv preprint arXiv:2303.12214.

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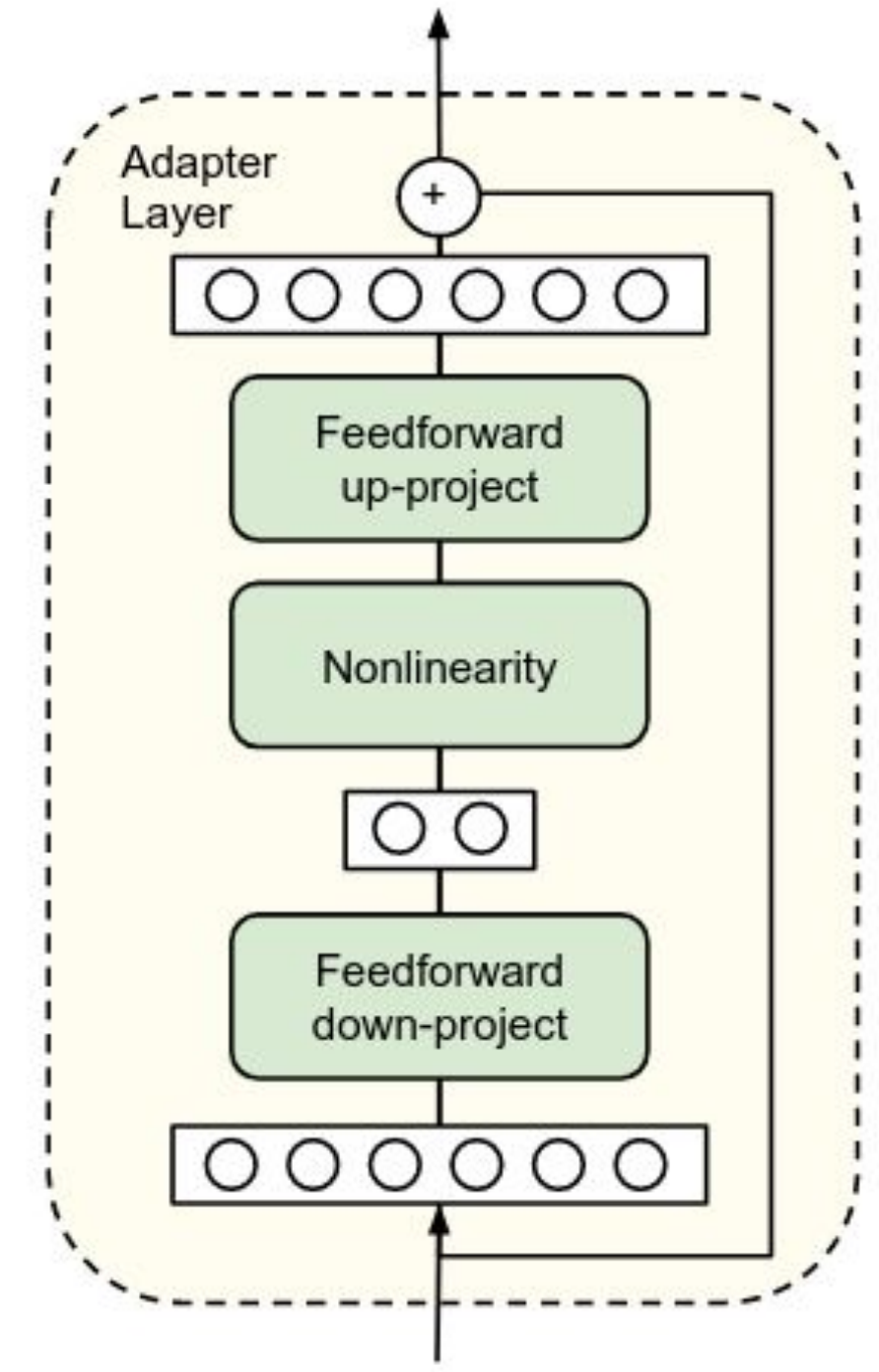
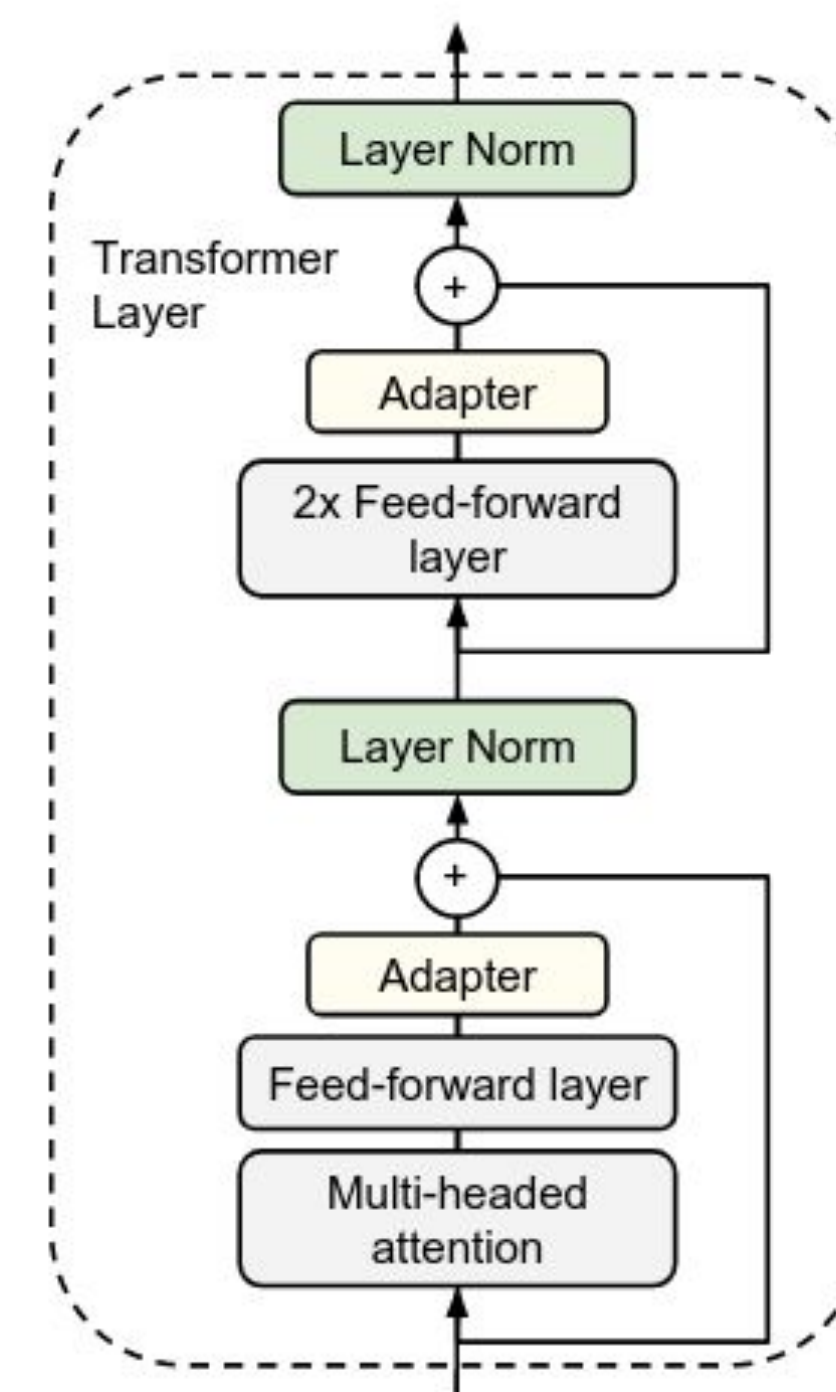
Adapters

- Houlsby et al 2019 introduced something called an **adapter for NLP**. Adapters serve the same purpose as fine-tuning but do it by **stitching in layers** to the main pre-trained model, and updating the weights of these new layers, freezing the weights of the pre-trained model.
- More efficient in terms of time and storage
- Match the performance of state-of-the-art fine-tuning methods.
- Benefits
 - Task-specific layer-wise representation learning
 - Small, scalable, shareable
 - Non-interfering composition of information



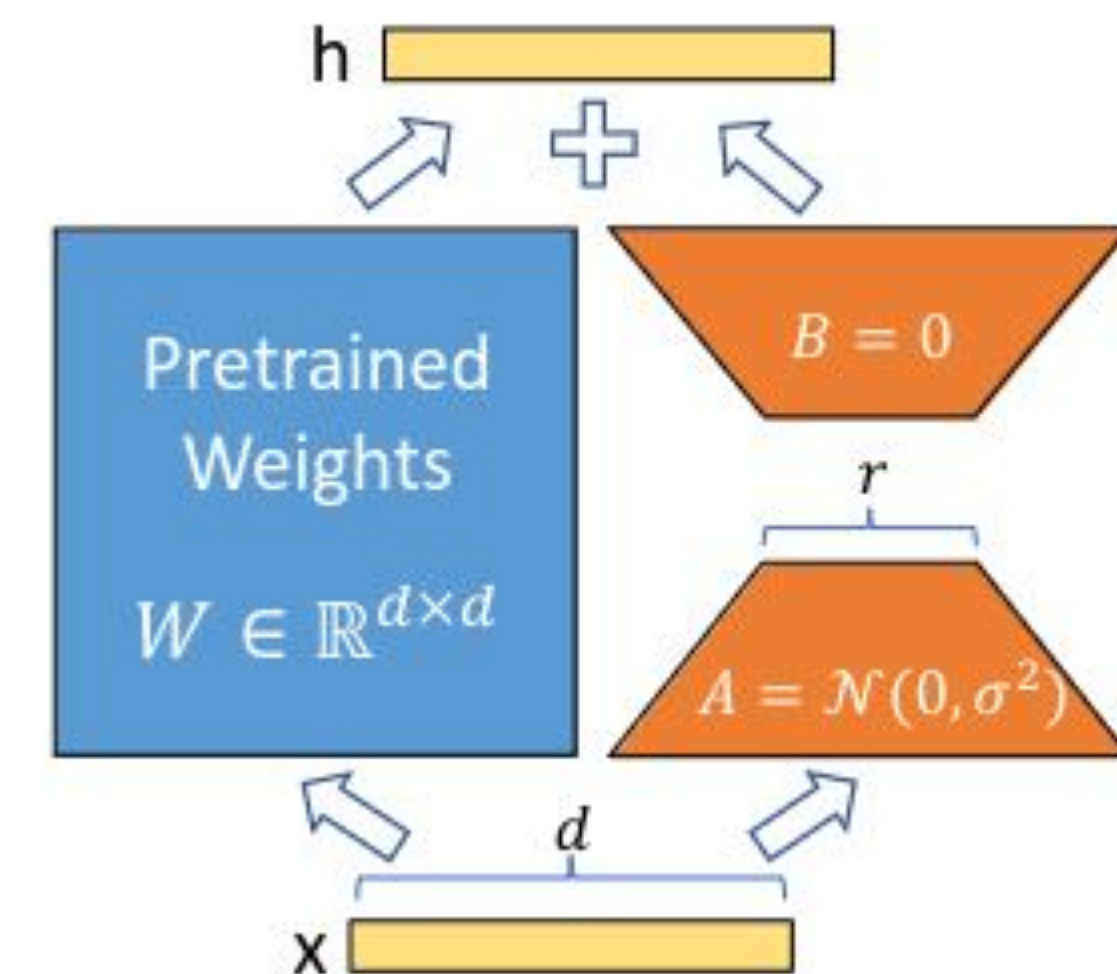
Bottleneck Adapter

- Houlsby et al 2019 integrated with the Transformer.
- Added twice to each Transformer layer. Only green layers are trained.
- The adapter consists of a bottleneck which contains few parameters relative to the attention and feedforward layers in the original model.
- There is a skip connection.
- On the GLUE dataset, bottleneck adapter.
 - Maintain within 0.4% of the performance of full fine-tuning
 - Adding only 3.6% parameters per task, compared to fine tuning 100% of the parameters per task.



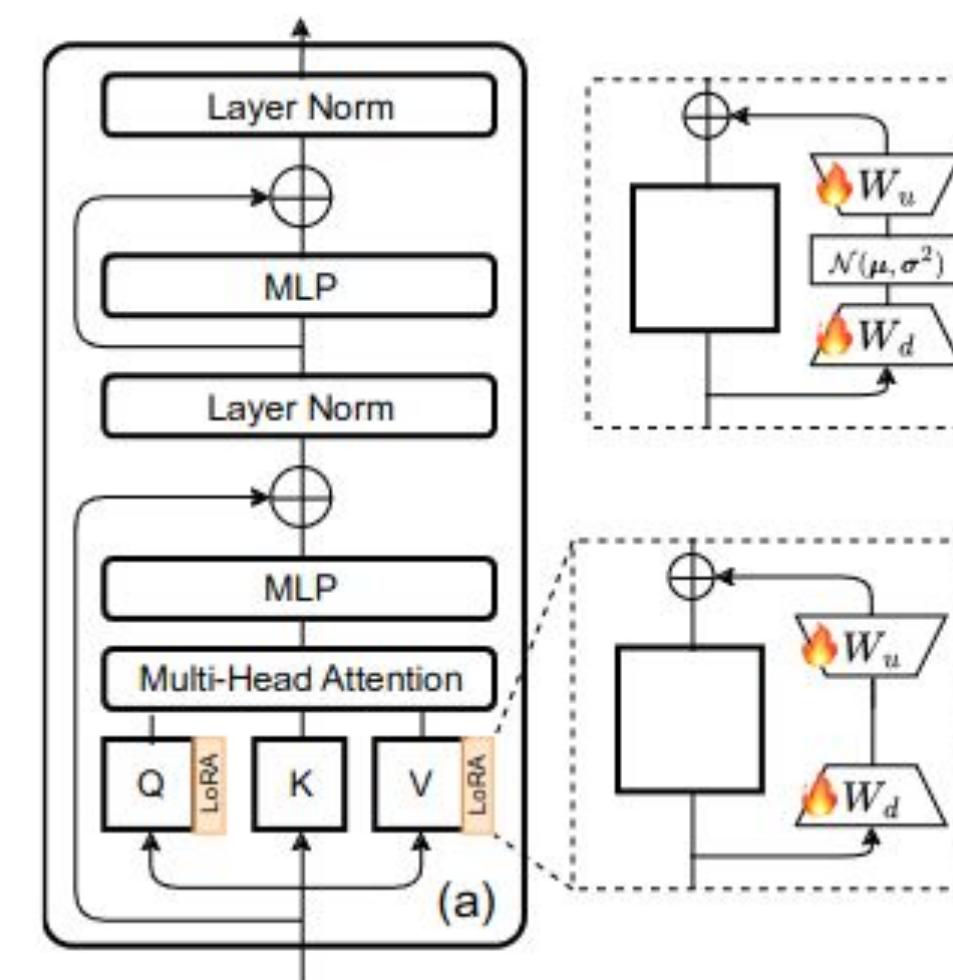
LoRA Adapter

- Hu et al 2021 proposed the low-rank adaptation for NLP.
- A pre-trained model can be shared and used to build many small LoRA modules for different tasks.
- LoRA makes training more efficient and lowers the hardware barrier to entry by up to 3 times when using adaptive optimizers. Instead, only the injected much smaller low-rank matrices are optimized.
- Orthogonal to many prior methods and can be combined with many of them, as prefix-training



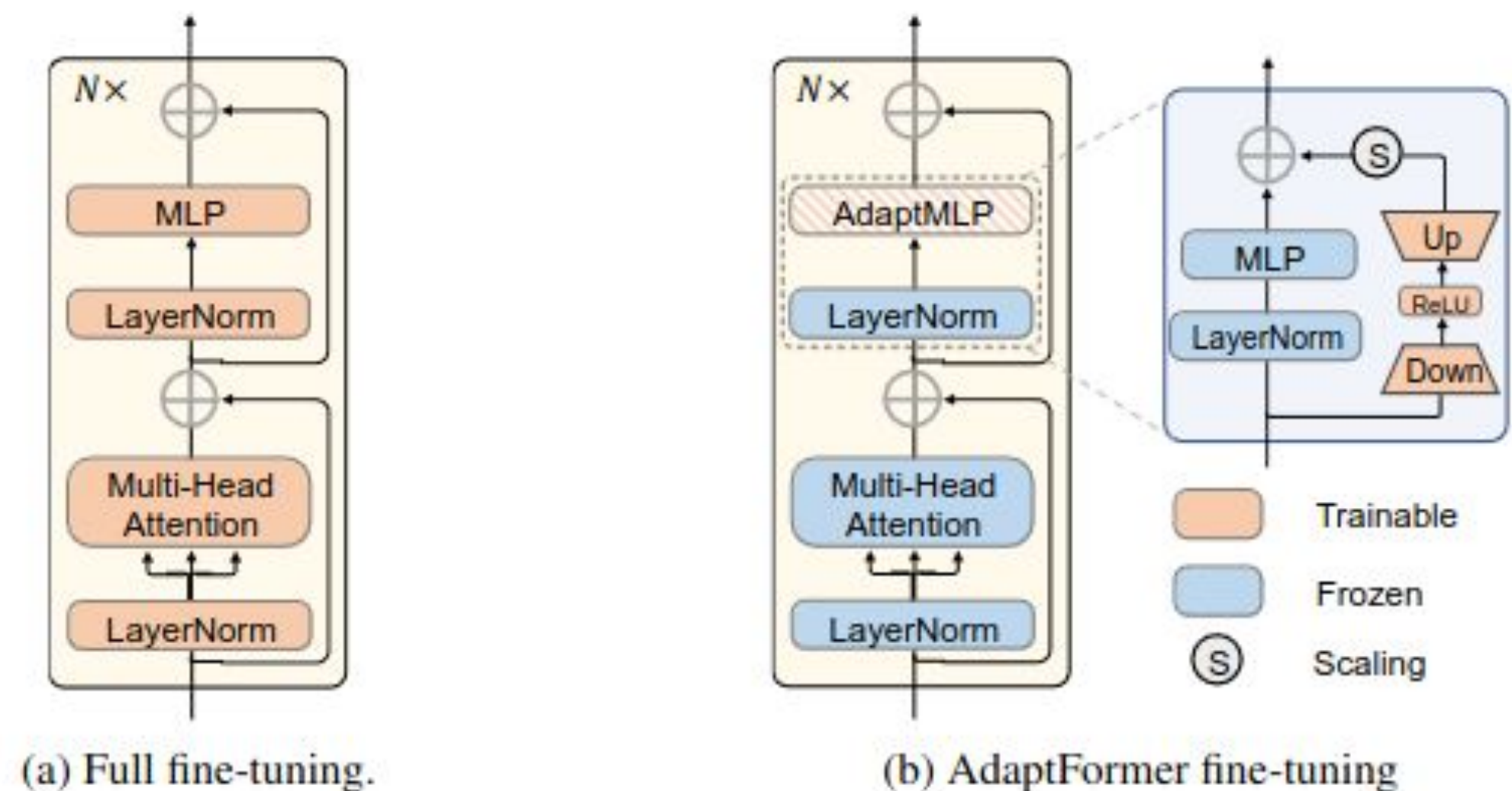
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- In the Transformer architecture, there are four weight matrices in the self-supervised module (W_q, W_k, W_v, W_o) and two in the MLP module
- There is an ablation to decide where to apply. The best performance when applied to W_q, W_v



AdaptFormer Adapter

- Chen et al 2022 proposed AdaptFormer, which can adapt the pre-trained ViTs into many different image and video tasks efficiently.
- Very light add less than 2% extra parameters to a ViT, while it increases the ViT's transferability without updating its original pre-trained parameters, significantly outperforming the existing fully fine tuned models.
- Plug-and-play in different Transformers and scalable to many visual tasks.

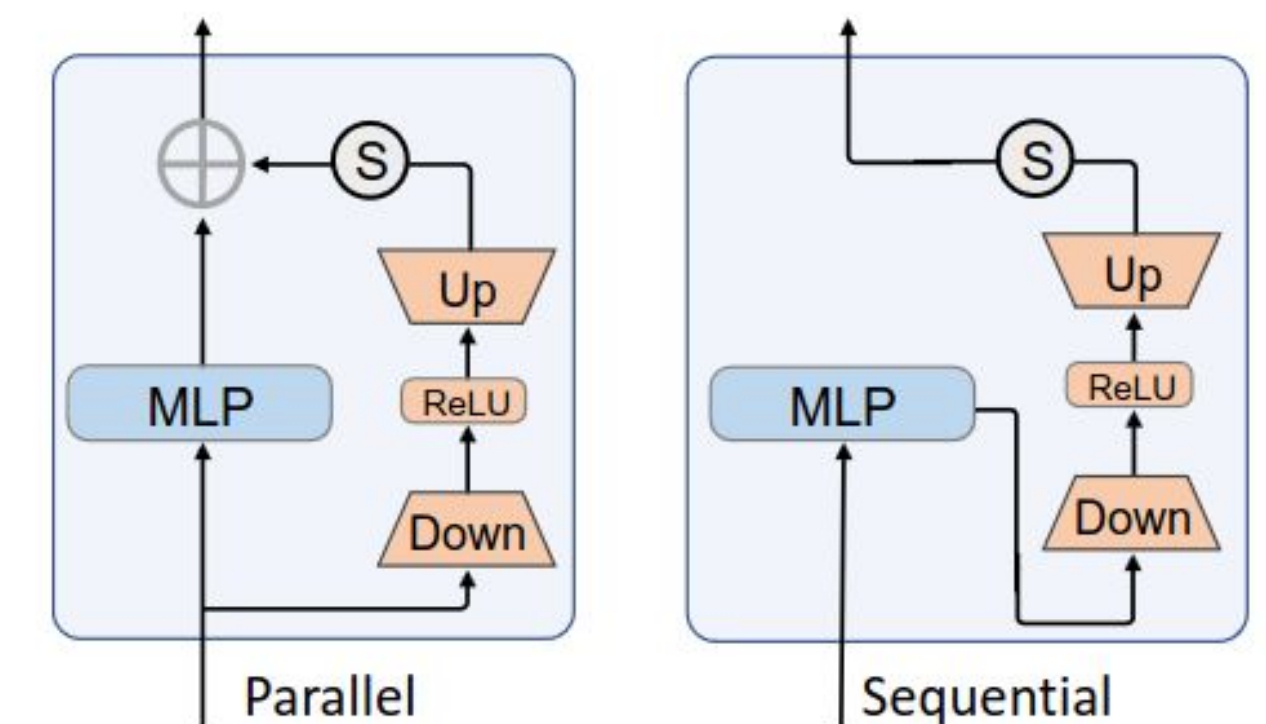


AdaptFormer Adapter

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Method	Avg. Params (M)	Image			Video	
		CIFAR-100	SVHN	Food-101	SSv2	HMDB51
Full-tuning	86.04 (100%)	85.90	97.67 [†]	90.09 [†]	53.97	46.41
Linear	0.07 (0.08%)	69.83 (-16.07)	66.91 (-30.76)	69.74 (-20.35)	29.23 (-24.74)	49.84 (+3.43)
VPT [51]	0.08 (0.09%)	82.44 (-3.46)	94.02 (-3.65)	82.98 (-7.11)	43.73 (-10.24)	52.67 (+6.26)
AdaptFormer-1	0.10 (0.12%)	83.52 (-2.38)	93.04 (-4.63)	83.64 (-6.45)	50.03 (-3.94)	51.68 (+5.27)
AdaptFormer-4	0.15 (0.17%)	84.83 (-1.07)	96.19 (-1.48)	85.42 (-4.67)	54.70 (+0.73)	51.81 (+5.40)
AdaptFormer-64	1.26 (1.46%)	85.90 (0.00)	96.89 (-0.78)	87.61 (-2.48)	59.02 (+5.05)	55.69 (+9.28)

- A lot of ablations with respect to the position.
- Parallel works better than sequential
 - The parallel design maintains the original feature using an independent branch.
 - The sequential design is equivalent to adding more layers, which might cause optimization difficulty.



Conclusions

- Deep Learning and AI will play a key role the next years in Precision Medicine
- Very active research area
- A number of challenges should be still addressed both from the engineering and medical society
- The collection of data is very important and define the use and selection of the algorithms
- Interpretability is very important for medical problems
- Fusion of models will be very important for clinical endpoints

